Lecture 3

Tuesday, 17 January 2023 7:08 AM

→ Term projects

→ Tutorial session (in 2 weeks)
(Need laptop)

Many Particles in a box — TISE (Time-independent), No particle interaction

N particles : $H = -\sum_{i=1}^{N} \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_i^2}$

$\Downarrow$

$-\sum_{i=1}^{N} \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_i^2} = E\psi$ $\qquad \iint\cdots\int \psi^*\psi \, dx_1 dx_2 \cdots dx_N = 1$

eg: 2 particles:

$\frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_1^2} - \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_2^2} = E\psi$

↓

Solving for ψ: Separation of variables : $\psi(x_1, x_2) = \phi_1(x_1)\,\phi_2(x_2)$

$\Rightarrow -\frac{\hbar^2}{2m}\phi_1''\phi_2 - \frac{\hbar^2}{2m}\phi_1\phi_2'' = E\phi_1\phi_2$

$\Rightarrow \underbrace{-\frac{\hbar^2}{2m}\frac{\phi_1''}{\phi_1}}_{f(x_1)} - \underbrace{\frac{\hbar^2}{2m}\frac{\phi_2''}{\phi_2}}_{f(x_2)} = E = \text{constant} = -\lambda^2$ (λ = real)

$\therefore\ -\frac{\hbar^2}{2m}\frac{d^2\phi_1}{dx_1^2} = (E-\lambda^2)\phi_1$; $-\frac{\hbar^2}{2m}\frac{d^2\phi_1}{dx_2^2} = \lambda^2\phi_2$

BC's: 1) $\psi(0, x_2) = 0$ 2) $\psi(L, x_2) = 0 = \psi(x_1, 0) = \psi(L, x_1)$

↳ $\phi_1(0) = \phi_1(L) = 0$; $\phi_2(0) = \phi_2(L) = 0$

$\therefore\ \phi_1(x_1) = A_1 \sin\left(\frac{2m(E-\lambda^2)}{\hbar^2}x_1\right) + \cancel{\cos(\cdots)}^{\,0}$

$\phi_2(x_2) = A_2 \sin\left(\sqrt{\frac{2m\lambda^2}{\hbar^2}}\,x_2\right) + \cancel{\cos(\cdots)}^{\,0}$

$\sqrt{\frac{2m(E-\lambda^2)}{\hbar^2}}\,L = n_1\pi$ $\qquad$ $\sqrt{\frac{2m\lambda^2}{\hbar^2}}\,L = n_2\pi$

$n_1 = 1, 2, 3 \ldots$ $\qquad$ $n_2 = 1, 2, 3, \ldots$

$E - \lambda^2 = \frac{n_1^2\pi^2\hbar^2}{2mL^2}$; $\lambda^2 = \frac{n_2^2\pi^2\hbar^2}{2mL^2}$

$$\boxed{E = \frac{(n_1^2 + n_2^2)\pi^2\hbar^2}{2mL^2} \qquad n_1, n_2 = 1, 2, \ldots}$$

Any linear combination of $\psi_{n_1 n_2}$'s is also a valid solution

$\psi(x_1, x_2) = \phi(x_1)\phi(x_2) = A_1A_2 \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)$

↳ Normalization : $A_1A_2 = \sqrt{\frac{2}{L}}\sqrt{\frac{2}{L}}$ $\Rightarrow$

$$\boxed{\psi_{n_1,n_2}(x_1, x_2) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_1}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_2}{L}\right)}$$

Q) What is the lowest energy state for a system of multiple particles (eg e⁻)?

Naive answer: $n_1 = n_2 = 1 \Rightarrow E_{GS} = \frac{\pi^2\hbar^2}{mL^2}$

$\psi_{1,1} = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x_1}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x_2}{L}\right)$

This analysis is correct for bosons (integer spin)

Relativistic QM, we know that fermions (half-integer spin) are e⁻

$\psi(x_1, x_2) = -\psi(x_2, x_1)$

(note: $\psi^*\psi(x_1, x_2) = \psi^*\psi(x_2, x_1)$)

1, 1 ↑ quantum no

If we take $\psi_{GS}(x_1, x_2) = \phi_{1,1}(x_1)\ \phi_{2,1}(x_2)$

$\phi_{1,1}(x_1)\,\phi_{2,1}(x_2) = \phi_{1,1}(x_2)\,\phi_{2,1}(x_1)$

(same because $\phi_1 = \phi_2 = \phi = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)$)

↳ Not true for fermions

For fermions w/o spin, we cannot have more than one particle at the lowest energy state

$\Rightarrow$ We can use Slater orbitals to ensure that linear combination of $\phi_{n_1}(x_1)$ & $\phi_{n_2}(x_2)$ are anti symmetric

We know: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = -\begin{vmatrix} c & d \\ a & b \end{vmatrix}$

S_0: $\Psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_1(x_1) & \phi_2(x_1) \\ \phi_1(x_2) & \phi_2(x_2) \end{vmatrix} = -\frac{1}{\sqrt{2}} \begin{vmatrix} \phi_1(x_2) & \phi_2(x_2) \\ \phi_1(x_1) & \phi_2(x_1) \end{vmatrix}$

n_1 n_2

Thus, GS for two fermions is:

$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}} \left[\phi_{n_1=1}(x_1)\, \phi_{n_2=2}(x_2) - \phi_{n_1=1}(x_2)\, \phi_{n_2=2}(x_1) \right]$$

In HW: Show

$$E = \frac{\int_0^L \int_0^L \Psi^* H \Psi \, dx_1 dx_2}{\iint \Psi^* \Psi \, dx_1 dx_2}$$

w/ Ψ being antisym

For many particle systems, we can make more progress w/ DFT

Idea: Total ground state e^- density of atoms or molecules uniquely determines all electronic structure properties (i.e. energies, positions, wavefunctions)

↓

TISE: $H\Psi = E\Psi$

$$H = \underbrace{\sum_{i,e^-} -\frac{\hbar^2}{2m_e} \nabla_i^2}_{\text{electron kinetic energy}\ (1)} + \underbrace{\frac{1}{2} \sum_{i}^{N} \sum_{j \neq i} \frac{e^2}{4\pi\varepsilon_0 |r_i - r_j|}}_{\text{Coulomb repulsion}\ (2)} + \underbrace{\sum_{i}^{e^-} \sum_{j}^{\text{nuclei}} \frac{Z_\alpha e^2}{4\pi\varepsilon_0 |r_i - R_j|}}_{\text{electron-nuclei interaction}\ (3)}$$

Bohn-Heimer approximation: assumes nuclear motion is separated (since it is much heavier)

Since Ψ gives all knowable properties of system, if we know H, then we know all properties of system

If H is completed determined by total ρ (e^- density), then ρ determines all properties

Total electron density defined as:

$$n(\underline{x}) = \rho(\underline{x}) = N \iint \cdots \iint \Psi^* \Psi \, d\underline{x}_2 \, d\underline{x}_3 \cdots d\underline{x}_N$$

} Probability that any of e^-'s will be found in a box

$\therefore \int \rho(\underline{x}) \, dx dy dz = N$ and thus determines (1) and (2) of H

Kohn & Hohenberg's 1st theorem (1964)

states that $\rho(\underline{x})$ also determines term (3)

(i.e. to determine H from any potential external to the e^-)

$\Longrightarrow$ H is completely determined by $\rho(x)$

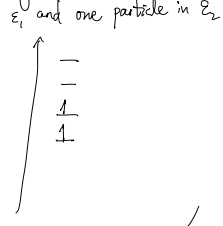
In principle, $\rho(x)$ determines E_n & Ψ_n (not just the GS)

→ $F[y(x)] = \int_a^b y(x) \cdot dx$ → gives given

energy level

$= \varepsilon_1 + \varepsilon_2$

The interpretation being one particle in ε_1 and one particle in ε_2



Simplified Pauli Exclusion principle

mmetric

$(dxdydz)$

$$F[\rho] = \int f(\underline{r}, \rho(\underline{r}), \nabla\rho(\underline{r})) \cdot d\underline{r}$$

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a function as

The functional derivation is

In principle, $\rho(\underline{r})$ determines E_n & ψ_n [illegible]

$F[y(x)] = \int_a^b y(x)\,dx \rightarrow$ gives given [illegible]

K & H's second theorem

States that there is a unique functional of $\rho(\underline{r})$ that, when minimized w.r.t $\rho(\underline{r})$, gives GS E and $\rho(\underline{r})$ of system: $\min_{\rho(\underline{r})} \{ E[\rho(\underline{r})] \} = E_{GS}$

Simple functional for 1 dimensional problems:

$E_{GS} = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_N$ $\quad N$ = total # of particles (Non-interacting fermions)

For PIB: $\varepsilon_{ni} = \frac{n_i^2 \pi^2 \hbar^2}{2mL^2}$ (ignore spin)

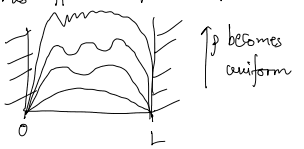
$$E = \sum_{i=1}^{N} \frac{\pi^2\hbar^2}{2mL^2} n_i^2 \quad ; \quad n_i = 1, 2, 3 \dots N$$

$$= \frac{\pi^2\hbar^2}{2mL^2} \left(\sum_{i=1}^{N} n_i^2 \right) \rightarrow \text{can be evaluated analytically}$$

$$E \approx \frac{\pi^2\hbar^2}{2mL^2} \left(\frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N \right) \approx \frac{\pi^2\hbar^2}{2mL^2} \frac{N^3}{3}$$

dominant for Large N

$\rho(x)$ becomes approx uniform in space



Thus if ρ = constant, $N = \int_0^L \rho(x)\,dx \approx \rho \int_0^L dx = \rho L$

$$\therefore \rho = \frac{N}{L}$$

So, $\boxed{\frac{E}{N} = \frac{\pi^2\hbar^2}{6mL^2} N^2 = \frac{\pi^2\hbar^2}{6m} \rho^2}$ → KE per particle (non-interacting fermions) ; $e_{kin}(\rho)$

out a number
a function as
ut

The functional derivation is

$$\frac{\delta F}{\delta \rho(r)} = \frac{\partial f}{\partial \rho} - \nabla \cdot \frac{\partial f}{\partial \nabla \rho}$$

and this can be used to find the ground state when functional derivative is set to zero

Lecture 4

Thursday, 19 January 2023 10:30 AM

Simple KE functional: $\frac{E}{N} = e_{kin}(\rho) = \frac{\pi^2\hbar^2}{6m}\rho^2$ (for non-interacting fermions in a box)

↓ energy per per particle

→ $\rho(x)\cdot dx$ is no. of particles in length dx

→ $e_{kin}(\rho)(\rho\cdot dx)$ is KE in length dx

→ Total energy is $E_{KE}[\rho(x)] = \int e_{kin}(\rho)\cdot\rho\,dx$ over all space } general KE expression

If there is potential energy or interactions, $E_{kin}[\rho(x)] = \int e_{tot}(\rho)\,\rho\cdot dx$

Ex: Reproduce PIB energies for N particles

$$E[\rho] = \int_0^L \frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx \quad ; \text{ Constraint: } \int_0^L \rho(x)\cdot dx = N$$

(to avoid trivial solution)

Use Lagrange multipliers:

define $F[\rho] \equiv E[\rho] - \mu\left(\int_0^L \rho(x)\cdot dx - N\right)$

↑ Lagrange multiplier

Note: $\frac{\partial}{\partial\rho}\int F(\rho,\nabla\rho)\cdot dr = \frac{\partial f}{\partial\rho} - \frac{\partial f}{\partial\nabla\rho}$

$F(\rho,\nabla\rho)$: function of ρ & grad ρ

Then minimize

$$\frac{\partial F(\rho)}{\partial\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 - \mu\left(\frac{\partial}{\partial\rho}\int\rho\,dx - \frac{\partial N}{\partial\rho}\right)$$

($\int\rho\,dx = 1$, $\frac{\partial N}{\partial\rho} = 0$)

$$\therefore \mu = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2$$

↓

$$\int_0^L \sqrt{\frac{2m\mu}{\pi^2\hbar^2}}\cdot dx = N \Rightarrow \sqrt{\frac{2m\mu}{\pi^2\hbar^2}} = \frac{N}{L}$$

$$\boxed{\rho_{min} = \frac{N}{L}}$$

$$\therefore \boxed{E_{min} = \frac{\pi^2 \hbar^2}{6m} \frac{N^3}{L^3} L}$$

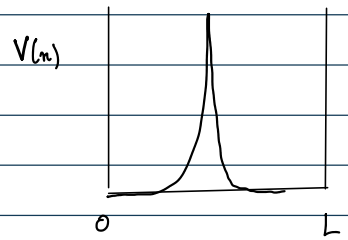
This approximation improves for large N:

N	E_{exact}	E_{DFT}	$\frac{DFT}{exact}$
1	1	1/3	0.33
4	30	64/3	8.71
20	2870	8000/3	0.93

Principle: DFT gives more accurate results for systems that are similar to the systems for which the functional was derived

However, results are surprisingly good for different system types, as well

Example 2



δ function spike

Properties of δ: 1) $\int_0^\infty \delta(x)\,dx = 1$ (thin & tall)

2) $\int_0^L f(x)\,\delta\left(x-\frac{L}{2}\right)dx = f\left(\frac{L}{2}\right)\int_0^L \delta\left(x-\frac{L}{2}\right)dx = f\left(\frac{L}{2}\right)$

New functional: $E[\rho] = \int_0^L \frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \underbrace{\alpha\int \delta\left(x-\frac{L}{2}\right)\rho\,dx}_{\text{external PE}}$

Lagrange functional:

$$F[\rho] \equiv E[\rho] - \mu\left(\int_0^L \rho\,dx - N\right)$$

$$\frac{\partial F}{\partial \rho} = -\mu + \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + \frac{\partial}{\partial \rho}\left(\rho\left(\frac{L}{2}\right)\cdot\alpha\right) \to 0$$

$$\therefore \rho_{min} = \sqrt{\frac{2m\mu}{\pi^2\hbar^2}} = \frac{N}{L} \text{ constant}$$

S does not affect density

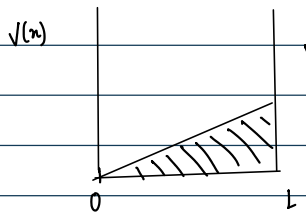
$$E_{min} = \frac{\pi^2\hbar^2}{6m}\left(\frac{N}{L}\right)^3 L + \frac{\alpha N}{L}\int_0^L \delta(x-\tfrac{L}{2})\cdot dx \quad (\int \delta = 1)$$

$$\therefore E_{min} = \frac{\pi^2\hbar^2}{6m}\left(\frac{N}{L}\right)^3 L + \frac{\alpha N}{L}$$

S does add $\frac{\alpha}{L}$ per particle → (α = constant)

physically an obstacle (or nucleus) in the center will act as a S function (it will affect the energy everytime an e^- crosses it in the particle in box example)

Example 3



$$V_{ext}(x) = \frac{V_{max}}{L}x$$

$$E_{tot}[\rho] = \frac{\pi^2\hbar^2}{6m}\rho^2 + \frac{x}{L}V_{max}$$

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\cdot dx + \int_0^L \frac{V_{max}}{L}x\cdot\rho\,dx$$

Lagrange multiplier:

$$\frac{\partial F}{\partial \rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + \frac{V_{max}}{L}x - \mu$$

$$\therefore \rho_{min} = \sqrt{\frac{2m}{\pi^2\hbar^2}\left(\mu - \frac{V_{max}}{L}x\right)}$$ (ρ_{min} is not constant)

Finding μ: $\int_0^L \rho_{min}\cdot dx = N$

$$\int_0^L \sqrt{\frac{2m}{\pi^2\hbar^2}\left(\mu - \frac{V_{max}}{L}x\right)}\cdot dx = N$$

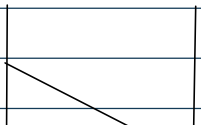
u-sub: $u = \mu - \frac{V_{max}}{L}x$

$$(\mu - V_{max})^{3/2} - \mu^{3/2} = -\frac{3}{2}\frac{V_{max}N}{L}\sqrt{\frac{\pi^2\hbar^2}{2m}}$$

Either solve numerically or solve for small $\frac{V_{max}}{\mu}$.

$$\mu \approx \frac{\pi^2\hbar^2}{2m}\frac{N^2}{L^2} + O(V_{max})$$

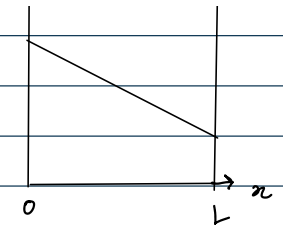
$$\rho_{min}[x] = \frac{N}{L}\left[1 - \frac{V_{max}x}{L}\frac{2mL^2}{\pi^2\hbar^2N^2}\right]$$



Electric field pushes on particle, less probability for it to be on

$$\rho_{min}[x] = \frac{N}{L}\left[1 - \frac{V_{max}\,x}{L}\,\frac{2mL^2}{\pi^2\hbar^2N^2}\right]$$

$\gamma = \frac{N}{L}$ $\quad \beta = \frac{V_{max}}{L}\frac{2mL^2}{\pi^2\hbar^2N^2}$



Electric field pushes on particle, less probability for it to be on that side of the box

$$\rho = \gamma(1-\beta x)$$

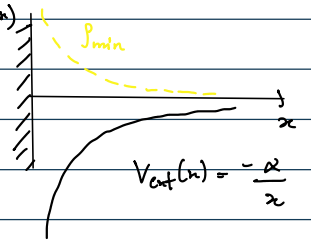
$$E[\rho_{min}] = \frac{\pi^2\hbar^2}{6m}\int_0^L \gamma^3(1-\beta x)^3\,dx + \frac{\gamma V_{max}}{L}\int_0^L x(1-\beta x)\,dx$$

$$= -\frac{1}{\beta}\frac{\pi^2\hbar^2}{6m}\frac{\gamma^3}{4}\underbrace{\left[(1-\beta L)^4 - 1\right]}_{\approx -4\beta L \text{ (for small } \beta)} + \frac{V_{max}}{L}\gamma\left(\frac{L^2}{2} - \beta\frac{L^3}{3}\right)$$

$$\boxed{E_{min} = \frac{\pi^2\hbar^2}{6mL^2}N^3 + \frac{1}{2}V_{max}N} \quad \text{for small } V_{max}$$

↳ So we add $\frac{1}{2}V_{max}$ to energy of each particle (average)

HW preview: V(x)



Coulomb potential

$\mu < 0$ → need absolute values. (replace $\mu = -|\mu|$)

Answer: $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^\infty \rho^3\,dx - \alpha\int_0^\infty \frac{\rho(x)}{x}\,dx$

$$0 = \frac{\partial F}{\partial \rho} \rightarrow \quad \rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\left(\mu + \frac{\alpha}{x}\right)^{1/2}$$

$$\Rightarrow N = \int_0^\infty \rho_{min}\,dx = \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\int_0^{\alpha/|\mu|}\left(\frac{\alpha/|\mu|}{x} - 1\right)^{1/2}dx\;|\mu|^{1/2}$$

ρ can never be imaginary!

So truncate domain to keep ρ_{min} real $\left(\frac{\alpha}{|\mu|}\right)$

$$\mu = -\frac{m}{2\hbar^2}\frac{\alpha^2}{N^2}$$

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}N^3 + \frac{\pi^2\hbar^2}{6m}\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}|\mu|^{\frac{3}{2}}\int_0^{\alpha/|\mu|}\left(\frac{\alpha/|\mu|}{x} - 1\right)^{3/2}dx$$

$$E_{min}[\rho_{min}] = \frac{\pi^2\hbar^2}{6mL^2}N^3 + \frac{\pi^2\hbar^2}{6m}\int_0^{\alpha/|\mu|}\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}|\mu|^{\frac{3}{2}}\left(\frac{\alpha/|\mu|}{x}-1\right)^{3/2}\cdot dx$$

diverges (since we are modelling an atom in 1D)

↓

Larger N means that particles will start to pack in the lower potential regime → nowhere for them to go

↓

unphysically packed

Thomas-Fermi Atom

Interactions b/w particles

1) Coulomb : classic electrostatic interactions

$E \sim \frac{e^2}{4\pi\varepsilon_0 x}$ repulsion b/w electrons

Electrostatic functional:

$$E_{coul} = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint\frac{\rho(x)\,\rho(x')}{(x-x')}\,dx\,dx'$$

dummy var

avoid double counting

2) Exchange : purely QM property/interaction (attractive and b/w charged particles)

↳ Fully accounted for in TISE, but DFT requires extra terms

Simple approximation w/ dimensional analysis

$$E_{exch} \sim \int\frac{e^2}{\varepsilon_0}\rho^n\cdot dx$$

$n=2$ gives dimensional consistency

Putting it together: (in 1D)

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint \frac{\rho(x)\rho(x')}{|x-x'|}\,dx\,dx' - C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx + \int V_{ext}\,\rho\,dx$$

V_{ext} could be $-\frac{\alpha}{x}$, $\frac{1}{2}kx^2$, ...

Lecture 5

Tuesday, 24 January 2023 10:30 AM

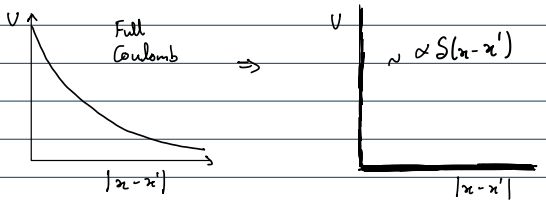
- Start thinking of term projects → List 2-3 thoughts
- Tutorial session → next thursday

One-dimensional Atom

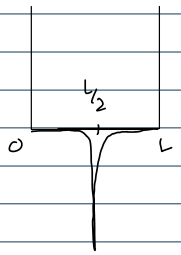
$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint \frac{\rho(x)\rho(x')}{|x-x'|}\,dx\,dx' - \underbrace{C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx}_{\text{Exchange energy}} + \int V_{ext}(x)\,\rho\,dx$$

Simplify Coulomb terms using delta functions:

$e^- e^-$ repulsion



$$V_{ext} = -\gamma\,\delta\left(x-\frac{L}{2}\right)$$



From an e^- perspective, nucleus is external.

Thus, our 1D atom becomes,

(e^--e^- interactions; e^- nucleus interaction)

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\,dx - C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int_0^L\int_0^L \delta(x-x')\rho(x)\rho'(x)\,dx'\,dx - \gamma\int_0^L \delta\left(x-\frac{L}{2}\right)\rho\,dx$$

$$= \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int_0^L [\rho(x)]^2\,dx - C_x\frac{e^2}{4\pi\varepsilon_0}\int_0^L \rho^2\,dx - \gamma\,\rho\left(\frac{L}{2}\right) - \mu\left(\int_0^L \rho\,dx - N\right)$$

$$\frac{\delta E}{\delta \rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + 2\frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{2} - C_x\right)\rho_{min} - \mu$$

($\frac{1}{2}$: Coulomb; C_x: exchange)

ρ_{min} = constant →

$$\int_0^L \rho_{min}\,dx = N$$

$$\rho_{min} = \frac{N}{L}$$

↙ Plugging back into energy functional

Electron Density

$$E_{e-e}[\rho] = \frac{\pi^2\hbar^2}{6}N^3 + \frac{e^2}{8\pi\varepsilon_0}\left(1-C_x\right)N^2 - \gamma N$$

Electron Density

energy functional

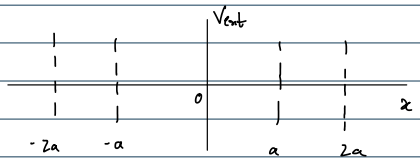
$$E_{GS} = E[\rho] = \underbrace{\frac{\pi^2\hbar^2}{6mL^2}N^3}_{KE} + \underbrace{\frac{e^2}{4\pi\varepsilon_0 L}\left(\frac{1}{2} - C_m\right)N^2}_{\text{interaction term}} - \underbrace{\frac{\gamma}{L}N}_{\text{attractions to nucleus}}$$

Note: These treatments work okay for atoms but not for molecules

DFT & TISE w/ periodic boundary conditions

eg: hot catalysts might have some periodicity

1) Basic idea: $V_{ext}(x) = V_{ext}(x+a)$



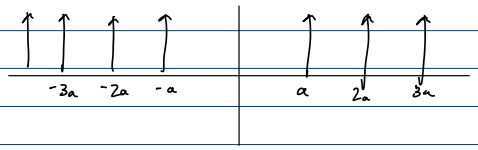
If $V_{ext}(x)$ is periodic, observable properties are also periodic: $\rho(x) = \rho(x+a)$

Approach $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{-na}^{+na}\rho^3\,dx + \int_{-na}^{na}V_{ext}(x)\,\rho\,dx$; n = some integer

↳ $\frac{\delta F}{\delta\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + V_{ext}(x) - \mu$

$\rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\left(\mu - V_{ext}(x)\right)\right)^{1/2}$ → periodic in x ✓

Cr: Dirac Comb:



$V_{ext}(x) = \alpha\sum\delta(x-na)$; n is a positive or negative integer

P periods are analyzed

$$E[\rho] = \int_{-\frac{P}{2}a}^{\frac{P}{2}a}\frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \int_{-\frac{Pa}{2}}^{\frac{Pa}{2}}\alpha\sum\delta(x-na)\,\rho\,dx$$

$\frac{\delta F}{\delta\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 - \mu$; $\rho_{min} = \text{constant} = \frac{N}{L}$ (N is total # of particles in all P periods)

$E[\rho_{min}] = \frac{\pi^2\hbar^2}{6m}\frac{N^3}{(Pa)^2} + \alpha\,\rho(na)\,P$; Let $N_{cell} = \frac{N}{P}$

($\rho(na) \to \frac{N}{L}$)

$$\boxed{\frac{E_{GS}}{P} = \frac{\pi^2\hbar^2}{6ma^2}N_{cell}^3 + \frac{\alpha}{a}N_{cell}}$$

energy per unit cell in a Dirac potential

Kohn-Sham DFT

+ shares many features w/ TISE treatment of non-interacting particles ⇒ Need to study periodic treatment of TISE

+ Shares many features w/ TISE treatment of non-interacting particles ⇒ Need to study periodic treatment of TISE

+ As w/ DFT, assume $V(x) = V(x+na)$ n = integer

+ All measurable quantities are also periodic $\rho(x) = \rho(x+a)$

$$|\Psi(x)|^2 = \Psi^*\Psi = |\Psi(x+a)|^2 \quad ; \text{ wavefunctions are complex so } \Psi(x) \neq \Psi(x+a)$$

+ Most general periodicity is: $\Psi(x+a) = \lambda\Psi(x)$ w/ $\lambda^*\lambda = 1$

+ If we have P total periods, and we assume periodic BC's (the system "wraps around" itself)

$$\Psi(x+Pa) = \lambda^P\Psi(x) = \Psi(x) \quad ; \quad \lambda^P = 1 = e^{2n\pi i} = \cos(2n\pi) + i\sin(2n\pi) \quad ; \quad n = 0, \pm1, \pm2, \ldots$$

$$\lambda = e^{\frac{2\pi n}{P}i} = e^{iKa} \quad ; \quad K = \frac{2\pi n}{Pa}$$

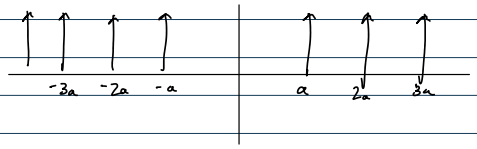
(K: wave number; Pa: total length of system)

wavefunction displaced by a period →

$$\therefore \underset{K}{\Psi}(x+a) = \Psi_K(x)e^{iKa} \quad ; \quad K = \frac{2\pi n}{Pa} \quad ; \quad n = 0, \pm1, \pm2, \ldots \quad ; \quad K \text{ is quantum number depending on periodicity}$$

} Bloch's Theorem

Ex: Dirac Comb:



For $0 < x < a$, $V(x) = 0$; so TISE is $-\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = E\Psi$

$$\frac{d^2\Psi}{dx^2} = -k^2\Psi \quad ; \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\Psi(x) = A\sin(kx) + B\cos(kx) \quad ; \quad 0 < x < a$$

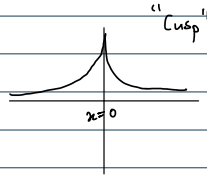
From Bloch's theorem for $-a < x < 0$,

$$\Psi(x+a) = e^{iKa}\left[A\sin(k(x+a)) + B\cos(k(x+a))\right]$$

Initial conditions: @ $x=0$, $\Psi_{0^-} = \Psi_{0^+}$ (Ψ is cont.)

$$\hookrightarrow B = e^{-iKa}\left[A\sin(ka) + B\cos(ka)\right]$$

Also, $\left.\frac{d\Psi}{dx}\right|_{0^+} - \left.\frac{d\Psi}{dx}\right|_{0^-} = \frac{2m\alpha}{\hbar^2}\Psi(0)$



$$kA - e^{iKa}k\left[A\cos(ka) - B\sin(ka)\right] = \frac{2m\alpha}{\hbar^2}B$$

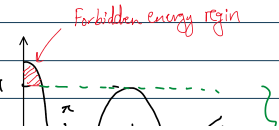
Eliminate A & B:

$$\boxed{\cos(Ka) = \cos(ka) + \frac{m\alpha a}{\hbar^2}\frac{\sin(ka)}{ka}}$$

Solve for k given K:

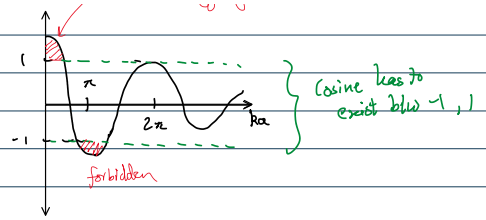
(graphically)

For $\frac{m\alpha a}{\hbar^2} = 1$



For $\frac{mEa}{\hbar^2} = 1$

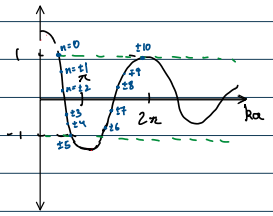
$K = \frac{2\pi n}{Pa}$



Certain values of $k (\sim E^{1/2})$ are forbidden by periodicity.
This causes band structure in periodic systems

K is quantized, as $P \to \infty$ K becomes continuous

↳ Try P = 10



States for P+1 non-interacting fermions in first band (ignoring spin) and P states for all other bands

To get total E, sum energies of particles (non-interacting)

$$E_{tot} = \sum_{particles} \varepsilon_n = \sum_{parti} \frac{\hbar^2}{2m} k_n^2 \quad ; \text{ If } N \text{ is large we can approximate sum w/ integral}$$

$$= \sum \frac{\hbar^2}{2m} k_n^2 \Delta n \quad ; \quad \Delta n = \frac{Pa}{2\pi} \Delta K$$

$$= \frac{Pa}{2\pi} \sum \frac{\hbar^2}{2m} k_n^2 \Delta K \approx \frac{Pa}{2\pi} \int \frac{\hbar^2}{2m} [k(K)]^2 \cdot dK$$

$$\boxed{\frac{E_{tot}}{P} \approx \frac{a}{2\pi} \int_{\substack{\text{occupied} \\ \text{bands}}} \frac{\hbar^2}{2m} [k(K)]^2 \, dK}$$

Energy per cell in periodic system of non-interacting particles ($P \to \infty$) K is cont.

Lecture 6

Thursday, 26 January 2023 10:30 AM

Email purdue username ; Send 2-3 preliminary ideas for term projects

Periodic TISE solution

$V(x+a) = V(x)$

$$\frac{E}{\rho} = \frac{a}{2\pi} \int_{\text{occupied levels}} \frac{\hbar^2}{2m} [k(K)]^2 dK \quad ; \quad K = \frac{2\pi n}{P_a} \quad ; \quad n = 0, \pm 1, \pm 2 \ldots$$

Algorithm to solve:

1) Pick values of K for numerical integration "K points"

2) At each K point, find k(K) from TISE solution or Kohn-Sham (K-S)

3) Advantage: only need TISE / K-S solutions for single unit cell

4) Small lattice constant "a" → larger range of K for integration → need more K points to converge the integral

large "a" → fewer k points

Rule of thumb: # K pts $\gtrsim \frac{2\pi}{0.3a}$ (Each dimension) } K point analysis

Kohn-Sham DFT

Normal DFT: extremely efficient, but kinetic energy functionals aren't good enough (except for low atomic #'s)

∴ Idea: Use a more accurate treatment of KE, at cost of (much) greater computational expense

↓

Introduced a fictitious, non-interacting gas of e^-'s that has the true, total e^- density $\rho(\underline{r})$ — Slater determinant?

$$\rho(\underline{r}) = \sum_{i=1}^{N} |\psi_i(\underline{r})|^2$$, where $\psi_i(\underline{r})$ are fictitious, single e^- orbitals that don't interact (note: $\rho(\underline{r}) = N \iiint \Psi^* \Psi \, dx_1 \, dx_2 \, dx_3 \ldots dx_N$)

Total Ψ is slater det of ψ_i's (→ anti-symmetric)

KE's for non-interacting particles:

$$\frac{-\hbar^2}{2m} \sum_{i=1}^{N} \int \psi_i^*(\underline{r}) \nabla^2 \psi_i(\underline{r}) \, d\underline{r}$$

{ in PIB: ψ ‖ $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E_{KE}\psi$; $\frac{-\hbar^2}{2m}\int \psi^* \frac{d^2\psi}{dx^2} dx = E_{KE}$ }

Total K-S functional:

$$E[\rho] = \underbrace{-\frac{\hbar^2}{2m} \int \psi_i^*(\underline{r}) \nabla^2 \psi_i(\underline{r}) \, d\underline{r}}_{T_s[\rho]} + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} \iint \frac{\rho(\underline{r})\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r} \, d\underline{r}' + \underbrace{E_{xc}[\rho]}_{\text{residual KE excess + exchange + corr.}} + \int V_{ext}(\underline{r}) \rho \, d\underline{r} \quad \ldots\ldots \text{(Eq 1)}$$

Now Lagrangian (w/ constraint of $\int \rho(\underline{r}) \cdot d\underline{r} = N$): $\frac{\partial F[\rho]}{\partial \rho} = \frac{\partial T_s[\rho]}{\partial \rho} + V_{ext}(\underline{r}) + \frac{\partial E_{xc}[\rho]}{\partial \rho} + \frac{e^2}{4\pi\varepsilon_0}\int \frac{\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}' - \mu$

Defined $V_{eff}(\underline{r}) = V_{ext}(\underline{r}) + \frac{\partial E_{xc}}{\partial \rho} + \frac{e^2}{4\pi\varepsilon_0}\int \frac{\rho}{|r-r'|} \cdot dr'$

$$\frac{\partial T_s[\rho]}{\partial \rho} + V_{eff}(\underline{r}) = \mu$$

$$\frac{\partial T_s[\rho]}{\partial \rho} + V_{ext}(\underline{r}) = \mu$$ Non-interacting systems

Since full TISE separates into single particle equations for non-interacting particles, K+S reasoned this must also be possible for their "fictitious" system

Kohn-Sham eqn's

$$\boxed{\left[-\frac{1}{2}\nabla_i^2 + V_{ext}(\underline{r}) + \frac{\partial E_{xc}}{\partial \rho} + \int \frac{\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}'\right]\psi_i = \varepsilon_i \psi_i \quad ; \quad i = 1 \cdots N}$$

Algorithm:
1) Guess fictitious ψ_i
2) Find $\rho = \sum |\psi_i|^2$
3) Evaluate $V_{eff}(\underline{r})$ ← Kohn-Sham potential
4) Solve K-S eqs to get new ψ_i's
5) Repeat 1-4

Once converged:
$\Psi(\underline{r}_{all})$ = slater det of ψ_i's

Then find total energy: $E[\rho_{min}] = \sum_{i=1}^{N} \varepsilon_i - \frac{1}{2}\iint \frac{\rho(\underline{r})\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}\, d\underline{r}' - \int \frac{\partial E_{xc}}{\partial \rho} \rho\, dr + E_{xc}[\rho]$

or use the earlier equation (1)

Example: Two fermions in a 1D box w/ "δ-function repulsion"

K-S energy is: $E[\rho] = -\frac{1}{2}\left(\int_0^L \psi_1^* \frac{d^2\psi}{dx_1^2} dx_1 + \int_0^L \psi_2^* \frac{d^2\rho}{dx_2^2} dx_2\right) + \frac{\alpha}{2}\left(\int_0^L\int_0^L \rho(x)\rho(x')\,\delta(x-x')\, dx_1\, dx_2\right)$

(ignore exchange & correlation) → Can only do this to simple problem → In real case we need to have guesses for these

$\therefore$ K-S Eqs are: $\left[-\frac{1}{2}\frac{d^2}{dx_i^2} + \alpha\rho(x)\right]\psi_i = \varepsilon_i\psi_i$

2 particles: $\left.\begin{aligned} -\frac{1}{2}\frac{d^2\psi_1}{dx_1^2} + \alpha\rho\psi_1 &= \varepsilon_1\psi_1 \\ -\frac{1}{2}\frac{d^2\psi_2}{dx_2^2} + \alpha\rho\psi_2 &= \varepsilon_2\psi_2 \end{aligned}\right\}\ \rho(x) = |\psi_1|^2 + |\psi_2|^2$

Basis expansion: $\psi_1 = \sum_{n=1}^{\infty} a_n \phi_{1n}(x_1) \qquad \psi_2 = \sum_{n=1}^{\infty} b_n \phi_{2n}(x_2)$

PIB solutions: $\phi_n(x) = \sqrt{2}\sin(n\pi x)$ [assume $L=1$]

$$-\frac{1}{2}\frac{d^2}{dx_1^2}\sum_{n=1}^{\infty} a_n \phi_{1n}(x_1) + \alpha\left[\left(\sum_{n=1}^{\infty} a_n\phi_{1n}\right)^2 + \left(\sum b_m \phi_{2m}\right)^2\right]\sum_{j=1}^{\infty} a_j \phi_{1j}(x_1) = \varepsilon_1 \sum_{n=1}^{\infty} a_n \phi_{1n}(x_1)\ ;\ \text{Similar for } x_2$$

Eigenvalue problem (to determine coefficients) & basis functions

$$\Rightarrow -\frac{1}{2}\frac{\pi^2}{2}\sum_{n=1}^{\infty} n^2 a_n \phi_{1n}(x_1) + \quad " \quad "$$

→ Basis functions are orthonormal ∴ Multiply by $\phi_{1k}(x_1)$ and integrate from 0 to 1

$$\frac{\pi^2}{2}k^2 a_k + \alpha\int_0^1 \sum_n\sum_m (a_n a_m + b_n b_m)\,\phi_{1n}\phi_{1m}\underbrace{\sum_{j=1}^{\infty} a_j \phi_{1j}\phi_{1k}}\,dx_1 = \varepsilon_1 a_k$$

represents particle interactions

Similar for ε_2, but don't need to solve that since particles are indistinguishible

Note: $$\int_0^1 \phi_{1n}\phi_{1m}\phi_{1j}\phi_{1k}\,dx = \frac{1}{2}\left[\delta_{n-m-j+k} + \delta_{n-m+j-k} - \delta_{n-m-j-k} - \delta_{n-m+j+k} - \delta_{n+m-j+k} - \delta_{n+m+j-k} + \delta_{n+m-j-k} + \delta_{n+m+j+k}\right]$$

$$\Rightarrow \frac{\pi^2}{2}k^2 a_k + \frac{1}{2}\alpha\sum_n\sum_m (a_n a_m + b_n b_m)\left(a_{n-m+k} + a_{k-n+m} - a_{n-m-k} - a_{m-n-k} - a_{n+m+k} - a_{k-n-m} + a_{n+m-k} + a_{-n-m-k}\right) = \varepsilon_1 a_k\ ;\ \text{Similar for } \varepsilon_2$$

Non-linear eigenvalue solved self-consistently w/ an initial guess for $\{a_i\}$ & $\{b_i\}$

If $n_{max}=2$, $\psi_1^\circ = a_1^\circ \phi_{11}(x_1) + a_2^\circ \phi_{12}(x_1)$

$\psi_2^\circ = b_1^\circ \phi_{21}(x_1) + b_2^\circ \phi_{22}(x_1)$

Plugging ↓ Guess $a_1^\circ = 1$, $a_2^\circ = 0$, $b_1^\circ = 0$, $b_2^\circ = 1$ $\Rightarrow$ $\rho^\circ = a_1^{\circ 2} + b_2^{\circ 2} = 2$

$$\Rightarrow \frac{\pi^2}{2}a_1' + \frac{1}{2}\alpha\left[1\cdot 3a_1' + 1\cdot 2a_1'\right] = \varepsilon_1 a_1' \ ;\ k=1$$

$$2\pi^2 a_2' + \frac{1}{2}\alpha\left[1\cdot 2a_2' + 1\cdot 3a_2'\right] = \varepsilon_1 a_2' \ ;\ k=2$$

from ρ°

$$\frac{\pi^2}{2}a_1 + \frac{5}{2}\alpha a = \varepsilon_1 a_1 \ ;\ 2\pi^2 a_2 + \frac{5}{2}\alpha a_2 = \varepsilon_1 a_2$$

$$\left(\frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1\right)a_1 + 0\cdot a_2 = 0 \ ;\ a_1 + \left(2\pi^2 + \frac{5}{2}\alpha - \varepsilon_1\right)a_2 = 0$$

$$\therefore \begin{bmatrix} \frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1 & 0 \\ 0 & 2\pi^2 + \frac{5}{2}\alpha - \varepsilon \end{bmatrix} = 0$$

$$\left(\frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1\right)\left(2\pi^2 + \frac{5}{2}\alpha - \varepsilon_1\right) = 0$$

$$\therefore \varepsilon_1 = \frac{\pi^2}{2} + \frac{5}{2}\alpha \quad ; \quad \varepsilon_{12} = 2\pi^2 + \frac{5}{2}\alpha$$

Solve for a_1 & a_2:

$$\varepsilon_1 = \frac{\pi^2}{2} + \frac{5}{2}\alpha \rightarrow a_2^1 = 0 \,,\; a_1^1 = 1$$
$$\varepsilon_1 = 2\pi^2 + \frac{5}{2}\alpha \rightarrow a_1^1 = 0 \,,\; a_2^1 = 1$$

} Iteration has converged

$$\therefore \varepsilon = \begin{cases} \frac{\pi^2}{2} + \frac{5}{2}\alpha \\ 2\pi^2 + \frac{5}{2}\alpha \end{cases}$$

$$\Psi_{min}(x_1, x_2) = \frac{1}{\sqrt{2}} \left\{ \phi_{11}(x_1)\,\phi_{21}(x_2) - \phi_{21}(x_2)\,\phi_{12}(x_1) \right\}$$

$$\rho_{min} = 2\sin^2(\pi x) + 2\sin^2(2\pi x)$$

$$E[\rho_{min}] = \frac{5}{2}\pi^2 + \frac{5}{2}\alpha$$

Not the sum one particle

Normal DFT $\Rightarrow \rho_{min} = \frac{N}{L} = 2$ (constant)

$\hbar = m = 1$

$$E[\rho_{min}] = \frac{4}{3}\pi^2 + 2\alpha$$

Exact TFSF: $E_{GS} = \frac{5}{2}\pi^2$

KS more accurate for Al+ atoms

↑

Al & beyond.

(but DFT approaches KS for smaller atomic sizes)

Lecture 7 (Thermo)

Tuesday, 31 January 2023 10:31 AM

Stationary states : solutions to TISE $\quad i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$

$\hat{H}\Psi = E_n \Psi_n$; $\Psi_n \rightarrow$ stationary state (physically observable properties are constant)

Basis Set

Basis functions to solve K-S eqns ; Larger basis set → more accurate solutions, but more computational time } w/ convergence tests [Consistent for all DFT Codes]

Basis set size given as energy cutoff :

PIB : $E_{pw_j} = \frac{\hbar^2 \pi^2}{2m}\left(\left(\frac{n_x}{L_x}\right)^2 + \left(\frac{n_y}{L_y}\right)^2 + \left(\frac{n_z}{L_z}\right)^2\right)$; pw → plane wave

$n_x, n_y, n_z = 1, \ldots \infty$

Choose total plane wave energy "cut off energy" → Determine no. & type of plane waves that fall in this E bound

→ In 1D, for a unit w/ d ~ 1nm, usually 4-6 basis functions for convergence → E ~ 10-15 eV } Codes can be sped up w/ 1000 of tricks

→ In 3D, unit cell V ~ $1\ nm^3$, thousands of planewaves needed ~ cutoff ≈ 400 eV

Recall K-S eqns:

$$\left[-\frac{1}{2}\nabla_i^2 + V_{ext}(r) + \int \frac{\rho(r')}{|r-r'|}dr' + \frac{\delta E_{xc}}{\delta \rho}\right]\Psi_i = \varepsilon_i \Psi_i$$

↑ This is where most approximations are made

Possible choices:

1) LDA → Local density approximation : (very expensive)

Similar to class functionals → function of ρ(r) only

Fit to QMonte Carlo calc. of homogenous e^- gas

$E_{xc}^{LDA} = E_{xc}^{LDA}[\rho]$ (Closed form)

+ Works best for bulk materials/systems w/ slowly varying (wrt space) ρ's.
+ Does not work well for surfaces or systems w/ sharp ρ gradients

2) Generalized Gradient Approximation (GGA)

All are explicit in ρ and $\nabla\rho$: $E_{xc}^{GGA} = E_{xc}^{GGA}(\rho, \nabla\rho)$

+ Standard for surfaces and interfaces
ex: PBE → Purdue-Burke Ernzerhoff
RPBE → Revised "
PW91 →
BEEF → Bayesian estimate

3) Meta-GGA

Incorporates second derivatives : $E_{meta} = E_{meta}[\rho, \nabla\rho, \nabla^2\rho]$

ex:
+ SCAN
+ Improvement in performance is relatively small

4) Hybrid/Hyper functionals

Based on GGA functionals, but add a fraction of exact exchange from Hartree Fock

Calculate HF exchange using Ψ_i from DFT

$E_{xc}^{hyb} = E_x^{GGA} + a\left(E_x^{HF} - E_x^{GGA}\right) + E_c^{GGA}$

↳ E_{xc} depends on Ψ_i's so leads to much longer computation times

$$E_{xc}^{hyb} = E_x^{GGA} + a\left(E_x^{HF} - E_x^{GGA}\right) + E_c^{GGA}$$

↳ E_{xc} depends on ψ_i's so leads to much longer computation times

Ex: B3LYP, M06, PBE0, HSE06
↳ a=0.25 ↳ Hybrid screen exchange (a=0.25) w/ Ex applied over short distances

+ Full hybrids don't work well on metals because e^- clouds screen out exchange effects
+ Screen hybrids are better but not great
+ Hybrids work well for oxides (esp. semiconductors)

Major limitations of existing DFT techniques:

1) Band gaps (HOMO-LUMO gaps) in semiconductors or molecules:

$\Delta = \varepsilon_{valence} - \varepsilon_{conduction}$

$\varepsilon_{LUMO} - \varepsilon_{HOMO}$

Typically, $\Delta_{GGA,\ LDA} < \Delta_{exp}$ because they don't localize e^- properly (i.e. errors in e^- localization)

2) Self-interaction error:

In DFT, e^- feel themselves;

Exact Coulomb: $V = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}\frac{e^2}{4\pi\varepsilon_0|\underline{r}_i - \underline{r}_j|}$

DFT: $\frac{1}{2}\iint \frac{\rho(\underline{r})\,\rho(\underline{r}')}{|\underline{r}-\underline{r}'|}\,d\underline{r}\,d\underline{r}'$ (no $j\neq i$ equivalent)

Therefore, the issue is DFT under localizes e^-; for metals errors cancel.
However, in oxides, redox processes give errors that do not cancel (as much)

Most imp. for d and f e^-'s which are highly localized

Two practical approaches:

(1) DFT + U (Hubbard U) → empirical term that penalizes delocalized orbitals; U depends on material

(2) Exact exchange functionals: SIE is smaller because of Hartree-Fock (SIE exactly cancels in HF)

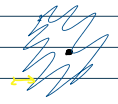
(3) Van der Waals interactions: In standard GGA calculations $\rho(r)$ decays exponentially from nucleus; } exponential is faster
vDW decays as $\frac{1}{r^6}$

intermolecular interactions are underestimated

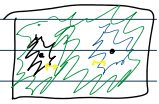
Empirical corrections:

(a) DFT-Dx ; x=1,2,3

+ Add $\frac{C_i}{r^6}$ terms to energy
+ Not great for metals (does not account for e^- screening)



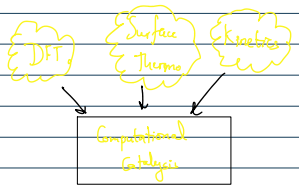
+ Correlated fluctuations in e^- density result in vdW
+ In metals, this is reduced due to the presence of metallic delocalized e^-



VdW is essential for zeolites, hydrocarbon interactions on surface.

(b) Explicit non-local functionals that include vdW like-terms (→ self-consistently incorporated into energies) → ok for metals

Eg: DFT-vdW, opt PBE, opt BBB, BEEF-vdW



Surface Thermodynamics

DFT + Thermo gives: ① What is state of cat. surface (structure, ads coverage, e^- properties) for given environmental conditions? (T, P_i, V)

② How can we determine adsorption isotherms for use in microkinetic models?

For surfaces, convinient free energy is "Grand Potential"

$$\Omega = \underbrace{E - TS}_{\text{Helmholtz energy (F)}} - \sum_i \mu_i N_i \quad ; \quad \text{FE: } d\Omega = -SdT - PdV - \sum N_i d\mu_i$$

(Ω, S, V, N are extensive)

T, V, μ_i are fundamental variables

Convinient free energy because P is hard to define for a surface

Also, surface is often in quasi-eq w/ gas or solid bulk phase → easier to find μ_i's

Consider:

If interphase is planar:

$T^\alpha = T^\beta$; $P^\alpha = P^\beta$; $\mu_i^\alpha = \mu_i^\beta$

$$\Omega^{total} = \Omega^\alpha + \Omega^\beta + \Omega$$

↳ excess grand potential from interface

$d\Omega = \underbrace{-SdT - PdV}_{dF} - \sum_i N_i d\mu_i \quad ; \quad \Omega(T,V,\mu_i)$

V^α | α / β | V^β — Const V_{tot} ; $\Omega^{tot} = \Omega^\alpha + \Omega^\beta + \Omega$

Free energy per volume in bulk α: $\omega^\alpha V^\alpha$

$d\Omega^t = d\Omega^\alpha + d\Omega^\beta + d\Omega$

$d\Omega^\alpha = -S^\alpha dT - PdV^\alpha - \sum_i N_i^\alpha d\mu_i \quad ; \quad d\Omega^\beta = -S^\beta dT - PdV^\beta - \sum_i N_i^\beta d\mu_i$

$T^\alpha = T^\beta = T^{interface}$
$P^\alpha = P^\beta = P^{int}$
$\mu_i^\alpha = \mu_i^\beta = \mu_i^{int}$

How N_i at V_i?

$N_{in} = N_t - \rho_m V_{m,bulk} - N_m^{vapor}$ ← excess amount of interfacial material

Including interfacial term: $d\Omega^t = -S^t dT - P\,dV^t + \underbrace{\gamma dA}_{\text{work associated w/ changes in interfacial area}} - \sum_i N_i^t d\mu_i$

Also: $S^t = S^\alpha + S^\beta + S$; $N_i^t = N_i^\alpha + N_i^\beta + N_i$; $V^t = V^\alpha + V^\beta$ (Couple of atomic layers thick)

$S^\alpha V^\alpha$ (per volume)

Boundary: Often placed where there is a maximum conc. gradient

$d\Omega = \left(-S^t dT - PdV^t + \gamma dA - \sum N_i^t d\mu_i\right) + \left(S^\alpha dT + P dV^\alpha + \sum N_i^\alpha d\mu_i\right) + \left(S^\beta dT + P dV^\beta + \sum N_i^\beta d\mu_i\right)$

$d\Omega = -\underbrace{(S^t - S^\alpha - S^\beta)}_{S} dT - P d(\underbrace{V^t - V^\alpha - V^\beta}_{0}) + \gamma dA - \sum_i (N_i^t - N_i^\alpha - N_i^\beta) d\mu_i$

No excess interfacial volume

$\therefore \boxed{d\Omega = -SdT + \gamma dA - \sum_i N_i d\mu_i}$ All quantities are excess interfacial quantities $\Omega = \Omega(T, A, \mu_i)$

Euler Integration → $\Omega = \gamma A$ → $\boxed{\gamma = \frac{\Omega}{A}}$ ← Surface tension is excess Ω per unit area

; Aside: $\Omega = \gamma A$
$d\Omega = \gamma dA + A d\gamma$
$A d\gamma = -SdT - \sum_i N_i d\mu_i$
$d\gamma = -\frac{S}{A} dT - \sum_i \frac{N_i}{A} d\mu_i$

Gibbs-Duhem for interfaces → Relates to Gibbs-Duhem Isotherm $\gamma = \gamma(T, \mu_i)$

Calculate γ at given $\{T, \mu_i\}$ and find structure w/ lowest γ
↳ Surface phase diagram

Idea: Calculate $\frac{\Omega}{A}(X)$, using DFT + stat mech, assuming surface is in equilibrium w/ bulk phases: (bc $\mu_i = \mu_i^{gas}$, $\mu_i = \mu_i^{bulk\ metal}$)

$\boxed{\gamma = (U - TS - \Sigma_i \mu_i N_i)\frac{1}{A}}$; Find lowest γ for desired T, μ_i

XS interfacial internal energy; XS entropy; XS moles of i

Algorithm: ① U can be approximated in many ways: (potential energy for atoms)

+ Ignore kinetic of surface molecules, and assume $U = E_{DFT}$

$U = E_{DFT} - \frac{P_{bulk} V^\alpha}{A}$ (Bulk energy density)

} Often people ignore this XS aspect of surface energy
↳ phase diagrams

Better + U ≈ E_{DFT} + ZPE
↳ zero point energy (Lowest vibrational energy state for molecules)

Note: typically neglect ZPE of surface itself

Term project idea:
+ Methods of app for Q
+ phonon interact in ZPE

Best + Calculate $Q = \sum_j e^{-\beta \varepsilon_j}$ → states of surface (have to make approximations since infinite) and use $U = \frac{-\partial \ln Q}{\partial \beta}$

partition function

② TS excess surface entropy approximations:

+ Assume TS = 0 (Surfaces are not bulk phases → reduced dimensionality Change decrease in entropy)

First approximation

+ Adsorbates: count all possible arrangements of adsorbate "i", then $S_i = k_B \ln \Omega$ (No. of states)

↳ Configurational entropy: mean field treatment

+ Partition function: $S = k_B \ln Q - \frac{\partial \ln Q}{\partial \beta}$ (Calculated for Stat mech)

↳ Adsorbates: count all possible arrangements of adsorbate i, then $S_i = k_B \ln \Omega$

↳ Configurational entropy: mean field treatment

+ Partition function: $S = k_B \ln Q - \frac{1}{T}\frac{\partial \ln Q}{\partial \beta}$ (Calculated for Stat. mech)

+ Use empirical correlations for S_i of surface adsorbates.

③ $\{\mu_i\}$

+ Leave these as undetermined independent parameters

+ Evaluate μ_i by equilibrium w/ a bulk phase $\rightarrow \mu_i = \mu_i^{gas}$

↳ Use measured data: $\mu_{gas}(T,P) = \mu_{gas}(T,P^\circ) + k_B T \ln\left(\frac{P}{P^\circ}\right)$ (i.g.)

$= h_{gas}(T,P^\circ) - TS_{gas}(T,P^\circ) + k_B T \ln\left(\frac{P}{P^\circ}\right)$

$= \underbrace{\left[h_{gas}(T,P^\circ) - h_{gas}(0,P^\circ)\right]}_{\int_0^T c_p\,dT \text{ @ } P^\circ} + \underbrace{\left[h_{gas}(0,P^\circ) - E^{DFT}_{gas}\right]}_{\text{Only KE at 0K is the zero point energy}} + \underbrace{\left[E^{DFT}_{gas} - TS_{gas}(T,P^\circ)\right]}_{\text{absolute vals. from tables}} + k_B T \ln\left(\frac{P}{P^\circ}\right)$ } Adding & subtracting various reference states

$\int_0^T c_p\,dT$ @ P° → entropolate, assume constant, etc.

$h_{gas}(0,P^\circ) - E^{DFT}_{gas} = ZPE$

E^{DFT}_{gas} → Potential energy from DFT calc. of gas phase

↳ $\mu_{gas} = h_{gas}(T,P) - TS_{gas}(T,P)$

↓

$h_{gas}(T,P) = U_{gas}(T,P) + \underbrace{k_B T}_{\text{if ig}}$; Find U_{gas} w/: (i) Equipartition theorem (high T)

Total DOF is $3N$ (the same) ↗

+ Each translational mode (3 total) gives $\frac{1}{2}k_B T$
+ Each vibrational mode ($3N-6$ non-linear; $3N-5$ linear) gives $k_B T$
+ Each rotational mode (3 non-linear; 2 linear) gives $\frac{1}{2}k_B T$

$\Rightarrow U_{gas} = 3k_B T + (3N-6)k_B T + E^{DFT}_{gas}(\text{non-linear})$

$= \frac{5}{2}k_B T + (3N-5)k_B T + E_{DFT,gas}(\text{linear})$

(ii) Real statistical mechanics for U_{gas} (ΔS_{gas}):

Vibrational: $Q_{vib} = \prod_i \frac{e^{-\frac{h\nu_i}{2}\beta}}{1 - e^{-h\nu_i \beta}}$; ν_i = vibrational freq. (measured or calc. from molec.); i = all normal modes of molecule

Rotational: $Q_{rot} = \frac{\sqrt{\pi}}{\sigma}\sum_{j=1}^{3}\left(\frac{2I_j}{\hbar^2\beta}\right)^{1/2}$ non-linear ; σ → symmetry factor; $\beta = \frac{1}{k_B T}$; $I_{1,2,3}$ → principle moments of inertia

$Q_{rot} = \frac{I}{\sigma\beta\hbar^2}$ (linear)

Translational: $Q_{trans} = \left(\frac{m}{2\pi\beta\hbar^2}\right)^{3/2} V$ (for 1 molecule); V = total volume of system

$U = \langle E \rangle = \frac{\sum E_i e^{-\beta E_i}}{Q}$; $e^{-\beta E_i}$ → probability of system in state

$= -\frac{\partial \ln Q}{\partial \beta}$

$= -\frac{1}{Q}\frac{\partial Q}{\partial \beta}$

$U = -\frac{\partial \ln Q}{\partial \beta}$

Translation, rotation, vibration

Q for ideal gases $\langle U \rangle = -\frac{\partial \ln Q}{\partial \beta}$ $\beta = \frac{1}{k_B T}$

$U_{vib} = \sum_i \left(\underbrace{\frac{1}{2} h\nu_i}_{\text{ZPE}} + \underbrace{\frac{h\nu_i}{e^{\beta h\nu_i} - 1}}_{\text{Thermal excitation}} \right)$ i = normal vib modes of molecules

$U_{rot} = \begin{cases} \frac{3}{2} k_B T & \text{non-linear} \\ k_B T & \text{linear} \end{cases}$ same as equipartition since Q_{rot} is already for "high" T

$U_{trans} = \frac{3}{2} k_B T$

} Add E_{DFT} to these terms

F = Helmholtz

$S_{gas}(T,P)$ can also be found from Q_{gas}:

$S = \underbrace{k \ln Q}_{-\frac{F}{T}} - \frac{1}{T} \frac{\partial \ln Q}{\partial \beta} = k \ln Q + \frac{U}{T}$

$\Downarrow$

$S_{vib} = \sum_i \left(\frac{-h\nu_i}{2T} - k_B \ln\left(1 - e^{-h\nu_i \beta}\right) \right) + \frac{1}{T} \sum_i \left(\frac{1}{2} h\nu_i + \frac{h\nu_i}{e^{\beta h\nu_i} - 1} \right)$

$= \int k_B \ln \left[\frac{\sqrt{\pi}}{6} \prod_{i=1}^{3} \left(\frac{2 I_i}{\hbar^2 \beta} \right)^{1/2} \right] + \frac{3}{2} k_B \rightarrow$ Non-linear

$= \int k_B \ln \left[\frac{I}{6 \beta \hbar^2} \right] + k_B \rightarrow$ linear

$S^{tot}_{trans} = Q^{tot}_{trans} = \frac{\left(Q^{moles}_{trans}\right)^{N_e}}{N!}$ N_e = # molecules

$N!$ ← Indistinguishable particles

$S^{tot}_{trans} = Nk \ln Q^{mol}_{trans} - k \ln N! + \frac{3}{2} k_B N$

$\approx Nk \ln Q^{mol}_{trans} - kN \ln N + kN + \frac{3}{2} k_B T$

$= Nk \ln \left[\left(\frac{m}{2\pi\beta\hbar^2} \right)^{3/2} \frac{N k_B T}{P} \frac{1}{N} \right] + \frac{5}{2} k_B T$

$\therefore \boxed{\frac{S^{tot}_{trans}}{N} = k_B \ln \left[\left(\frac{m}{2\pi\beta\hbar^2} \right)^{3/2} \frac{k_B T}{P} \right] + \frac{5}{2} k_B T \rightarrow \text{per molecule}}$

fugacities from an EOS

(iii) Real gases:

$\mu_{gas}(T,P) = \mu_{gas}(T,P^\circ) + k_B T \ln \left(\frac{f(T,P)}{f(T,P^\circ)} \right)$

If P° is 1 bar, use i.g. approximations

(1) H^* on Ni(111) ; Bulk stability limit : $Ni(s) + \frac{1}{2} H_2(g) \rightleftharpoons NiH(s)$

[QE: $\frac{1}{2} H_2(g) + * \rightleftharpoons H^*$

$\frac{1}{2} \mu_{H_2(g)} = \mu_{H^*}$]

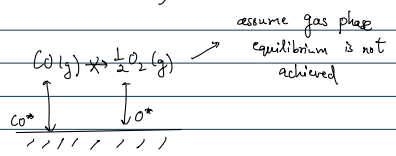
$\Delta G_{molar} = \mu_{NiH} - \frac{1}{2} \mu_{H_2} - \mu_{Ni}$

$\boxed{\Delta G_{molar} = \mu_{NiH} - \mu_H - \mu_{Ni}}$

For $\Delta G_f > 0$, $\boxed{\mu_H < \mu_{Ni:H(bulk)} - \mu_{Ni(bulk)}}$

w/ DFT $\mu_{bulk} \approx E_{DFT,\,bulk}$

(2) CO oxidation on RuO_2 (110) :



$\mu_{CO(g)} \rightleftharpoons \mu_{CO*}$

$\frac{1}{2}\mu_{O_2(g)} \approx \mu_{O*}$

Bulk stability: $Ru(s) + O_2(g) \rightleftharpoons RuO_2(s)$

$\Delta G_f = \mu_{RuO_2} - \mu_{O_2(g)} - \mu_{Ru} < 0$

$$\boxed{\frac{1}{2}\mu_{O_2(g)} = \mu_{O*} > \frac{1}{2}\left(\mu_{RuO_2(s)} - \mu_{Ru(s)}\right)}$$

Zeolites

Bulk zeolite are not equilibrium structures (bulk SiO_2 is equilibrium)
(metastable → decays at high temp)

↳ True excess quantities are not useful

$$\gamma = \frac{U_{cell} - TS - \sum \mu_i N_i - U_{clean}}{A}$$

- U_{cell}: DFT energy includes adsorbates
- TS: entropy in zeolites
- $\sum \mu_i N_i$: Not true interfacial free energy
- U_{clean}: DFT of clean zeolite
- A: Area in pores

(1) U_{cell} and U_{clean} are taken from DFT energies + Thermal corrections (as before)

(2) μ_i are evaluated similarly unless significant mass transport barriers exist

(3) Entropy effects are similar (see upcoming lecture)

(4) No point in estimating bulk stability ranges

(5) If pores have wetting/non-wetting transition, both energies and entropies are best evaluated using Ab-initio molecular dynamics

Dealing w/ nanoparticles → need to know what the surface looks like:

1) Simple case: surface energy in a vacuum

$$\gamma = \left(U^{xs} - TS^{xs} - \sum_i \mu_i N_i^{xs} \right) \frac{1}{A}$$

- $U^{xs} = U_{\underset{DFT}{surf}} - \sum_{metal} \mu_{\underset{bulk}{metal}} N_{\underset{DFT}{metal}}$

For an ordered surface; $N^{xs}_{metal} \approx 0$

$$\therefore \quad \gamma = \frac{1}{A}\left(U_{\underset{DFT}{surf}} - \cancel{TS^{xs}}^{\,\to 0} - \sum \mu_{metal} N_{metal} \right)$$

(N_{metal}: total)

Typically, $\mu_{metal} = g_{metal} \approx e^{\underset{DFT}{bulk}}_{metal}$

(DFT energy per bulk metal atom)

Lecture 10 (Wulff & Langmuir)

Tuesday, 14 February 2023 10:30 AM

Surface Thermodynamics / Free energy

$$\gamma = \frac{1}{A}\left(U_{DFT} - TS_{excess} - \sum \mu_{metal} N^{metal}_{slab} \right) \quad ; \quad \mu_{metal} = g^{metal}_{bulk} \approx e^{DFT}_{bulk}$$

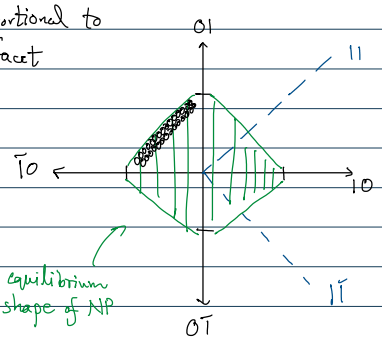
γ: Free energy change

$TS_{excess} \approx 0$ (first approx.)

Approach: Calculate γ's for different surfaces [(111), (100), (110), ...] → goal is to figure out which surface is present at equilibrium for catalyst nanoparticle.

Wulff Theorem (facet will be proportional to energy)

Distance from center of NP to a given facet is proportional to energy of that facet



$\gamma_{10} = 1$

$\gamma_{11} = 3/4$

$\gamma_{10} = \gamma_{01} = \gamma_{\bar{1}0} = \gamma_{0\bar{1}}$

$\gamma_{11} = \gamma \ldots$

(edges are the surfaces) — equilibrium shape of NP

Ref: Schall book Chp 2., 4.4

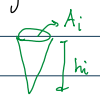
→ Doesn't say anything about how the particles are formed

"Proof":

$$G = \sum_i \gamma_i A_i \quad \text{(over facets)} \quad ; \text{ constant } V \rightarrow \text{const. bulk Energy}$$

Constraint: $V = \sum_i \frac{1}{3} h_i A_i$ (sum of a series of cones)

h_i: distance to center of NP



Lagrangian functional:

$$F = G - \lambda\left(\sum \tfrac{1}{3} h_i A_i - V\right)$$

$$\delta F = \delta G - \lambda \delta\left(\sum \tfrac{1}{3} h_i A_i\right)$$
$$= \delta\left[\gamma_i A_i - \lambda \tfrac{1}{3} \sum h_i A_i\right]$$
$$= \sum \delta\left[\gamma_i A_i - \tfrac{1}{3}\lambda h_i A_i\right] = 0$$

→ If the A_i can be varied independently → so each term is zero

$$\therefore \quad \gamma_i A_i = \frac{1}{3} h_i A_i$$

$$h_i = \frac{3\gamma_i}{\lambda} \quad \rightarrow \text{from constraint:} \quad \lambda = \sum_i A_i \gamma_i$$

$h_i = \frac{3\gamma_i}{\lambda}$ → from constraint: $\lambda = \frac{\sum_i A_i \gamma_i}{V}$

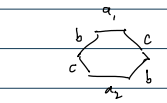
$$\boxed{h_i = \frac{3V}{\sum_j A_j \gamma_j}\gamma_i} \quad *$$

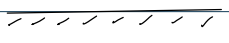
★ $h_i \propto \gamma_i$

*Not a full proof since we do not consider directions

Generalize Wulff to supports ⇒ adsorbates

1) Supports: start w/ a suspended NP





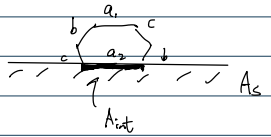
$$\Omega = 2\gamma_a\gamma_a + 2\gamma_b A_b + 2\gamma_c A_c + \underbrace{A_s\gamma_s}_{\text{constant}}$$

No interaction

$$\Omega - \gamma_s A_s = \underbrace{2\gamma_a A_a + 2\gamma_b A_b + 2\gamma_c A_c}_{\text{Standard Wulff construction}}$$

(⇒ we get same shape as unsupported case)

Interaction



$$\Omega = A_{a_1}\gamma_{a_1} + A_{a_2}\gamma_{a_2} + 2\gamma_b A_b + 2\gamma_c A_c + \gamma_{int} A_{int} + \gamma_c(A_s - A_{int})$$

$$\Omega - A_s\gamma_s = \gamma_{a_1} A_{a_1} + \gamma_{a_2} A_{a_2} + 2\gamma_b A_b + 2\gamma_c A_c + A_{int}(\gamma_{int} - \gamma_s)$$

$\gamma_{int} A_{int} = E_{int} - TS - \sum \mu_i N_i$ } XS quantities at interface

$\approx E_{int}^{DFT} - \mu_s N_s^{DFT} - \mu_m N_m^{DFT}$ } Total quantities in calculation

(support) $m = a_2$

Freestanding surfaces:

$$A_{int}\gamma_s = E_s^{DFT} - \mu_s N_s^{DFT}$$
$$A_{int}\gamma_m = E_m^{DFT} - \mu_m N_{m(a_2)}^{DFT}$$
} Freestanding surfaces

$$\Rightarrow A_{int}(\gamma_{int} - \gamma_{m,a_2} - \gamma_s) = E_{int}^{DFT} - E_s^{DFT} - E_m^{DFT} = \text{Adhesion Energy}$$

$$\Omega - A_s\gamma_s = \gamma_{a_1} A_{a_1} + 2\gamma_c A_c + 2\gamma_b A_b + A_{int}\left(\gamma_{m,a_2} + \frac{\Delta E_{adh}}{A_{int}}\right)$$

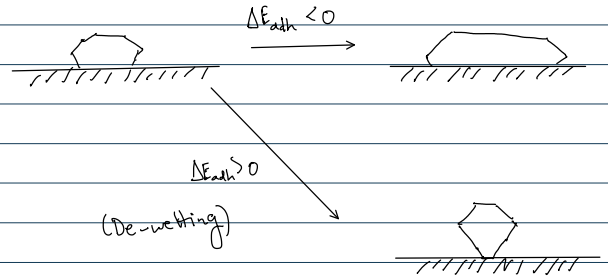
Wulff theorem

Effectively, replace γ with $\gamma + \frac{\Delta E_{adh}}{A_{int}}$

Effectively,

replace γ_{m,a_2} w/ $\gamma_{m,a_2} + \frac{\Delta E_{adh}}{A_{int}}$

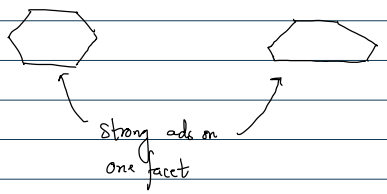
Eg:



(2) Adsorbates – "easy"; just replace $\gamma_{m,vacuum}$ w/ γ calculated from an ab-initio phase diagram

$$\gamma(\Theta_i) \rightarrow \gamma(\mu_i) \rightarrow \gamma(T, P_i)$$

Ex:



→ Re optimizing the shape of NP depending on the T, P

→ Assumption → ② NP equilibrium

Why eq has high energy facets → Volume constraint!

(Need other facets or else will have higher bulk free energy → penalized)

Isotherm Development

Pros: DFT calcs not necessary; better treatment of surface entropies

Cons: must parameterize surface energies and adsorption energies

(1) Langmuir isotherm → classic isotherm

Key assumptions: · no interaction b/w adsorbates
· uniform surface sites
· constant adsorption energies

Can be occupied or unoccupied

+ Minimize Ω for given values of μ and T: $\Omega_A(\mu, T, N_s)$

+ Minimize Ω for given values of μ and T : $\Omega(\mu, T, N_s)$ (A)

(Grand potential)

$$\Omega = U - TS - \mu N = -k_B T \ln \Xi$$

Ξ ← grand canonical partition function

$$\Xi = \sum_j e^{-\beta(E_j - \mu N_j)} = \sum_j e^{-\beta\left[\varepsilon(\sigma_1 + \sigma_2 \cdots \sigma_{N_s}) - \mu(\sigma_1 + \sigma_2 + \cdots \sigma_{N_s})\right]}$$

ε ← binding energy of 1 molecule

N_s → total sites

$\sigma_i = \begin{cases} 0 \\ 1 \end{cases}$ unoccupied / occupied So, $\sum_i^{N_s} \sigma_i = N$

$$\therefore \Xi = \sum_{\sigma_1=0}^{1} e^{-(\varepsilon-\mu)\sigma_1} \sum_{\sigma_2=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma_2} \cdots \sum_{\sigma_{N_s}=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma_{N_s}}$$

$$= \prod_{i=1}^{N_s} \sum_{\sigma=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma}$$

$$= \prod_{i=1}^{N_s} \left(1 + e^{-\beta(\varepsilon-\mu)}\right)$$

$$\boxed{\Xi = \left[1 + e^{-\beta(\varepsilon-\mu)}\right]^{N_s}}$$

$$\therefore \Omega = -k_B T N_s \ln\left(1 + e^{-\beta(\varepsilon-\mu)}\right)$$

$$\langle N \rangle = -\frac{\partial \Omega}{\partial \mu} = \frac{e^{-\beta(\varepsilon-\mu)}}{1 + e^{-\beta(\varepsilon-\mu)}} \cdot N_s$$

$$\therefore \frac{\langle N \rangle}{N_s} = \theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1 + e^{-\beta(\varepsilon-\mu)}}$$

If surface is in equil. w/ ideal gas:

$$\mu = \mu_{gas} = \mu^\circ_{gas} + k_B T \ln\left(\frac{P}{P^\circ}\right)$$

$$\theta = \frac{\frac{P}{P^\circ} e^{-\beta(\varepsilon-\mu^\circ)}}{1 + \frac{P}{P^\circ} e^{-\beta(\varepsilon-\mu^\circ)}}$$

Note: This assumes that S is only configurational entropy

ε can be replaced w/ surface free energy w/o losing generality

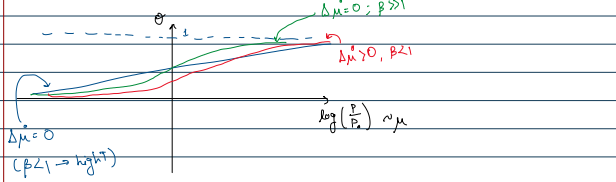
$$\varepsilon \rightarrow \varepsilon + ZPE + \text{thermal vibrations} - TS^\circ_{loc}$$

$\simeq \mu^\circ_{surf}$

S°_{loc} ← Not configurational

$$\therefore \theta = \frac{\frac{\rho}{\rho^\circ} e^{-\beta \Delta\mu^\circ}}{1 + \frac{\rho}{\rho^\circ} e^{-\beta \Delta\mu^\circ}}$$

(1) Langmuir : $\frac{\langle N\rangle}{N_s} = \theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}$ * Ignores interactions



(2) Interactions b/w adsorbates :

→ Use Ising Isotherm : assume interactions have a constant value and only exist b/w nearest neighbors.

→ Can analyze analytically in 1-D : $E = \underbrace{\sum_{i=1}^{N_s} \sigma_i \sigma_{i+1} J}_{\text{Interaction}} + \underbrace{\varepsilon \sum_{i=1}^{N_s} \sigma_i}_{\text{Langmuir part}}$; $\sigma_i = 0, 1$ (Occupied or unoccupied)

J = interaction ($J<0$ → attraction) (is a constant)

As before, $\Xi(\mu, T, N_s) = \sum_{\text{all states}} e^{-\beta(E_j - \mu N_j)}$

Total over all states on surface $= \sum_{N=0}^{N_s} \sum_{\text{states w/ } N} e^{-\beta E + \beta\mu N}$

$= \sum_{N=0}^{N_s} \sum e^{-\beta\left[\sum_i^{N_s} (\sigma_i \sigma_{i+1} J + \varepsilon\sigma_i - \mu\sigma_i)\right]}$; $\sum \sigma_i = N$

Writing as a nested sum:

$= \sum_{\sigma_1=0}^{1} \sum_{\sigma_2=0}^{1} \cdots \sum_{\sigma_{N_s}=0}^{1} e^{-\beta\left[\sum_{i=1}^{N_s} (\sigma_i \sigma_{i+1} J + \frac{\sigma_i + \sigma_{i+1}}{2}(\varepsilon-\mu))\right]}$

Introduce : $P = \begin{bmatrix} P_{00}: 1 & P_{01}: e^{-\frac{\beta}{2}(\varepsilon-\mu)} \\ P_{10}: e^{-\frac{\beta}{2}(\varepsilon-\mu)} & P_{11}: e^{-\beta(J+\varepsilon-\mu)} \end{bmatrix}$ Symmetric matrices are diagonalizable

(Transfer matrix for lattice statistical mechanics problem)

$\Xi = \sum_{\sigma_1=0}^{1} \sum_{\sigma_2=0}^{1} \cdots \sum_{N_s=0}^{1} \langle\sigma_1|P|\sigma_2\rangle\langle\sigma_2|P|\sigma_3\rangle \cdots \langle\sigma_{N_s}|P|\sigma_1\rangle$; where $\sigma_i = \begin{pmatrix}1\\0\end{pmatrix}$ if "i" is unoccupied

↑ periodic BC's $\sigma_i = \begin{pmatrix}0\\1\end{pmatrix}$ if "i" is occupied

? - - - | x x | x x | x x | x x | - - - -

Eg : $N_s = 2$,

$\Xi = P_{00}P_{00} + P_{01}P_{10} + P_{10}P_{01} + P_{11}P_{11}$ ($P_{01} = P_{10}$)

sum of the eigenvalues of matrix $P:P = P^2$ ← dyadic product?

P_{00} → element (0,0) of matrix P

In general, Ξ = largest eigenvalue of P^{N_s}

$P^{N_s} = P:P:P:P:\ldots$ N_s times

= (eigenvalue of P)^{N_s}

$\begin{vmatrix} 1-\lambda & e^{-\frac{\beta}{2}(\varepsilon-\mu)} \\ e^{-\frac{\beta}{2}(\varepsilon-\mu)} & e^{-\beta(J+\varepsilon-\mu)} - \lambda \end{vmatrix} = 0$; $\Xi \approx \lambda_+^{N_s} = \left[\frac{1 + e^{-\beta(J+\varepsilon-\mu)} + \sqrt{(1+e^{-\beta(J+\varepsilon-\mu)})^2 - 4(e^{-\beta(J+\varepsilon-\mu)} - e^{-\beta(\varepsilon-\mu)})}}{2}\right]^{N_s}$

↑ largest evalue

$\langle N\rangle = -2\frac{\partial\Omega}{\partial\mu} = \frac{\partial}{\partial\mu}(k_B T \ln \Xi)$

$\therefore \frac{\langle N\rangle}{N_s} = \theta = \frac{e^{-\beta(J+\varepsilon-\mu)} + \frac{-e^{-\beta(J+2\varepsilon-2\mu)} + e^{-2\beta(\varepsilon+J-\mu)} + 2e^{-\beta(\varepsilon-\mu)}}{\left[(1-e^{-\beta(J+\varepsilon-\mu)})^2 + 4e^{-\beta(\varepsilon-\mu)}\right]^{1/2}}}{1 + e^{-\beta(J+\varepsilon-\mu)} + \left[(1-e^{-\beta(J+\varepsilon-\mu)})^2 + 4e^{-\beta(\varepsilon-\mu)}\right]^{1/2}}$

If $J=0$, $\theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}$

If $|\beta J| << 1$,

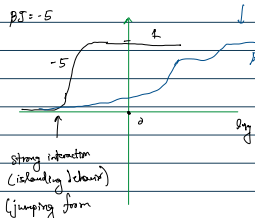
$$\theta \simeq \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}\left[1 - \frac{2e^{-\beta(\varepsilon-\mu)}}{(1+e^{-\beta(\varepsilon-\mu)})^2}\beta J\right]$$

→ In 2D, partial analytical solution exists for $\theta = \frac{1}{2}$

Onsager (1942)

→ For more complex interactions, use: + cluster expansion method (power series) } Term project idea → different interactions
+ machine learning potentials

strong repulsion (Two langmuir isotherms → Bimodal langmuir behavior)



strong interaction (disturbing behavior) (jumping from 0 → 1)

(change w/ very small P change)

S → shape (phase separation)

βJ → order of kT

@ low Pressure → empty / vacant sites
@ P ↑ → filling sites randomly (w/ no nearest neighbor
@ P=0.5 → give max no. of adsorbate w/ no nearest neighbor (get a plateau)

Alternate filling

Have to increase the P by a tremendous amount to get more populated sites.

Second langmuir isotherm (i.e. now you basically have two diff sites)

For 2D case → The binding starts to minimize repulsion. (i.e. the βJ will increase by

The βJ will increase by 4

given any surface you can intuitively get the shape / plateaus for adsorption

H_2O adsorption on electrodes?
↳ quite attractive H-bonding
↳ complex structure

Def. of monolayer can be different to that of how you define what the surface sites are (and how many bonds they make)

Kinetics w/ DFT

1) Discuss TST and calculating k's from DFT data

2) Review of basic principles of microkinetic modelling

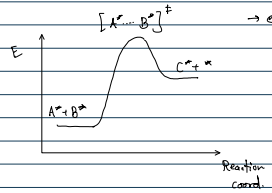
3) Scaling relationships, BEP reations, volcano plots, trends in catalytic properties

Transition State Theory (TST)

Strategy for estimating rate constants from DFT (exp.) results

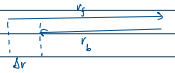
Elementary surface reaction

$A^* + B^* \rightleftharpoons C^* + *$



Two key assumptions:

1) To a first order perturbation, reactants are in (quasi) equilibrium w/ T.S.



↳ or else there would be no rate (if exactly at equilibrium)

2) Rate of conversion of TS to products governed by attempt frequency, ν

↳ (vibrational frequency that dictates how it "vibrates" from

(i) $K^\ddagger = e^{\frac{-\Delta\mu^{o\ddagger}}{k_B T}} = \frac{a_{TS}}{a_{A^*} a_{B^*}}$

$K^\ddagger$ → standard state chemical potential for transition state

$\Delta\mu^{o\ddagger} = \mu^o_{TS} - \mu^o_{A^*} - \mu^o_{B^*}$

Recall: $\mu_i = \mu_i^o + k_B T \ln a_i$; $a_i = \frac{f_i}{f_i^o}$; μ_i^o depends on T

(a) For ideal gases, $a_i = \frac{P_i}{P^o}$

(b) For liquid species,

$a_i \approx$ { 1 if conc. (usually solvent); $\frac{c_A}{c_A^o}$ if dilute "i"; c_A^o ← infinitely dilute solute extrapolated to 1M (of Henry's law) }

(c) Surface species: Use partition functions and divide into purely local part and a configurational part

Note: $a_{A^*} \approx \theta_A$

(This usually gives right answer but is not that rigorous)

Assume: non-interacting adsorbates on lattice N_s sites. Adsorption energy is ε for each adsorbate. Each adsorbate has local energy characterized by q_{loc} (vibration, rotation)

$Q_{tot}(T, N_s, N) = \frac{N_s!}{(N_s - N)!\,N!} \left(e^{-\beta\varepsilon} q_{loc}\right)^N$ ↳ canonical

$F = -k_B T \ln Q_{tot}$ extensive

$\ln Q_{tot} = -\beta\varepsilon N + N \ln q_{loc} + \ln N_s! - \ln(N_s - N)! - \ln N!$; $\ln x! = x \ln x - x$

$= N_s \ln\left(\frac{N_s}{N_s - N}\right) + N \ln\left(\frac{N_s - N}{N}\right) - \beta\varepsilon N + N \ln q_{loc}$

$F = -k_B T \ln Q$, $\mu = \left.\frac{\partial F}{\partial N}\right|_{T, N_s} = \underbrace{\varepsilon - k_B T \ln q_{loc}}_{\text{adsorption free energy for an isolated molecule}} + \underbrace{k_B T \ln \frac{\theta}{1-\theta}}_{\text{differential configurational entropy}}$; $\theta = \frac{N}{N_s}$

} only approximation is that they are non-interacting

If more than one adsorbate is present, $\mu_i = \varepsilon_i - k_B T \ln q_{trc,i} + k_B T \ln \frac{\theta_i}{\theta_*}$; $\theta_* = 1 - \theta_i - \theta_B - \theta_C - \dots$

We say that $\mu_i^\circ = \varepsilon_i - k_B T \ln q_{trc,i}$; $a_i = \frac{\theta_i}{\theta_*}$

Standard state : isolated molecule w/ no configurational entropy

↳ For reaction: $A^* + B \overset{TS}{\rightleftharpoons} C^* + *$, $K^\ddagger = \frac{q_{TS}^\ddagger}{q_{A^*} q_{B^*}} = \frac{\theta_{TS}\,\theta_*}{\theta_A\,\theta_B}$; $\ln K^\ddagger = \frac{\mu_{TS}^\circ - \mu_{A^*}^\circ - \mu_{B^*}^\circ}{k_B T}$

Second assumption: $r \approx \nu \theta_{TS} \theta_*$ (assumes immediate TS (Langmuir Hinshelwood))

$\therefore r = \nu K^\ddagger \theta_A \theta_B = \underbrace{\nu e^{\frac{\Delta\mu^\ddagger}{k_B T}}}_{\text{what about this?}} \theta_A \theta_B$ (Law of mass action)

For particles on surface, $\mu_i^\circ = \varepsilon_i - k_B T \ln q_{trc}$

$= -k_B T \ln e^{-\beta \varepsilon_i} - k_B T \ln q_{trc}$

$= -k_B T \ln \left[q_{trc,i} e^{-\frac{\varepsilon_i}{k_B T}} \right]$

q_i (includes local kinetic & potential energies)

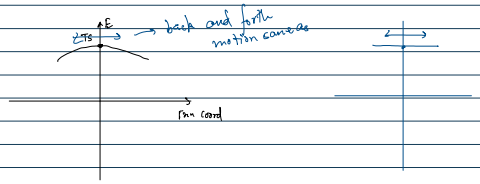
$\therefore \mu_i^\circ = -k_B T \ln q_i$

$\Delta\mu^{\circ\ddagger} = \mu_{TS}^\circ - \mu_{A^*}^\circ - \mu_{B^*}^\circ = -k_B T \ln \left(\frac{q_{TS}}{q_{A^*} q_{B^*}} \right)^*$ $\Rightarrow$ $\boxed{r \approx \nu \frac{q^\ddagger}{q_{A^*} q_{B^*}} \theta_A \theta_B}$; $q^\ddagger = q_{TS}$

To find ν: write q_i's as : $q_i = \sum e^{-\beta E_{vj,i}} = e^{-\beta E_{DFT,i}} \cdot q_{vib,i}$ (like ε; Assume only harmonic (parabolic energy potential) vib modes)

$= e^{-\beta E_{DFT,i}} e^{-\beta (ZPE)_i} \prod_{j=1}^{3N} \left(\frac{1}{1 - e^{-h\nu_j / k_B T}} \right)$

For mode "j" along reaction coord ($A^* \cdots B^*$)



Generally assumed we can treat this as a curvature of zero, but attempt frequency is accurate

$\frac{1}{\left(1 - e^{-h\nu / k_B T}\right)} \approx \frac{k_B T}{h \nu_j}$

$\therefore q^\ddagger = e^{-\beta E_{TS}^{DFT}} \underbrace{q_{vib}^{\not rc}}_{\text{all modes except rxn coord.}} \cdot \frac{k_B T}{h \nu_j}$

$r \approx \frac{\nu \frac{k_B T}{h \nu_j} e^{-\beta E_{TS,DFT}} q_{vib,TS}^{\not \nu}}{e^{-\beta E_{A^*DFT}} e^{-\beta E_{B^*DFT}} q_{vib,A^*} q_{vib,B^*}}$; If $\nu = \nu_j$,

$\boxed{r = \frac{k_B T}{h} \frac{q_{vib,TS}^{\not x}}{q_{A^*,vib} q_{B^*,vib}} e^{\beta \Delta E_a^{ZPE}} \theta_A \theta_B}$

(does not include ZPE; Includes potential E & ΔZPE)

This can be written as $r = \nu e^{\frac{-\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B$

$= \nu \frac{k_B T}{h \nu} e^{\frac{-\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B$

$\boxed{r = \frac{k_B T}{h} e^{\frac{-\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B}$

k (rate constant)

3) Rate is proportional to the coverage of TS on substrate

Transition State Theory w/ gaseous species

3) Rate is proportional to the coverage of TS on substrate

k (rate constant)

Transition State Theory w/ gaseous species

$A(g) + B^* \xrightarrow{TS^*} C^*$ (treat TS as a surface species which is often a good assumption)

$$K^\ddagger = \frac{a_{TS}^*}{a_A a_B^*} = \frac{\theta_{TS}/\theta_*}{\frac{P_A}{P^\circ}\frac{\theta_B}{\theta_*}} \Rightarrow \theta_{TS} = K^\ddagger \theta_B \frac{P_A}{P^\circ}$$

Thus, $r = \nu \theta_{TS} = \nu K^\ddagger \theta_B \frac{P_A}{P^\circ} = \frac{k_B T}{h} K^\ddagger \theta_B \frac{P_A}{P^\circ}$

Now including rxn coord.

Lecture 13

Tuesday, 28 February 2023 10:32 AM

Thursday MT → 2 questions, open book?

TST for gas-solid reactions

Eley-Rideal : $A_{(g)} + B^* \overset{TS}{\rightleftharpoons} C^*$

Assuming TS is close to surface (adsorbed)

$$\Rightarrow K^* = \frac{a_{TS^*}}{a_A\, a_{B^*}} = \frac{\theta_{TS}/\theta_*}{\frac{p_A}{p^*}\frac{\theta_B}{\theta_*}} = \frac{\theta_{TS}}{\frac{p_A}{p^*}\theta_B}$$

$$\theta_{TS} = K^* \theta_B \frac{p_A}{p^*} \qquad \text{(Only 1 adsorbed species)}$$

TST assumption (rate prop. to coverage of TS)

$$r = \nu \theta_{TS} = \nu K^{\ddagger} \theta_B \frac{p_A}{p^*}$$

$$\boxed{r = \frac{k_B T}{h} K^{\ddagger} \theta_B \frac{p_A}{p^\circ}}$$

No reaction coord.

$$K^{\ddagger} = e^{\frac{-\Delta\mu^{\circ\ddagger}}{k_B T}} \quad \text{where} \quad \Delta\mu^{\circ\ddagger} = \mu^{\circ\ddagger} - \mu^\circ_{A_{(g)}} - \mu^\circ_{B^*}$$

(Adsorbed TS) $\mu^{\circ\ddagger} = -k_B T \ln q^\circ_{vib,TS} \cdot e^{-\beta(E_{DFT_{TS}} + ZPE_{TS})}$

$$\mu^\circ_{B^*} = -k_B T \ln q^\circ_{vib,B^*} \cdot e^{-\beta(E_{DFT_{B^*}} + ZPE_{B^*})}$$

$$\mu^\circ_A = -k_B T \frac{\partial}{\partial N_{A_{(g)}}} \left[\ln \left\{ Q_{trans,A}\, q^{N_A}_{vib,A}\, q^{N_A}_{rot,A}\, e^{-\beta N_A (E_{DFT_A} + ZPE_A)} \right\} \right]$$

$$= -k_B T \ln q^\circ_{vib,A} - k_B T \ln q^\circ_{rot,A} - k_B T \frac{\partial}{\partial N_A}\left(\ln Q^\circ_{trans,A}\right) + E_{DFT_A} + ZPE_A$$

$$\therefore K^{\ddagger} = \frac{q_{vib,TS^*}}{q_{vib,B^*}\, q^\circ_{vib,A}\, q^\circ_{rot,A}\, e^{\frac{\partial \ln Q^\circ_{trans,A}}{\partial N_A}}} \cdot e^{-\beta(\Delta E_{DFT} + \Delta E_{ZPE})}$$

↓ activation energy

$A_1, A_2 \Longleftrightarrow A_2 A_1$ (Same?)

$$Q^\circ_{trans} = \frac{q^{\circ N}_{trans}}{N!}$$

$$e^{\frac{\partial \ln Q^\circ_{trans,A}}{\partial N_A}} = \left(\frac{q_{trans,A}}{N}\right)^1$$

$$\boxed{r = \frac{k_B T}{h} \frac{q^\circ_{vib,TS^*}}{q_{vib,B^*}\, q^\circ_{vib,A}\, q^\circ_{rot,A} \left(\frac{q^\circ_{trans,A}}{N}\right)} e^{-\beta(\Delta E + \Delta E_{ZPE})} \frac{p_A}{p^\circ}\theta_B}$$

Note:

1) All q_{vib}'s have ZPE part removed
2) q° for vib/rot partition functions is that of isolated species (gas or surface)
3) P° is often lumped into rate constant (careful of units)
4) $q_{electronic}$ is part of the E_{DFT} expression

Alternate TST w/o Statistical Mechanics

In general, $k = \frac{k_B T}{h} e^{\frac{-\Delta\mu^{\ddagger}}{k_B T}}$ ← No rxn coord.

$$\Delta\mu^{\circ\ddagger} = \Delta\left(\frac{\partial F}{\partial N_i}\right)^{\circ\ddagger} = \Delta \frac{\partial}{\partial N_i}\left(U_i - TS_i\right) \qquad (H_i)$$

$$= \Delta\bar{U}^{\circ\ddagger} - T\Delta\bar{S}^{\circ\ddagger}$$

↖ partial molar quantities

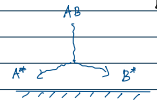
$$\boxed{k = \frac{k_B T}{h} e^{\frac{\bar{S}^{\circ\ddagger}}{k_B}} \cdot e^{\frac{-\bar{U}^{\circ\ddagger}}{k_B T}}}$$

For pure species (i.e. @ partial P_i), or non-interacting surface-species, partial molar is same as "normal" molar quantity

Many ways to calc. $\bar{S}^\circ$:
+ experimental values
+ Stat. mech.
+ Correlations (for surface species)
+ $\Delta S^{\ddagger} \approx 0$ (if reaction is entirely on the surface)

Ex: AB ? Dissociative adsorption

Ex: Dissociative adsorption



a) Simple assumption : $S^\circ_{TS} \approx 0$, thus $\Delta S^{\circ\ddagger} = S^\circ_{TS} - S^\circ_{AB(g)} \approx -S^\circ_{AB(g)}$

and $k = \frac{k_B T}{h p^\circ} e^{\frac{-S^\circ_{AB}}{k_B}} e^{\frac{-\Delta H^{\circ\ddagger}}{k_B T}}$

$e^{\frac{-S^\circ_{AB}}{k_B}}$ ← Thermo tables

Collision Theory (rates of adsorption/desorption of ideal gases w/o dissociation)

Kinetic Theory of Gases : Flux of molecule hitting a surface $= \frac{P_i}{\sqrt{2\pi m k_B T}}$ [?] $\frac{1}{m^2 \cdot s}$

On a per site basis, $\boxed{k^\circ_{ads} = \frac{1}{\sqrt{2\pi m k_B T}\, \rho}}$; $\rho \equiv \frac{\text{sites}}{\text{area}}$ (or n_s)

↳ If adsorption is activated, add a Boltzmann factor and scale by number of vacant sites:

$$k = k^\circ_{ads} e^{\frac{-\Delta E^{act}}{k_B T}}$$

$$r_{ads} = k P_i \theta_*^n ; \quad n = 1 \text{ or } 2$$

($n=1$: simple adsorption; $n=2$: dissociative adsorption)

↳ Same expression from TST: assume TS is a 2D ideal gas

+ Calculate rate from 3D to 2D gas

Partition functions: $Q_{2D} = \frac{q_{2D}^N}{N!} = \left(\frac{2\pi m k_B T}{h^2} A\right)^N \frac{1}{N!}$; $\ln Q_{2D} = N \ln\left[\frac{2\pi m k_B T}{h^2} A\right] - N \ln N + N$

$Q_{3D} = \frac{q_{3D}^N}{N!} = \left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V\right]^N \frac{1}{N!}$; $\ln Q_{3D} = N \ln\left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V\right] - N \ln N + N$

$F_{2D} = -k_B T \ln Q_{2D}$; $\mu_{2D}^{(TS)} = \frac{\partial F_{2D}}{\partial N} = -k_B T \left[\ln\left(\frac{2\pi m k_B T A}{h^2}\right) - \ln N\right]$

$F_{3D} = -k_B T \ln Q_{3D}$; $\mu_{3D} = \frac{\partial F_{3D}}{\partial N} = -k_B T \left[\ln\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V - \ln N\right]$

@ quasi-equil, $\mu_{2D}^{TS} = \mu_{3D}$

$$\frac{2\pi m k_B T}{h^2} \frac{A}{N_{2D}} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} \frac{V}{N_{3D}}$$

$$\left(\frac{N}{A}\right)_{2D} = \frac{P}{k_B T} \sqrt{\frac{h^2}{2\pi m k_B T}}$$

From TST,

$$r = \gamma \left(\frac{N}{A}\right)_{2D} = \frac{k_B T}{h} \left(\frac{N}{A}\right)_{2D} = \frac{P}{\sqrt{2\pi m k_B T}}$$

So, $\boxed{k^\circ_{ads} = \frac{1}{\sqrt{2\pi m k_B T}\, \rho}}$ ← Same as collision theory

Thermodynamic consistency of rate constants

Seek to relate kinetic and thermo parameters for elementary reactions

Ex: $A(g) + B^* \rightleftharpoons C^*$

@ equil., $K = \frac{a_{C^*}}{a_A a_{B^*}} = \frac{\theta_C}{\frac{P_A}{P^\circ} \theta_B}$

& $r_f = r_r$ ($r_{net} = 0$)

$$k_f P_A \theta_* = k_r \theta_c \Rightarrow \frac{k_f}{k_r} = \frac{\theta_c}{P_A \theta_B}$$

$$\therefore \frac{\theta_c}{\frac{P_A}{P^\circ}\theta_B} = \frac{k_f P^\circ}{k_r} = K$$

$$\therefore \boxed{k_r = \frac{k_f P^\circ}{K}}$$

Microkinetics

Overview (Langmuir-Hinshelwood):

Consider reaction mechanism:

1) $A_2(g) + 2* \rightleftharpoons 2A^*$ } might be QE
2) $B_2(g) + 2* \rightleftharpoons 2B^*$ }
3) $A^* + B^* \rightleftharpoons AB^* + *$
4) $AB^* + B^* \rightleftharpoons AB_2^* + *$
5) $AB_{(g)} + * \rightleftharpoons AB^*$ } Simple adsorption
6) $AB_{2(g)} + * \rightleftharpoons AB_2^*$ }

Make balances for each adsorbed species: (rates must be $[L]S^{-1}$ (turn over frequency))

$$A^*: \frac{d\theta_A}{dt} = 2r_1 - r_3$$

$$B^*: \frac{d\theta_B}{dt} = 2r_2 - r_3 - r_4$$

$$AB^*: \frac{d\theta_{AB}}{dt} = r_3 - r_4 + r_5$$

$$AB_2^*: \frac{d\theta_{AB_2}}{dt} = r_4 + r_6$$

r's are net rates (can be <0)

@ SS: $\frac{d\theta_i}{dt} = 0 \Rightarrow$

$2r_1 - r_5 = 0$
$2r_2 - r_3 - r_4 = 0$
$r_3 - r_4 + r_5 = 0$
$r_4 + r_6 = 0$ } express in terms of product rates, r_5 and r_6

$\Downarrow$

$r_4 = -r_6 \qquad r_2 = -\frac{1}{2}r_5 - r_6$
$r_3 = -r_5 - r_6 \qquad r_1 = -\frac{1}{2}r_5 - \frac{1}{2}r_6$

} Two linearly independent rates [6 unknowns, 4 equations]

$\Downarrow$

Need additional info (Use TST-like rate expressions)

$$r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A^2$$

$$r_2 = k_2 P_{B_2} \theta_*^2 - k_{-2} \theta_B^2$$

$$r_3 = k_3 \theta_A \theta_B - k_{-3} \theta_{AB} \theta_*$$

$$r_4 = k_4 \theta_{AB} \theta_B - k_{-4} \theta_{AB_2} \theta_*$$

$$r_5 = k_5 P_{AB} \theta_* - k_{-5} \theta_{AB_2}$$

Lecture 14

If k's and P_i's are known, we have 5 variables: $\theta_A, \theta_B, \theta_{AB}, \theta_{AB_2}, \theta_*$ } 5 eqs, 5 unknowns

Use $\theta_* = 1 - \theta_A - \theta_B - \theta_{AB} - \theta_{AB_2}$

If k_i's and P_i's are known, we have 5 variables: $\theta_A, \theta_B, \theta_{AB}, \theta_{AB_2}, \theta_*$

4 eq's + $\theta_* = 1 - \theta_A - \theta_B - \theta_{AB} - \theta_{AB_2}$ } 5 eqs, 5 unknowns

Comments:

+ solution of eqns for θ_i's gives rigorous rate. Can always do this
+ Assumption: L-H kinetics is valid

More practical: Assume some steps are rate limiting, and some are quasi-equilibrated

↳ ex: Assume 1, 2, 5, 6 are QE → why choose 4? → simple adsorption are often QE

$$K_1 = \frac{\theta_A^2}{\frac{P_A}{P^\circ}\theta_*^2} \Rightarrow \theta_A = \sqrt{K_1 \cdot \frac{P_{A_2}}{P_0}}\,\theta_*$$

$$\theta_B = \sqrt{K_2 \frac{P_{B_2}}{P^\circ}}\,\theta_* \quad , \quad \theta_{AB} = K_5 \frac{P_{AB}}{P^\circ}\theta_* \quad ; \quad \theta_{AB_2} = K_6 \frac{P_{AB_2}}{P^\circ}\theta_*$$

(Expressing everything in terms of θ_*)

From using site balance:

$$\theta_* = \frac{1}{1 + \sqrt{K_1 \frac{P_{A_2}}{P^\circ}} + \sqrt{K_2 \frac{P_{B_2}}{P^\circ}} + K_5 \frac{P_{AB}}{P^\circ} + K_6 \frac{P_{AB_2}}{P^\circ}}$$

(A^*) (B^*) (AB^*) (AB_2^*)

Finally, substitute into r_i's that are not QE:

$$r_3 = k_3 \sqrt{K_1 \frac{P_{A_2}}{P^\circ}} \sqrt{K_2 \frac{P_{B_2}}{P^\circ}}\,\theta_*^2 - k_{-3} K_3 \frac{P_{AB}}{P^\circ}\theta_*^2$$

$$r_4 = k_4 \sqrt{K_2 \frac{P_{A_2}}{P^\circ}} K_5 \frac{P_{AB}}{P^\circ}\theta_*^2 - k_{-4} K_6 \frac{P_{AB_2}}{P^\circ}\theta_*^2$$

Product production rates:

$$r_{AB_{(g)}} = -r_5 = r_3 - r_4$$

$$r_{AB_{2(g)}} = -r_6 = r_4$$

Explore simpler case in more detail:

1) $A_{(g)} + * \rightleftharpoons A^*$

2) $A^* \rightleftharpoons B^*$

3) $B_{(g)} + * \rightleftharpoons B^*$

$$\frac{d\theta_A}{dt} = 0 = r_1 - r_2 \quad ; \quad \frac{d\theta_B}{dt} = 0 = r_2 + r_3$$

$$r_1 = k_1 P_A \theta_* - k_{-1}\theta_A$$

$$r_2 = k_2 \theta_A - k_{-2}\theta_B \quad ; \quad \theta_* = 1 - \theta_A - \theta_B$$

$$r_2 = k_3 P_B \theta_* - k_{-3}\theta_B$$

Case 1: (1) & (2) are QE

$$\theta_A = K_1 \frac{P_A}{P^\circ}\theta_* \quad ; \quad \theta_B = K_2 \frac{P_B}{P^\circ}\theta_*$$

$$\theta_* = \frac{1}{1 + K_1 \frac{P_A}{P^\circ} + K_2 \frac{P_B}{P^\circ}}$$

$$r_{B(g)} = -r_3 = r_3 = k_3 K_1 \frac{P_A}{P^\circ}\theta_* - k_{-3} K_3 \frac{P_0}{P^\circ}\theta_*$$

Case 2: (1) is QE

$\theta_A = K_1 \frac{P_A}{P^\circ} \theta_*$; Find θ_B from stoichiometric SS eqn:

$0 = r_2 + r_3$

$k_2 \theta_A - k_{-2} \theta_B = k_{-3} \theta_B - k_3 P_B \theta_*$

$\Downarrow$

$\theta_B = \frac{k_2 K_1 \frac{P_A}{P^\circ} + k_3 P_B}{k_{-2} + k_{-3}} \theta_*$

$\theta_* = 1 - \theta_A - \theta_B$

$r_{avg} = -r_3 = r_2$ (either works here) ← can use any rate eq since neither are QE

Case 3: Only (3) is QE (exercise)

Case 4: No steps are QE (No assumption about kinetics apart from Langmuir Hinshelwood)

$r_1 = r_2 \Rightarrow k_1 P_A \theta_* - k_{-1} \theta_A = k_2 \theta_A - k_{-2} \theta_B$

$r_2 = -r_3 \Rightarrow k_2 \theta_A - k_{-2} \theta_B = k_{-3} \theta_B - k_3 P_B \theta_*$

$\theta_A + \theta_B + \theta_* = 1$

} SS balance to get θ_A, θ_B

$\theta_A = \frac{k_1 P_A \theta_* + k_{-2} \theta_B}{\underbrace{k_{-1} + k_2}_{\Delta k_a}}$; $\theta_B = \frac{k_2 \theta_A + k_3 P_B \theta_*}{\underbrace{k_{-2} + k_{-3}}_{\Delta k_b}}$

$\theta_B = \frac{k_2}{\Delta k_b} \left[\frac{k_1}{\Delta k_a} P_A \theta_* + \frac{k_{-1}}{\Delta k_a} \theta_B \right] + \frac{k_3}{\Delta k_b} P_B \theta_*$; $\theta_B = \frac{\frac{k_1 k_2}{\Delta k_a \Delta k_b} P_A + \frac{k_3}{\Delta k_b} P_B}{1 - \frac{k_2 k_{-2}}{\Delta k_a \Delta k_b}} \theta_*$

$\theta_A = \left[\frac{\frac{k_1 k_2 k_{-2}}{\Delta k_a^2 \Delta k_b} P_A + \frac{k_2 k_3}{\Delta k_a \Delta k_b}}{1 - \frac{k_2 k_{-2}}{\Delta k_a \Delta k_b}} + \frac{k_1}{\Delta k_a} P_A \right] \theta_*$

By taking various asymptotic limits of k_i's, we can explore the meaning of QE

Non QE = RLS
- Can have multiple RLS
- Each step can have different degree of RLS (Campbell score)
- Campbell: 0.0001 → QE; 1 → RLS

Q2 on HW

Lecture 14

Thursday, 2 March 2023 10:42 AM

1) $A_2(g) + 2* \rightleftharpoons 2A^*$

2) $B_2(g) + 2* \rightleftharpoons 2B^*$

3) $A^* + B^* \rightleftharpoons AB^* + *$

4) $AB^* + B^* \rightleftharpoons AB_2^* + *$

→ Contd on previous lecture notes.

5) $AB_{(g)} + * \rightleftharpoons AB^*$

6) $AB_2(g) + * \rightleftharpoons AB_2^*$

Scaling Relations :

1) $N_2(g) + 2* \rightleftharpoons 2N^*$

2) $N^* + H^* \rightleftharpoons NH^* + *$ QE

3) $NH^* + H^* \rightleftharpoons NH_2^* + *$

4) $NH_2^* + H^* \rightleftharpoons NH_{3(g)} + 2*$ QE

5) $H_2(g) + 2* \rightleftharpoons 2H^*$ QE

These don't happen in any specific sequence

$2r_1 = r_2$; $r_2 = r_3$, $r_3 = r_4$

↳ $2r_1 = r_3$

$$2\left(k_1 P_{N_2} \theta_*^2 - k_{-1}\theta_N^2\right) = k_3 \theta_{NH}\theta_H - k_{-3}\theta_{NH_2}\theta_*$$

$$K_2 = \frac{\theta_{NH}\theta_*}{\theta_N \theta_H} \quad ; \quad K_4 = \frac{\theta_*^2 \dfrac{P_{NH_3}}{p^\circ}}{\theta_{NH_2}} \quad ; \quad K_5 = \frac{\theta_H^2}{\theta_*^2 P_{H_2}/p^\circ}$$

"full solution"

$$\frac{\theta_N \theta_H}{\theta_{NH_2}\theta_H} \quad , \quad \frac{p^\circ}{} \quad , \quad \theta_*^2 \, p_{H_2}/p^\circ \Big)$$

Limiting Analysis:

Case 1 : (1) is RLS , (3) is QE

$$K_2 = \frac{\theta_{NH_2}\theta_H}{\theta_{NH}\theta_N} \; ; \; r_{ov} = 2r_1 = 2\left(k_1 P_{N_2} \theta_*^2 - k_{-1}\theta_N^2\right)$$

$$\theta_H = \sqrt{K_5 \frac{P_{H_2}}{p^\circ}}\,\theta_* \; ; \; \theta_{NH_2} = \frac{P_{NH_2}/p^\circ}{K_4\sqrt{K_5 \frac{P_{H_2}}{p^\circ}}}\,\theta_* \qquad P_i' = \frac{P_i}{P_0}$$

$$\theta_{NH} = \frac{\theta_{NH_2}}{K_3\sqrt{K_5 \frac{P_{H_2}}{p^\circ}}} = \frac{P_{NH_3}/p^\circ}{K_3K_4K_5\frac{P_{H_2}}{p^\circ}}\,\theta_* \; ; \; \theta_N = \frac{P_{NH_3}'}{K_2K_3K_4\left(K_5 P_{H_2}'\right)^{3/2}}\,\theta_*$$

$$r = 2k_1 P_{N_2}\theta_*^2\left[1 - \frac{k_{-1}}{k_1}\frac{P_{NH_3}^2}{P_{N_2}K_2^2K_3^2K_4^2K_5^2P_{H_2}^3}\right]$$

$$r_{1(lim)} = \frac{2k_1 P_{N_2}(1-\gamma)}{\left(1+\sqrt{K_5 P_{H_2}'} + \frac{P_{NH_3}'}{K_4\sqrt{K_5P_{H_2}'}} + \frac{P_{NH_3}'}{K_3K_4K_5P_{H_2}'} + \frac{P_{NH_3}'}{K_2K_3K_4\left(K_5P_{H_2}'\right)^{3/2}}\right)^2}$$

$$\gamma = \frac{1}{k_1K_2^2K_3^2K_4^2K_5^2}\,\frac{P_{NH_3}'^2}{P_{H_2}'^3P_{N_2}'} \; ; \; 0<\gamma<1$$

Case 2 : (3) is RLS , (1) is QE

$$r_{overall} = r_3 = k_3\theta_{NH}\theta_H - k_{-3}\theta_{NH_2}\theta_* \; ; \; K_1 = \frac{\theta_N^2}{\theta_*^2\frac{P_{H_2}}{p^\circ}} \Rightarrow \theta_N = \sqrt{K_1 P_{H_2}}\,\theta_*$$

$$\theta_{NH_2} = \frac{P'_{NH_3}\,\theta_*}{K_4\sqrt{K_5 P'_{H_2}}} \quad ; \quad \theta_N = \sqrt{K_5 P'_{H_2}}\;\theta_*$$

$$\theta_{NH} = K_2\sqrt{K_1 P'_{H_2}}\;\sqrt{K_5 P'_{H_2}}\;\theta_*$$

$$r_{3_{lim}} = \frac{k_3 K_2 K_1^{1/2} K_5 \sqrt{P'_{N_2}}\; P'_{H_2}\,(1-\sqrt{\gamma})}{\left(1+\sqrt{K_5 P'_{H_2}}+\sqrt{K_1 P'_{H_2}}+K_2\sqrt{K_1 P'_{N_2}}\sqrt{K_5 P'_{H_2}}+\frac{P'_{NH_3}}{K_4\sqrt{K_5 P'_{H_2}}}\right)^2}$$

Principle: many catalysts follow either $r_{1_{lim}}$, $r_{3_{lim}}$ expressions.

Lecture 15

Tuesday, 7 March 2023 10:40 AM

Project topic: HER on Pt(111) → aq. acidic solution

Volcano plots for heterogenous catalysis

ex: 1) $A_2(g) + 2* \rightleftharpoons 2A^*$ appropriate if θ_B is small ; If $\frac{kP_B}{p^\circ} \ll 1$ → we can assume θ_B is small

2) $A^* + B(g) \rightleftharpoons AB(g) + *$

$B(g) + * \rightarrow B^*$

$\theta^* = \frac{1}{1 + kP_B + \dots}$

$r = r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A'^2$

$K_2 = \frac{\theta_* P_{AB}'}{\theta_A' P_B'}$; $r = k_1 P_{A_2} \theta_*^2 - \frac{k_{-1}}{K_2^2} \left(\frac{P_{AB}'}{P_B'}\right)^2 \theta_*^2$

$= k_1 P_{A_2} \theta_*^2 \left(1 - \frac{k_{-1}}{k_1 K_2^2} \frac{P_{AB}'^2}{P_{A_2} P_B'^2}\right)$

$= P_0/K_1$

$r = k_1 P_{A_2} \theta_*^2 \left(1 - \frac{1}{K_1 K_2^2} \frac{P_{AB}'^2}{P_B'^2 P_{A_2}'}\right)$

γ = approach to equilibrium (set by conditions in reactor)

Site balance:

$\theta_* + \theta_A' = 1$

$\theta_* + \theta_* \frac{P_{AB}'}{K_2 P_B'} = 1$

Vacant sites

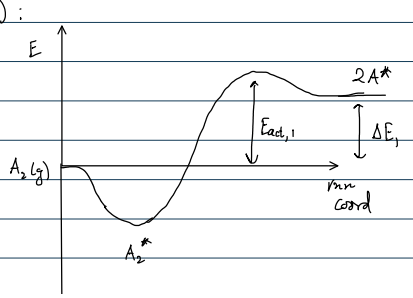
$$r = \frac{k_1 P_{A_2}' (1-\gamma)}{\left(1 + \sqrt{k_1 P_{A_2}'}\,\gamma\right)^2}$$

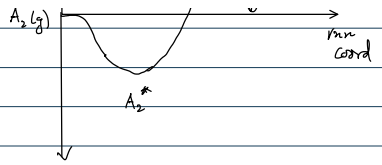
1^2 ← ignoring difference b/w H and E

$K_1 = e^{\Delta S_1^\circ / k_B} e^{-\Delta E_1 / k_B T}$

$k_1 = \frac{k_{1o}}{p^\circ} e^{-\frac{\Delta E_{act,1}}{k_B T}}$ } Find k_{1o} from TST

Energy diagram for (1):





If rxn follows a BEP relation: $E_{act,1} = \alpha_1 \Delta E_1 + B_1$

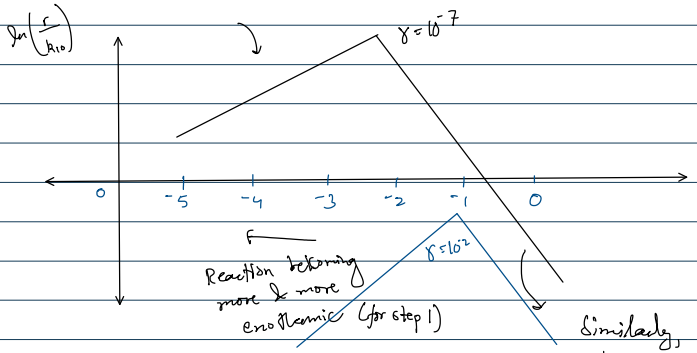
$$\ln\left(\frac{r}{k_{10}}\right) = -\frac{\alpha_1 \Delta E_1 + B_1}{k_B T} + \ln\left(\frac{P_{A_2}}{P^\circ}\right) + \ln(1-\theta) - 2\ln\left[1+\sqrt{e^{\frac{\Delta S_1^\circ}{k_B}} e^{\frac{-\Delta E_1}{k_B T}} \frac{P_{A_2}}{P_0}\gamma}\right]$$

With the RPBE functional,

$\alpha_1 = 0.9$, $B_1 = 1.34\ eV$; $\frac{P_{A_2}}{P^\circ} \approx 25$

$\Delta S_1^\circ = -200\ \frac{J}{mol\ K}$

} We assume everything else is independent of catalyst surface
↳ i.e. variations in catalyst are given by ΔE_1



(Rate decreases)
↓
Small no. of vacant sites
(A is bounding very strongly)
↳ θ_A approaches 1

Similarly, from BEP relations, $E_{act} \propto \Delta E_1$
(The activation energy increases

1) There are two physical processes that limit rate (even w/ 1 RLS)

2) Rate decreases as $\gamma \to 1$

3) Volcano maximum moves to higher ΔE_1 } $r = \frac{k_1 P_{A_2}}{\left(1 + K_1 \frac{P_{A_2}}{P^\circ}\gamma\right)^2}(1-\gamma)$

As $\gamma \to 1$, more A^* on surface
so catalyst compensates by weaker binding of A^*

4) Decrease of P_{A_2} shifts max to stronger binding of A^* (to compensate for reduced θ_*)

5) If TT, max. moves to stronger binding (same idea)

5) If T↑, max. moves to stronger binding (same idea)
(Langmuir isotherm → decrease θ_A on surface)

For a given catalyst, it will have just one E_r → the volcano plot actually tells you which one/catalyst would be optimal at the new T, P.

} Also, assuming that the RLS is the same across these changing conditions.

Ex 2: Same idea, but assume θ_{A_2} is not small

1) $A_2(g) + * \rightleftharpoons A_2^*$

2) $A_2^* + * \rightleftharpoons 2A^*$ RLS

3) $A^* + B_{(g)} \rightleftharpoons AB_{(g)} + *$

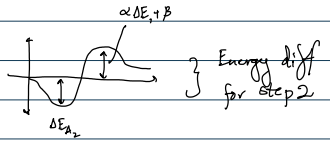
$r = r_2 = k_2 \theta_{A_2} \theta_* - k_{-2} \theta_A^2$; $K_1 = \frac{\theta_{A_2}}{P'_{A_2} \theta_*}$; $K_3 = \frac{P'_{AB} \theta_*}{P'_B \theta_A}$

$$\theta_* = \frac{1}{1 + K_1 P'_{A_2} + \frac{P_{AB}}{P_B} \frac{1}{K_3}}$$

(*) (A₂*) (A*)

$$r = \frac{k_2 K_1 \frac{P_{A_2}}{P^\circ} (1-\gamma)}{\left(1 + K_1 \frac{P_{A_2}}{P^\circ} + \sqrt{K_1 k_2 \frac{P_{A_2}}{P^\circ} \gamma}\right)^2} \quad ; \quad \gamma = \frac{1}{K_1 K_2 K_3^2} \left(\frac{P_{AB}^2}{P_B^2 \, P_{A_2}/P^\circ} \right)$$

$K_1 = e^{\frac{\Delta S_1^\circ}{k_B}} e^{\frac{-\Delta E_{A_1}}{k_B T}}$



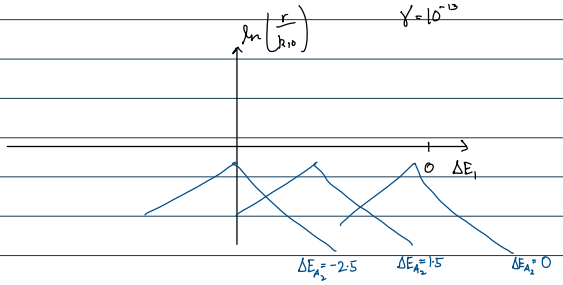
} Energy diff for step 2

$K_2 = e^{\circ} e^{\frac{-(-\Delta E_{A_2} + \Delta E_1)}{k_B T}}$

$k_2 = k_{20} e^{-(\alpha \Delta E_1 + \beta - \Delta E_{A_2})/k_B T}$

$$\ln\left(\frac{r}{k_{20}}\right) = -\frac{\alpha_1 \Delta E_1 + \beta_1}{k_B T} + \frac{\Delta S_1^\circ}{k_B} + \ln\left(\frac{P_{A_2}}{P^\circ}\right) + \ln(1-\gamma) - 2\ln\left[1 + K_1 \frac{P_{AB}}{P^\circ} + \sqrt{K_1 K_2 \frac{P_{A_2}}{P^\circ} \gamma}\right]$$

With some parameters,



↳ Why does max shift as ΔE_{A_2} increases?

↳ What happens if ΔE_1 & ΔE_{A_2} are correlated?

Lecture 16

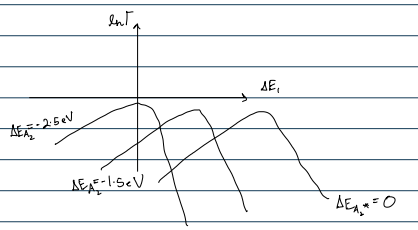
Tuesday, 21 March 2023 10:32 AM

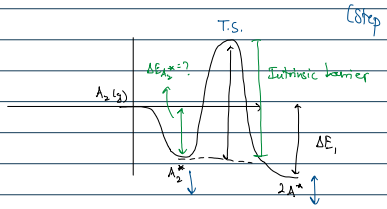
Recall:

1) $A_2(g) + * \rightleftharpoons A_2^*$

2) $A_2^* + * \rightleftharpoons 2A^*$

3) $A^* + B(g) \rightleftharpoons AB(g) + *$





(Step 2 is RLS)

Moving A_2^* will increase the E for TS → So ΔE_1 has to move proportionally down to keep the same barrier

What if there is more than 1 rate controlling step?

1) $A_2(g) + 2* \rightleftharpoons 2A^*$

2) $A^* + B(g) \rightleftharpoons AB(g) + *$

A_2^* → corresponds to the binding energy → If A_2^* goes really high up then the RLS will change

$$r_1 = k_1 \theta_*^2 P_{A_2} - k_{-1} \theta_A^2$$

$$r_2 = k_2 \theta_A P_B - k_{-2} \theta_* P_{AB}$$

$$r_1 = k_1 \theta_*^2 P_{A_2}(1-\gamma_1) \quad ; \quad \gamma_1 = \frac{\theta_A^2}{\theta_*^2 P_{A_2}' K_1}$$

$$r_2 = k_2 \theta_A P_B(1-\gamma_2) \quad ; \quad \gamma_2 = \frac{\theta_* P_{AB}'}{\theta_A P_B' K_2}$$

$$\gamma_{overall} = \gamma_1 \gamma_2^2$$

Steady state for A^*: $2r_1 = r_2$

Site balance: $\theta_* + \theta_A = 1$

$$2k_1 P_{A_2} \overbrace{(1-\theta_A)}^{\theta_*}{}^2 - 2k_{-1}\theta_A^2 = k_2 \theta_A P_B - k_{-2} P_{AB} \overbrace{(1-\theta_A)}^{\theta_*}$$

(quadratic in θ_A)

$$\theta_A = \frac{1 + k_{-2} P_{AB}\left(1 + K_2 \frac{P_B}{P_{AB}}\right)}{4k_1 P_{A_2}} \Bigg/ \left(1 - \frac{1}{K_1 P_{A_2}'}\right)$$

$$\left[1 - \sqrt{1 - \left(1 - \frac{1}{K_1 P_{A_2}}\right) \frac{\left(1 + \frac{k_{-2} P_{AB}}{2k_1 P_{A_2}}\right)}{\left(1 + \frac{k_{-2} P_{AB}\left(1 + K_2 \frac{P_B}{P_{AB}}\right)}{4k_1 P_{A_2}}\right)^2}} \right]$$

$$r = r_1 = k_1 P_{A_2}(1-\theta_A)^2 - k_{-1}\theta_A^2$$

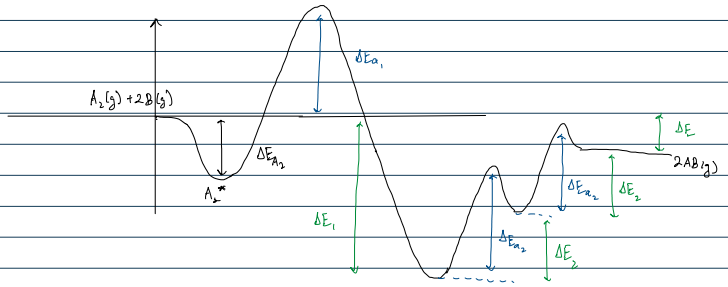
$$= k_1 P_{A_2}(1-\theta_A)^2 \left[1 - \frac{\frac{\theta_A^2}{(1-\theta_A)^2 P_{A_2}'}}{K_1} \right]$$

Exercise: Show that θ_A reduces to QE exp when

a) $\frac{k_{-2}}{k_1} \ll 1$ b) $\frac{k_{-2}}{k_1} \gg 1$

Step 2 is RLS Step 1 is RLS

Overall Energy Diagram



$$k_1 = k_{1_0} e^{-\frac{\Delta E_{a_1}}{k_B T}} \quad ; \quad k_{1_0} = \frac{k_B T}{h P_0} e^{\frac{\Delta S_1^{o\ddagger}}{k_B}} \quad ; \quad \Delta S_1^{o\ddagger} \approx -200 \frac{J}{mol\,K}$$

$$K_1 = e^{\Delta S_1^\circ / k_B} e^{-\Delta E_1 / k_B T}$$

$$\Delta E_{a_1} = \alpha_1 \Delta E_1 + \beta_1 \quad ; \quad K_2 = e^{\Delta S_2^\circ / k_B} e^{-\Delta E_2 / k_B T}$$

$$k_2 = \frac{k_B T}{h P^\circ} e^{\Delta S_2^{o\ddagger} / k_B} e^{-\Delta E_{a_2} / k_B T} \quad ; \quad k_{-2} = \frac{k_2}{K_2} \quad ; \quad \Delta S_2^{o\ddagger} \approx -200 \frac{J}{mol\,K}$$

$$\Delta E_{a_2} \approx \alpha_2 \Delta E_2 + \beta_2 \qquad \alpha_2 < 0$$

$$\alpha_2 \approx -\frac{1}{2} \quad ; \quad \beta_2 \approx 0.8\ eV$$

} Values can vary in the literature

where does this come from? physical meaning?

$$\Delta E_2 = \frac{\Delta E_{ov} - \Delta E_1}{2} \quad ; \quad \text{choose } \Delta E_{ov} \approx -1\ eV$$

$$\Delta S_2^\circ \propto S_{AB(g)}^\circ - S_{B(g)}^\circ \approx 0 \quad ; \quad \gamma = \frac{(P_{AB}/P_B)^2}{k_1 k_2^2 P_{A_2}'} \qquad \text{Assume, } P_{A_2}' = 25$$

↑ Standard entropy change for step 2

$$\Downarrow$$

$$\frac{P_{AB}}{P_B} = \sqrt{k_1 k_2 P_{A_2}' \gamma} \quad ; \quad P_B' \approx 50$$

↑ treat as a parameter (or add a reactor model)

→ design equations for PFR, PBR

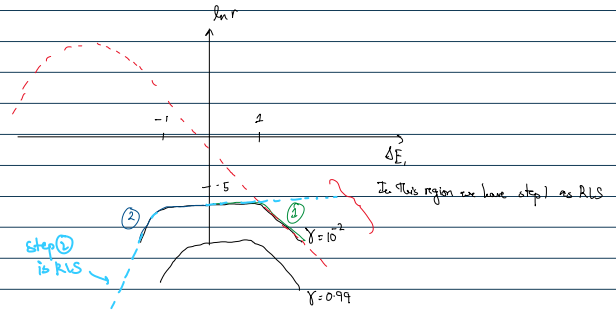
} Can solve for microkinetic eq & reactor eq to solve for partial pressures across the reactor

Note: ① Values of $S_s^{o\prime}$, P_s', γ have a big effect on rate, but a smaller effect on volcano maximum

② BEP relationships affect everything!

Volcano:

Assuming that first step is a lumped step; → Don't need binding/activation energy of step 2!



Different "legs" of volcano (① & ③) correspond (approx.) to different assumptions about RLS's

Leg ①: Assume step 1 is RLS (as before)

$$r = r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A^2 \quad ; \quad \theta_A = k_1^{-1} \frac{P_{AB}}{P_B} \theta_*$$

{ γ → overall approach to equilibrium

Leg ①: Assume step 1 is RLS (as before)

$r = r_1 = k_1 P_{A_2} \theta_\alpha^2 - k_{-1} \theta_A^2$; $\theta_A = k_2^{-1} \frac{P_{A_2}}{P_B} \theta_\alpha$

$\gamma \rightarrow$ overall approach to equilibrium

$r = k_1 P_{A_2} \theta_\alpha^2 (1-\gamma)$

$r_① = \frac{k_1 P_{A_2}}{\left(1 + k_2^{-1} \frac{P_{A_2}}{P_B}\right)^2} (1-\gamma)$; Sabatier: assume max rates $\hookrightarrow \theta_\alpha = 1$

$r_{cat} \approx k_1 P_{A_2} (1-\gamma)$ (upper bound)

↓ Sketch on above graph

$r_②$: assume step 2 is RLS:

$r = \frac{1}{2} r_② = \frac{1}{2} (k_2 P_B \theta_A - k_{-2} P_{A_2} \theta_\alpha)$

Step 1 is QE: $\theta_A = \sqrt{k_1 P_{A_2}'} \, \theta_\alpha$; $\theta_A + \theta_\alpha = 1$

$r_② = \frac{1}{2} k_2 P_B \theta_A (1 - \sqrt{\gamma})$

$r_③ = \frac{\frac{1}{2} k_2 \sqrt{K_1 P_{A_2}'}}{1 + \sqrt{K_1 P_A^2}} (1 - \sqrt{\gamma})$ < upper bound $\frac{1}{2} k_2 P_B \ldots$

Note: 1) The legs (sides) are determined by assuming different RLS's. The max of the volcano is a transition between different RLS

[Each RLS has its own volcano - competition b/w rate constants and coverage effects]

2) The limiting expressions approximate the true volcano except where volcanos cross

3) Sabatier rates are useful if $\gamma \ll 1$

4) Volcanos assuming only 1 RLS can wildly overestimate rates

↳ Volcanoes w/ > 2 steps may have many legs!

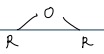
Lecture 17

Thursday, 30 March 2023 10:30 AM

Term Projects:

Epoxidation: $-C=C- \rightarrow$ (Epoxide) $-C(-O-)C-$

Ether:



General:

1) Literature survey: key papers

2) Calculations: mostly structural & thermodynamics but basic transition state for exp. students

HER on Pt (111) surface; + No water

1) $H^+ + e^- + * \leftrightarrow H^*$ (Don't need the kinetics but just the thermodynamics)

2) $2H^* \longleftrightarrow H_2(g) + 2*$ (Long-Hinshelwood step)

BE_H vs. θ_H (coverage) (Binding energy of H) $\Rightarrow$ Calculate H_2 formation barrier vs θ_H

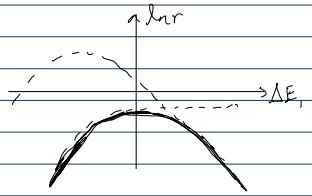
+ Surf-graph algorithm (Help w/ symmetry)

Then make a micro-kinetic model !!

Scaling Relationships & Volcano plots

1) $A_2(g) + 2* \longleftrightarrow 2A^*$

2) $A^* + B(g) \longleftrightarrow AB(g) + *$



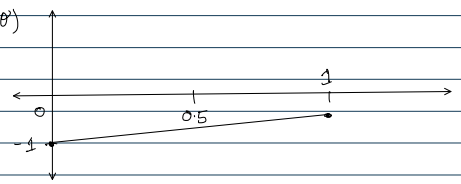
Interactions b/w adsorbates:

+ Surface phase diagram

+ Isotherms

+ Difficult to find relationships that are general enough to describe all of a volcano plot

+ Coverage-dependent binding energies (if deviations from mean field are small):



$$\Delta E_1 = \frac{E(N) - E(N-1)}{1} - E_{A(g)}$$

differential energy (given that there are already some adsorbates on the surface)

More weakly adsorbed @ higher coverages (more endothermic due to repulsion)

ex: $\Delta E_1(\theta) = \Delta E_1(\theta = 0) + \underbrace{0.75\theta}_{\text{Arbitrary slope}}$

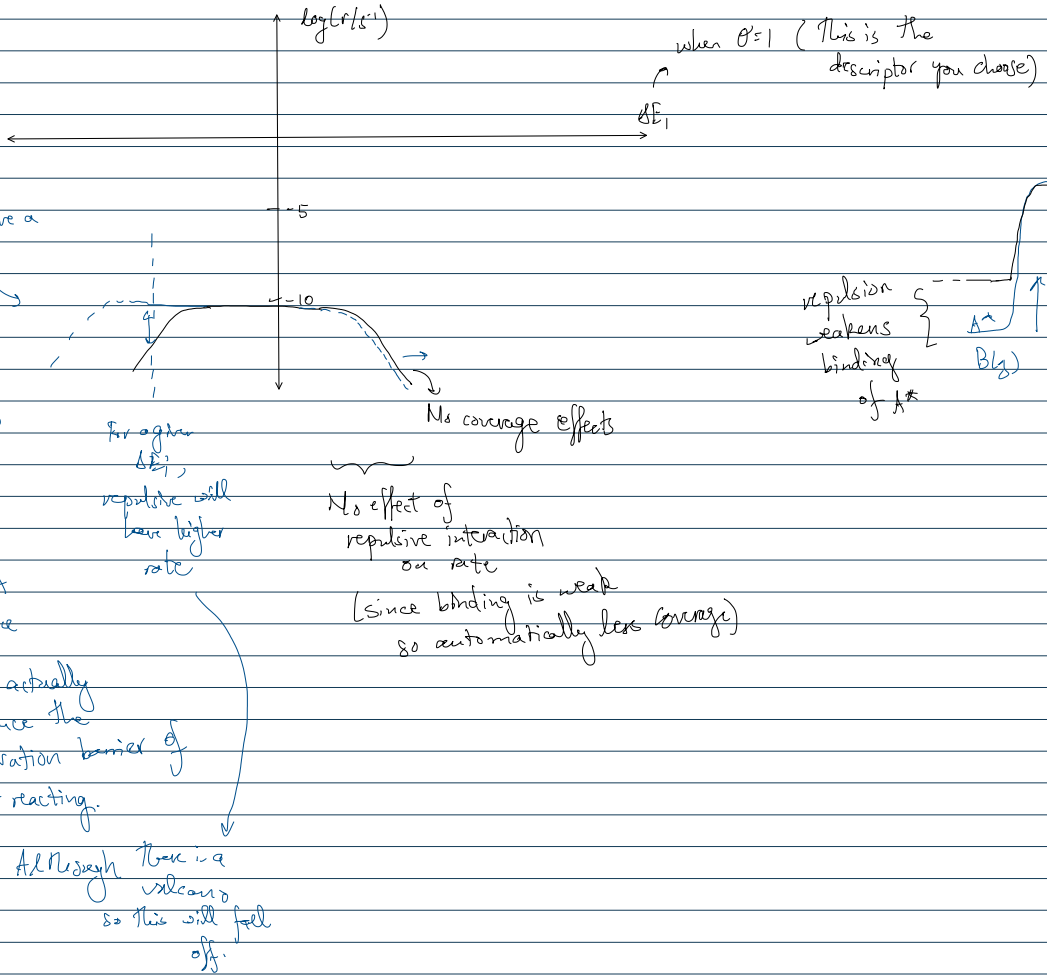
If we assume that BEP does not change w/ coverage (slope & intercept are same):

$$E_{a_1}(\theta) = \alpha_1 \Delta E_1(\theta) + \beta_1$$

If we assume that BEP does not change w/ coverage (slope & intercept are same):

$$E_{a_i}(\theta) = \alpha_i \Delta E_i(\theta) + \beta_i$$

Do not depend on coverage (although not always true)



Usually consider interactions b/w MASI & other intermediates on surface

Ammonia Synthesis:

1) $N_2(g) + 2* \leftrightarrow 2N^*$
2) $N^* + H^* \leftrightarrow NH^* + *$
3) $NH^* + H^* \leftrightarrow NH_2^* + *$
4) $NH_2^* + H^* \leftrightarrow NH_3(g) + 2*$
5) $H_2(g) + 2* \leftrightarrow 2H^*$

2 competing RLS

Limiting Analysis

overall reaction

1) 1 is RLS:

$$r = 2k_1 P_{N_2} (1-\theta)$$

Kinetic Analysis

1) 1 is RLS :
(from earlier analysis)

$$\Gamma_1 = \frac{2k_1 P_{H_2}(1-\theta)}{\left[1+\sqrt{K_5 P'_{H_2}}+\frac{P_{NH_3}}{K_4\sqrt{K_5 P_{H_2}}}+\frac{P'_{NH_3}}{K_3K_4K_5P'_{H_2}}+\frac{P'_{NH_3}}{K_3K_4K_5(K_5P'_{H_2})^{3/2}}\right]^2}$$

2) Step 3 is RLS:

$$\Gamma_3 = \frac{k_3K_2K_1^{1/2}K_5\,P_{N_2}^{1/2}\,P_{H_2}\,(1-\sqrt{\theta})}{\left[1+\sqrt{K_5P_{H_2}}+\sqrt{K_1P_{N_2}}+K_2\sqrt{K_1P_{H_2}}\sqrt{K_5P_{H_2}}+\frac{P_{NH_3}}{K_4\sqrt{K_5P_{H_2}}}\right]^2}$$

Parameters: $P'_{H_2} = 75$, $P'_{N_2} = 25$

$S^\circ_{H_2} \approx 192$ J/mol K ; $S^\circ_{NH_3(g)} \approx 193$ J/mol K
(From tables or partition function approaches)

↳ (Ignore S of surface species): $\Delta S^\circ_1 \approx -192$, $\Delta S^\circ_2 \approx \Delta S^\circ_3 \approx 0$, $\Delta S^\circ_4 = 193$, $\Delta S^\circ_5 \approx -131$

$E_{a_1} \approx 0.9\,\Delta E_1 + 1.34$ eV (RPBE)
↳ The defects/steps are taken as active sites

↳ 4 adsorption intermediates
↳ Can we express all energies in terms of ΔE_1?

Many parameter are correlated:

$\Delta E_{NH_2} = 0.37\,\Delta E_N - 1.26$
$\Delta E_{NH} \approx 0.67\,\Delta E_N - 1.04$
} DFT RPBE

Note: $\Delta E_{NH^*} \equiv \Delta E_{NH^*} - E_* - E_{NH_3(g)}$ ← Definition

These relations result from bond order conservation: Thus, $K_2 = e^{\frac{-\Delta E_2 - T\Delta S^\circ_2}{k_B T}}$ (need to find ΔE_2 expression)

↓

ΔE for reaction 2

$\Delta E_2 \equiv E_{NH^*} + E_* - E_{H^*} - E_{H^*}$ } expression in terms of ΔE_1

Lecture 18

Tuesday, 4 April 2023 10:33 AM

Tutorial on Thursday; Last week final project/presentation due

Ammonia synthesis as model reaction (Norskov group)

1) $N_2(g) + 2* \rightleftharpoons 2N^*$ → ΔE_1 ,

2) $N^* + H^* \rightleftharpoons NH^* + *$ → $\Delta E_{NH} = E_{NH^*} - E_* - E_{NH(g)}$

How many independent descriptors do we need to describe trends in activities?

3) $NH^* + H^* \rightleftharpoons NH_2^* + *$ → ΔE_{NH_2}

4) $NH_2^* + H^* \rightleftharpoons NH_3^* + 2*$

ΔE_H

5) $H_2(g) + 2* \rightleftharpoons 2H^*$

Scaling Relationships (real relationships that must be accounted for) → most accurate for species that bind using the same atom → H & N are linked via scaling relationship

$\Delta E_{NH} = 0.67\,\Delta E_N - 1.04$ (RPBE, eV)

$\Delta E_{NH_2} = 0.37\,\Delta E - 1.26$

Intercepts will change w/ functional (But slopes usually don't change)

$\Delta E_2 = E_{NH^*} + E_* - E_{N^*} - E_{H^*}$

$= -\Delta E_N - \Delta E_H + \Delta E_{NH} + \underbrace{(E_{NH} - E_N - E_H)_{gas}}_{\Delta E_{gas}}$; Can also use DFT to calculate energy of isolated species in gas phase

Plugging in ΔE_{NH} scaling relationship:

$\Delta E_2 = (0.67 - 1)\,\Delta E_N - 1.04 + \Delta E_{gas} - \Delta E_H$;

$\Delta E_H \approx -2.6$ eV (approximate this because H ads does not vary too much across different surfaces)

Recall:

$\Delta E_1 = (2E_N - E_{N_2})_{gas} + 2\Delta E_N$

$\Delta E_N = \frac{\Delta E_1}{2} + \frac{(E_{N_2} - 2E_H)_{gas}}{2}$

→ Alternatives: ① Finding a scaling relation b/w ΔE_H and ΔE_N

② Keep ΔE_H as an independent descriptors

$\therefore \Delta E_2 = -0.16\,\Delta E_1 - 0.16\underbrace{(E_{N_2} - 2E_H)_g}_{-9.47\text{eV}} - 1.04 - (-2.6) + \underbrace{(E_{NH} - E_N - E_H)_g}_{-3.76}$

$\Delta E_2 = -0.16\,\Delta E_1 - 0.69$

$\Delta E_6 = \underbrace{(E_{H_2} - 2E_H)_{gas}}_{4.58\text{eV} \rightarrow \text{constant for all catalysts}} + 2\Delta E_H = -0.62$ eV

$\Delta E_3 = -0.15\,\Delta E_1 - 0.22$; $\Delta E_4 = -0.19\,\Delta E_1 + 0.97$ → Slopes are similar due to the similarity of the elementary step.

Combining BEP and TST (Literature: J. Cat v192, p31, 2000)

→ $k_1 = \frac{1.2\times10^6}{\sqrt{T}}\left(\frac{\sqrt{K}}{bar}\right)\exp\left[\frac{-0.9\,\Delta E_1 + 1.34}{k_B T}\right]$

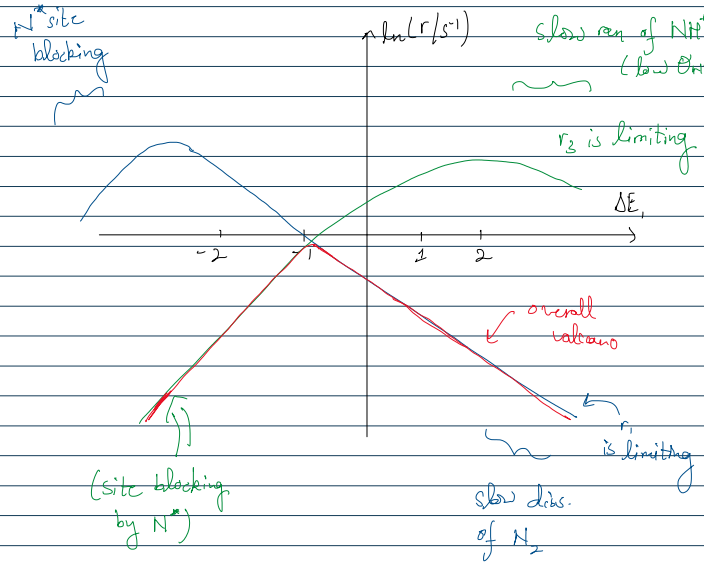
BEP relation

→ (Lit. J. Cat, V220, p273, 2003)

E_{a_3} (Ru step) = 1.35 eV

ΔE_3 (Ru step) = −0.83 eV

Assume, $\Delta E_{a_3} = \alpha_3 \Delta E_3 + \beta_3$; $\beta_3 = 1.51$ eV (0.9, −0.15δE − 0.32) } Very approx. results



(Caution): Most papers often assume that step 1 is always rate-limiting ↓ (not always true)

Q: How can you have r_3 limiting & still be varying w/ ΔE_1 descriptor ????

Physical basis of scaling relationships

+ Usually results from bond-order conservation principle

Bond order conservation

eg: ΔE_{NH} vs. ΔE_N ; N can form 3 bonds ; BOC: When N-containing species adsorb the maintain valency by making an appropriate no. of bonds to surface

Thus, $\Delta E_N \approx 3 e_{N-M} + \text{const.} = (E_{N^*} - E_* - E_{N(g)})$

(e_{N-M}: average strength of N-metal bond (forms 3 bonds to the surface))

$\Delta E_{N-H} \approx 2 e_{N-M} + \text{const.}$; $e_{N-M} = \frac{1}{3}\Delta E_N \Rightarrow$ $\boxed{\Delta E_{NH} = \frac{2}{3}\Delta E_N + \text{const.}}$ ★

(2 bonds to the surface (since 1 is bonded to H))

This is very robust on metal surfaces, simple oxides (reducible), zeolites, 3-phase boundary ↓ if charge balanced

In general, $\boxed{\Delta E_{A B_n} = \frac{n_{max,A} - x}{n_{max,A}} \Delta E_A + \text{const.}}$ → first order approximation using just simple bond order analysis as done previously

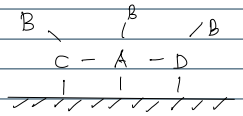
(B_n: binds to surface; B is often H)

One data point will give const.: $\text{const.} = \Delta E_{AB_n/Pt} - \frac{n_{max} - x}{n_{max,A}} \Delta E_{A/Pt}$ Pt

$$\therefore \Delta E_{AB_x} - \Delta E_{AB_x/Pt} = \frac{n_{max} - x}{n_{max,A}} \left(\Delta E_A - \Delta E_{A/Pt} \right)$$

Slope is relatively insensitive to facet, but intercept is sensitive

Consider multidentate species:

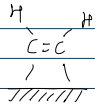


(eg)

Contribution from binding through C:

$$\Delta E_{CAB_2} - \Delta E_{CAB_2/Pt} = \frac{n_{max,} - n_{CAB_2}}{n_{max,C}} \left(\Delta E_C - \Delta E_{C/Pt} \right)$$

Assume contributions from C, A, D are independent → use linear sum } All bonds contribute a single bond order → This fails if we have multiple bonds b/w adsorbed species



$$\Delta E - \Delta E_{Pt} = \sum \frac{(n_{max,i} - n_i)}{n_{max,i}} \left(\Delta E_i - \Delta E_{i/Pt} \right)$$

But for eg if you have aromatics

In practice, higher order bonds are broken on surface, so scaling still holds (exceptions: Au, Ag, Cu) } Observation: Once in a while, species can form more than "one effective" bonds to the surface

Scaling is more accurate when correlating binding energies of species bound to same site geometries

↳ create separate scaling relations for diff site types.

Lecture 19

Tuesday, 11 April 2023 10:35 AM

Start project!, Send email about bonus lecture

Alternate Scaling Relationships

Traditionally: correlate binding of 2 adsorbates across many surfaces

↳ Correlate binding energies of many adsorbates on a single surface.
(cf group contribution methods)

Bond-order parameter: $\frac{\chi_{max} - \chi}{\chi_{max}}$ → Use this to correlate many adsorbates:

$$BE_{\text{(``any'' adsorbate)}} = \sum_{\substack{i \text{ atoms} \\ \text{that bind} \\ \text{covalently} \\ \text{to surface}}} P_i \left(\frac{\chi_{max} - \chi}{\chi_{max}} \right) + \text{const.}$$

P_i: parameters fit to DFT data (different for each surface)

} Simple regression analysis

w/ a small no. of DFT calcs, we can estimate BEs of many species

↳ Can add higher order corrections: $BE = \sum_i P_i \left(\frac{\chi_{max} - \chi}{\chi_{max}} \right) + \sum_{\substack{i,j \\ \text{nearest} \\ \text{neighbor} \\ \text{atoms}}} P_{ij} \left(\frac{\chi_{max,i} - \chi_i}{\chi_{max,i}} \right) \left(\frac{\chi_{max,j} - \chi_j}{\chi_{max,j}} \right)$

P_{ij}: interactions b/w atoms

Classic Example: Scaling/BEP analysis (Mavrikakis et al., JACS (2004))

Consider ethanol partial oxidation on Transition Metal Surfaces

$$C_2H_5OH_{(g)} + * \rightleftharpoons C_2H_5OH^*$$

$$C_2H_5OH^* + * \overset{C\text{-}H}{\rightleftharpoons} C_2H_4OH^* + H^*$$

$$\overset{O\text{-}H}{\searrow\!\!\nearrow} \quad C_2H_6O^* + H^* \overset{C\text{-}H}{\rightleftharpoons} C_2H_4O^* + H^*$$

$$C_2H_4OH^* + * \overset{C\text{-}H}{\rightleftharpoons} C_2H_3OH^* + H^*$$

$$\overset{O\text{-}H}{\searrow\!\!\nearrow} \quad C_2H_4O^* + H^*$$

$$\vdots$$

More dehydration can occur, down to CCO^* (never achieved in practice)

Intermediates:

$$C_2H_3OH^* + * \overset{C\text{-}C}{\rightleftharpoons} CH^* + CH_2OH^*$$

$$\overset{C\text{-}C}{\rightleftharpoons} CH_2^* + CHOH^*$$

Similar steps for other intermediates

$\overset{C-C}{\rightleftharpoons} CH_3^* + COH^*$

$\overset{C-O}{\rightleftharpoons} C_2H_3^* + OH^*$

To get maximum H_2 yield, suppress C-O bond scission (usually gives alkanes + H_2O)

How can one analyze such a complex mechanism?

Surface adsorption energies and surface coverages

↳ starting w/ a complete database of binding energies on Ru(0001), use scaling relationship

$$\Delta E - \Delta E_{Ru} = \sum_i \left(\frac{\chi_{max,i} - \chi_i}{\chi_{max,i}} \right) (\Delta E_i - \Delta E_{i/Ru}) \quad ; \quad i = C, O \text{ atoms}$$

↳ From these values, estimate θ's of all intermediates on all surface by assuming QE w/ appropriate gas species:

$$C_xH_y^* \rightleftharpoons C_3H_6(g) + \frac{y}{2} H_2(g) + *$$
$$C_xH_yO_z^* \rightleftharpoons C_2H_5OH(g) + \frac{z}{2} H_2(g) + *$$
$$CH_x^* \rightleftharpoons CH_4(g) + \frac{x}{2} H_2(g) + *$$

} Assume not rate limiting

⇓

For all metals except Cu, Ag, Au ; CO^* is the MASI

These had * as the MASI

} This is an approximation as well but assuming that C-H bond breaking is QE?

→ Since CO^* is at high coverage, account for interactions of species w/ CO^*

$BE_{CO}(\theta_{CO}) = -1.78 + 0.0065\, e^{4.79\theta_{CO}} + 0.314\, \theta_{CO}\, e^{4.79\theta_{CO}}$ } Pt(111) [PW91, differential BE = $\frac{dE}{d\theta}$]

↳ Crude first approximation: assume θ_{CO}-dependent terms for all adsorbates on all metals.

(b) Kinetics: + Use a BEP relationship to estimate barriers (energies of TS) for all possible reactions w/ C-C or C-O cleavage.

2 approaches: (i) correct reactant / product BE for θ_{CO}, then apply BEP

(ii) Apply BEP w/o θ_{CO}, then correct TS & reactant/product separately for θ_{CO}

} Barriers

+ Use coverages from thermo analysis

+ Calculate rates on each metal according to most stable TS for C-C & C-O bond breaking

$$R = 10^{13}\, s^{-1}\, \theta_{TS}\, \theta_*$$

$$\left(\approx \frac{k_B T}{h} K_{TS}\, \theta_{TS}\, \theta_* \right)$$

(c) Steps / Structure Sensitivity

+ general approach is to calculate ΔE to dissociate CO^* on steps

→ if $\Delta E < 0$, then steps will be poisoned w/ C^* ↔ do not contribute

→ if $\Delta E > 0$, steps must be considered

→ Redo calcs on steps

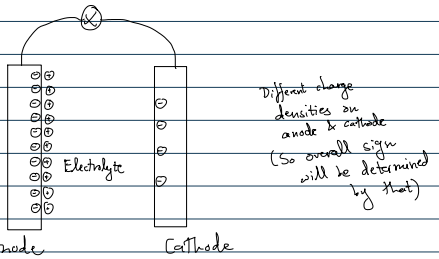
Con: not stated which is better

Lecture 20 (CHE)

Tuesday, 18 April 2023 10:32 AM

Electrochemistry

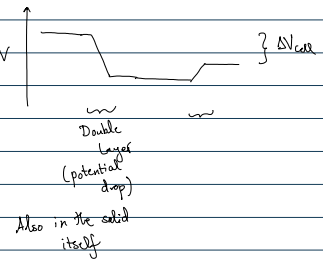
- electrocatalyst
- batteries
- capacitors
- devices
- Corrosion



Different charge densities on anode & cathode (So overall sign will be determined by that)

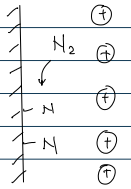
Key features: ① Solvent/solid interfaces
water - electrocatalysis
organic - batteries

② Existence of a charged double layer near the surface of an electrode that establishes potential drops across surfaces



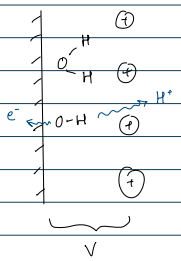
Reactions @ electrified interfaces

Chemical:



+ No e^- transfer
+ Small voltage dependence (rearrangement of anions)
↓
Can use normal DFT (possibly including solvent molecules)

Electrochemical:



$* + H_2O \rightarrow OH^* + H^+ + e^-$

→ Generate positive/negative charges and so are hugely affected by voltage

Analyzing thermodynamics.

+ Calculating ΔG:

$$\Delta G = \mu_{OH^*} + \mu_{H^+} + \mu_{e^-} - \mu_* - \mu_{H_2O}$$

$$\mu_{OH^*} \approx E_{OH^*} - TS_{OH^*} + ZPE_{OH^*} + \dots$$

$$\mu_* \approx E_*$$

$$\mu_{H_2O} \approx E_{H_2O} - TS_{H_2O} + ZPE_{H_2O}$$

$\mu_{H^+} + \mu_{e^-}$: assume proton entropy (activity) is $\sim k_B T \ln[H^+]$ } Assuming that H^+ don't interact w/ each other → even at pH 0, 1 H^+ per 55 H_2O molecules [So pretty good assumption unless negative pH's]

+ Assume electrostatic energy to separate charges across double layer is $\sim qV = -eV$

Thus, $\mu_{H^+} + \mu_{e^-} = \text{const.} + k_B T \ln[H^+] - eV$

(const.: undetermined)

Thus, $\mu_{H^+} + \mu_{e^-} = \text{const.} + k_B T \ln[H^+] - eV$

(const. ← undetermined)

Standard conditions: $\mu^\circ_{H^+} + \mu^\circ_{e^-} = \text{const.} + k_B T \ln[H^+]_0 - eV$

$$\therefore \boxed{\mu_{H^+} + \mu_{e^-} = \mu^\circ_{H^+} + \mu^\circ_{e^-} + k_B T \ln \frac{[H^+]}{[H^+]_0} - e(V - V_0)}$$

V_0 is an arbitrary reference (std state) for voltage; often chosen to be: $-eV_0 \equiv \frac{1}{2}\mu^\circ_{H_2(g)} - \left(\mu^\circ_{H^+} + \mu^\circ_{e^-}\right)$ } Standard hydrogen electrode reference

↳ Experimentally, use a counter electrode where $[H_2^\circ]$ and $[H^+]_0$ are in equilibrium w/ electrode surface

$H^\circ_{2(g)}$: typically 1 bar

$H^{+\circ}$: typically 1M

We get, $$\boxed{\mu_{H^+} + \mu_{e^-} = \frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0} - eV_{SHE}} \quad ; \quad \mu^\circ_{H_2} = E_{H_2(g)} - TS^\circ_{H_2(g)} + ZPE_{H_2(g)}$$

($[H^+]_0$ ← related to pH)

V_{SHE} has the μ_{H^+} dependence

$\Delta G_2 = \Delta G_2(V, pH, BE)$

Example: ORR

1) $O_{2(g)} + 2* \leftrightarrow 2O*$

2) $O* + H^+ + e^- \leftrightarrow OH*$

3) $OH* + H^+ + e^- \leftrightarrow * + H_2O_{(l)}$

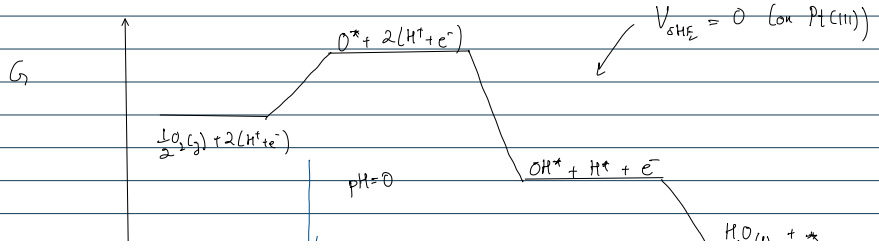
$\Delta G_1 = 2\mu_{O*} - 2\mu_* - \mu_{O_{2(g)}}$ [Independent of V]

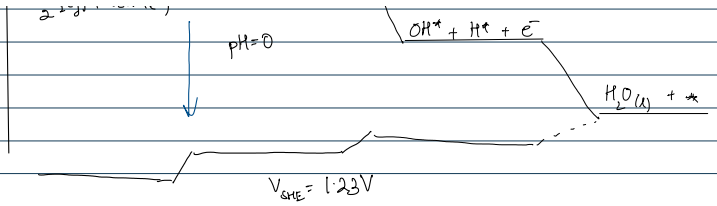
$\Delta G_2(V, pH) = \mu_{OH*} - \mu_{O*} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0}\right)$

$= \Delta G_2(0,0) - k_B T \ln \frac{[H^+]}{[H^+]_0} + eV_{SHE}$

$\Delta G_3(V, pH) = \mu_{H_2O(l)} + \mu_* - \mu_{OH*} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0} - eV_{SHE}\right)$

All quantities can be estimated w/ DFT → make a free energy diagram





V_{SHE} 0V : O_2 dissociation is RLS

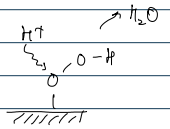
V_{SHE} 1.23V : conversion of OH^* to water may be RLS

} RLS can change w/ voltage in electrochemical systems. [RLS may be voltage dependent]

The potential at which all electrochemical steps are thermodynamically energetic is called the onset potential } approximation of V where rate becomes non-negligible

↳ We know this because when you lower the voltage from 1.23V there will be a point where the plateaus are level. (doesn't mean that they are at equilibrium)

Alternate ORR mechanism:



1) $O_{2(g)} + H^+ + e^- + * \longleftrightarrow OOH^*$

2) $OOH^* + H^+ + e^- \longleftrightarrow O^* + H_2O_{(l)}$

3) $O^* + H^+ + e^- \longleftrightarrow OH^*$

4) $OH^* + H^+ + e^- \longleftrightarrow H_2O_{(l)} + *$

Overall: $O_{2(g)} + 4(H^+ + e^-) \longleftrightarrow 2H_2O$

$$\Delta G_1(V_{SHE}, pH) = \mu_{OOH^*} - \mu_* - \mu_{O_2(g)} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln\frac{H^+}{H^+_\circ} - eV_{SHE}\right)$$

$$= \Delta G_1(0,0) + eV_{SHE} \quad @\ pH=0$$

$$\Delta G_2 = \mu_{O^*} + \mu_{H_2O(l)} - \mu_{OOH^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

$$\Delta G_3 = \mu_{OH^*} - \mu_{O^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

$$\Delta G_4 = \mu_{H_2O(l)} + \mu_* - \mu_{OH^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

Ex: $\Delta G^\circ_2(0,0) = \mu^\circ_{O^*} + \mu^\circ_{H_2O(l)} - \mu^\circ_{OOH^*} - \frac{1}{2}\mu^\circ_{H_2}$

$\mu^\circ_{H_2O(l)} = \mu^\circ_{H_2O(g)} - \Delta G^\circ_{vap}$ ↳ from exp.

$= E_{H_2O(g)} - TS^\circ_{H_2O(g)} + ZPE_{H_2O} - \Delta G^\circ_{vap}$

; $\mu^\circ_{O^*} = E_{O^*} + ZPE_{O^*} - TS^\circ_{O^*}$

$\mu^\circ_{OOH} = E_{OOH^*} + ZPE_{OOH^*} - TS^\circ_{OOH^*}$

↳ from exp.

$= E_{H_2O(g)} - TS^\circ_{H_2O(g)} + ZPE_{H_2O(g)} - \Delta G^\circ_{vap}$

$\mu^\circ_{OOH} = E_{OOH^*} + ZPE_{OOH^*} - TS^\circ_{OOH^*}$

$\mu^\circ_{H_2} = E_{H_2(g)} + ZPE_{H_2(g)} - TS^\circ_{H_2(g)}$

$$\Delta G_2^\circ = \left(E_{O^*} + E_{H_2O(g)} - E_{OOH^*} - \frac{1}{2}E_{H_2(g)}\right) + \left(ZPE_{O^*} + ZPE_{H_2(g)} - ZPE_{OOH^*} - \frac{1}{2}ZPE_{H_2(g)}\right)$$
$$- T\left(S^\circ_{O^*} + S^\circ_{H_2O(g)} - S^\circ_{OOH^*} - \frac{1}{2}S^\circ_{H_2(g)}\right) - \Delta G^\circ_{vap}$$
$$= \Delta E^\circ_{DFT} + \Delta ZPE - T\Delta S^\circ - \Delta G^\circ_{vap}$$

Often, define O^* and OOH^* w/ respect to a common reference: $\Delta E_O \equiv E_{O^*} + E_{H_2(g)} - E_{H_2O(g)} - E_*$

$[H_2O(g) + * \leftrightarrow O^* + H_2(g)]$

$\Delta E_{OOH} = E_{OOH^*} + \frac{3}{2}E_{H_2(g)} - 2E_{H_2O(g)} - E_*$

$\left[2H_2O(g) + * \leftrightarrow OOH^* + \frac{3}{2}H_2(g)\right]$

$$\boxed{\Delta G_2^\circ = \Delta E_O - \Delta E_{OOH} + \Delta ZPE_2 - T\Delta S_2^\circ - \Delta G^\circ_{vap}}$$

PCCP: 2007

$\Delta E_{OOH} = 0.53\,\Delta E_O + 3.18\ eV$

$\Delta E_{OH} = 0.5\,\Delta E_O + 0.05\ eV$ } RPBE

$\Delta G_2^\circ = \underbrace{0.47\,\Delta E_O}_{\text{depends on catalyst}} \underbrace{- 3.18\ eV + \Delta ZPE_2 - T\Delta S_2^\circ - \Delta G^\circ_{vap}}_{\text{ind. of catalyst}}$

$\Delta G_1^\circ = 0.53\,\Delta E_O + 3.18\ eV + \left(2E_{H_2O(g)} - E_{O_2(g)} - 2E_{H_2(g)}\right) + \Delta ZPE_1 - T\Delta S_1^\circ$

$\Delta G_3^\circ = -0.5\,\Delta E_O + 0.05\ eV + \Delta ZPE_3 - T\Delta S_3^\circ$

$\Delta G_4^\circ = -0.5\,\Delta E_O - 0.05\ eV + \Delta ZPE_4 - T\Delta S_4^\circ - \Delta G^\circ_{vap}$

Final report: 20 page report; decent literature review
(key papers, intro, why the interest,
methods + results, discussion, conclusion)

Geometrical alg to match surface of two solids
(w/ diff. lattice constants & periodicities).

Electrocatalysis Cont'd

ORR: 1) $O_2(g) + H^+ + e^- + * \longleftrightarrow OOH^*$

2) $OOH^* + H^+ + e^- \longleftrightarrow O^* + H_2O(g)$

3) $O^* + H^+ + e^- \longrightarrow OH^*$

4) $OH^* + H^+ + e^- \longleftrightarrow H_2O(g) + *$

Find ΔG_1, ΔG_2, ΔG_3, ΔG_4 in terms of ΔE_o

Use "limiting legs" analysis, combined w/ Sabatier Analysis (assume idealized surface energies)

Step 1 is RLS: (potential)

$$r = r_1 = k_1(U)\, P_{O_2}\, C_{H^+}\, \theta_* - k_{-1}(U)\, \theta_{OOH}$$

$$= k_1 P_{O_2} C_{H^+} \theta_* \left[1 - \underbrace{\frac{k_{-1}}{k_1} \frac{\theta_{OOH}}{P_{O_2} C_{H^+} \theta_*}}_{\gamma_1} \right]$$

Assume maximum rates: irreversibilities & idealized θ's = 1

Similarly:

$$r_1^{max} = k_1 P_{O_2} C_{H^+}$$

$$r_2^{max} = k_2 C_{H^+}$$

$$r_3^{max} = k_3 C_{H^+}$$

$$r_4^{max} = k_4 C_{H^+}$$

Example for k_1: voltage-dependent (use BEP relation)

$$\rightarrow k_1 = \frac{k_{1o}}{c_{H^+}^\circ P^\circ} \exp\left(-\alpha \left(\frac{\Delta G_1^\circ + eU}{k_B T} \right) \right)$$

(ΔG_1°: voltage dependent; usually $\frac{1}{2} < \alpha < 1$; k_{1o}: Pre-factor, Double layer)

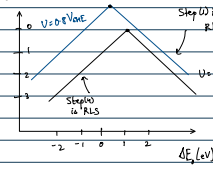
$\Downarrow$ assume $\alpha = 1$ for now

ΔG_1° expressed in terms of ΔE_o

$$-k_B T \ln\left(\frac{r_1}{k_{1o}}\right) = \Delta G_1^\circ + e V_{SHE} - k_B T \ln\left(\frac{P_{O_2} C_{H^+}}{P^\circ C_{H^+}^\circ}\right)$$

k_{1o} is a pre-exponential factor for H^+ transfer through double layer. Use TST or fit to experiment

Plot activity vs. ΔE_o for given U_{SHE}:



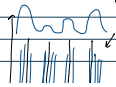
+ U affects location of max a little
+ Rate increases as U decreases (but cell efficiency decreases)

You want the potential difference b/w the anode & cathode to be the highest (but the rate is slow for when operating at 1.23 V)

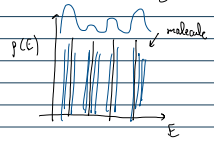
D-band Analysis

KS equations; KS wave functions → eigenvalues (ε_i) → app to energies of e^- orbitals.

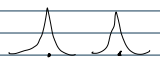
Density of states → electronic properties



Density of states → electronic properties



Norskov d-band theory: tight binding (molecular-orbital theory)



$\psi_{AB} = c_A \psi_A + c_B \psi_B$

↳ Can have more orbitals than particles also be involved.

$H_{AB} = -\frac{1}{2}\nabla^2 + v_A + v_B$

1 e^- Hamiltonian (approximate) → Matrix form & the set det. equal to zero → Formal way of understanding molecular orbital theory.

Newns-Anderson theory

↳ esp. for metals (using density of states instead of metal orbitals)

So, not completely ignoring s & p bands; d-band acts like a "molecule"

↳ just prioritizing d-band?

$\int, V_{ad}^2$ } bc/pm binding parameters

E_0 → estimate (interaction w/ s-p band)

$f = 1$ (full d-band)

↳ post-tran

mete

compression is same as molecular orbital theory emp. (ΔE)

ε_d → d-band center (i.e. independent of ads.)

ε_a →

Lecture 22 (Zeolites)

Tuesday, 25 April 2023 10:31 AM

K-point density in reciprocal space (higher unit cell requires less k-points)

↳ worse-case (over sampling k-pints for larger unit cell)

↳ Discuss this in the presentation

Solvents in microporous materials

S-L has higher complexity

Modelling accuracy: + Ignoring the solvent
+ Purely thermodynamic (chemical potentials) & phase diagram
+ H-bonding in an average way
+ Static solvent molecules (simulated annealing to find minima energy)
+ AIMD

Water in zeolite beta: how defects affect structure of water in zeolite pore

AIMD simulation
↳ for a few femtoseconds
& then relax w/ DFT

β w/ closed Sn sites → chemi-adsorb water molecules

β w/ hydrolized Sn site (OH from water & O from SiO_2) } Si-OH & OH from Sn →

Silanol Nests (OH-β)

AIMD approaches for Entropy: S = Solid-like + Gas-like

Vibrational density of states

Solid-like vibrations

Entropy increase upon formation of water chains → Approach to bulk liquid entropy at higher water density

S ↑ Gas

chain Bulk

Water Density

Repeat in calc

Water → gas
→ vibrating solid

Entropy from AIMD
↳ Vib. spectra
↳ we know gas phase (Sackur)
& substract

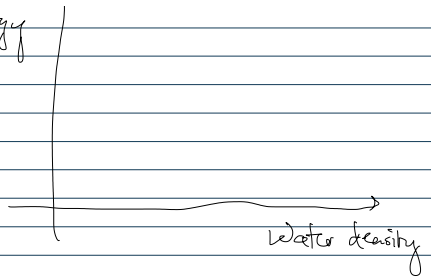
Not long enough to sample all cases (randomly new initialized structures)

↳ Decent values for faster time-scales.

Grand free energy (eV) ↑

↳ Decent values for faster timescales.

↳ Run 10 times as long or average the results over multiple runs.



TST & DFT for Kinetic Modelling

Lower frequency modes contribute the most to entropy ← But also are not accurate

→ Approximate partition function

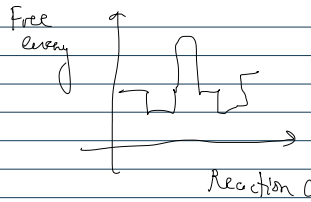
→ Frustrated rotational, translational

→ LA, LB zeolites (Butadiene synthesis)

Charlie Campbell

Campbell – Sellers Model for entropy (of weakly bonded adsorbates)

Microkinetic



(Each elementary step barrier is modelled) using DFT

Determine ROS → Find rate expression

Lecture 23

Thursday, 27 April 2023 10:30 AM

→ HW's: Can ask for solutions

→ Course eval → Friday deadline

$0.1\ eV \sim 10 k_B T_{mol}$

Zeolite continuation: DEE formation (Diethyle ether)

Microkinetic model → degree of rate control (sensitivity analysis)

(Using higher accuracy functional to get better elementary step endothermic value)

} K's changed a lot? → Maybe other steps were RDS / imp.? → Need to compare to the experimental values/data

MARI would depend on changing T, P & reactor conditions

Review of surface and gas phase chemical potential analysis

① (Ideal) gas treatments: divide them into translational, rotational, vib, electronic (DFT) parts

$$q_{vib} = \prod_i^{many} \frac{e^{-h\nu_i\beta/2}}{1-e^{-h\nu_i\beta}}$$ (per molecule) — includes ZPE

$$q_{rot} = \frac{I}{\sigma\beta\hbar^2}$$ (linear) per molecule

$$q_{trans} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V$$ (per molecule)

$$q_{el} = e^{-E_{DFT}/k_B T}$$

$$Q_{tot} = \frac{q_{trans}^N}{N!} q_{vib}^N q_{rot}^N e^{-E_{DFT}/k_B T}$$ (E_{DFT}: total energy)

$$F_{tot} = -k_B T \ln Q_{tot}, \quad \mu = \frac{\partial F_{tot}}{\partial N} = -k_B T \frac{\partial (\ln Q_{tot})}{\partial N}$$

$\ln N! \approx N \ln N - N$

$$\boxed{\mu = -k_B T \ln\left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}\left(\frac{V}{N}\right)\right] - k_B T \ln q_{vib} - k_B T \ln q_{rot} + \frac{\partial E_{DFT}}{\partial N}}$$

$\left(\frac{V}{N}\right) = P/k_B T$

Standard state: $\mu^\circ = \mu(P = P^\circ)$

$$\mu^\circ = -\frac{3}{2} k_B T \ln\left(\frac{2\pi m k_B T}{h^2}\right) - k_B T \ln\left(\frac{P^\circ}{k_B T}\right) - k_B T \ln q_{vib} - k_B T \ln q_{rot} + \frac{\partial E_{DFT}}{\partial N}$$ (includes ZPE) → Use to calculate K's in microkinetic model

$$\therefore \mu = \mu^\circ + k_B T \ln\left(\frac{P}{P^\circ}\right)$$

$$\mu^\circ = \varepsilon_{DFT} + ZPE - T\bar{S}, \quad \bar{S} = \frac{\partial S}{\partial N}$$

$$\frac{\partial E_{DFT}}{\partial N} = \frac{\partial}{\partial N}\left(N \varepsilon_{DFT}\right) = \varepsilon_{DFT}$$ (per molecule)

$$\frac{\partial E_{DFT}}{\partial N} = \frac{\partial}{\partial N}\left(N\,\varepsilon_{DFT}\right) = \varepsilon_{DFT} \qquad \text{(per molecule)}$$

(2) Liquid phase treatments – complicated

(3) Surface processes:

(a) First order treatment: calculate harmonic modes on surface ($3N_{atoms}$ modes) and use harmonic partition function:

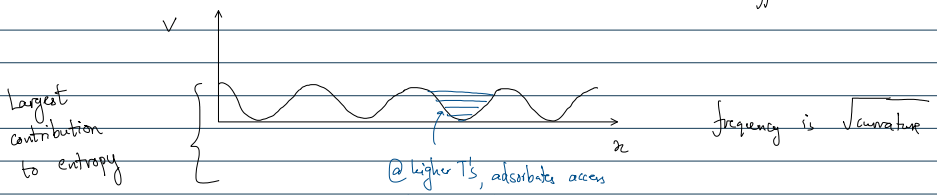
$$Q_{tot} = q_{vib}^{N}\, e^{-E_{DFT}/k_BT}\, \frac{N_s!}{N!\,(N_s-N)!}$$

$$\mu = \frac{\partial E_{tot}}{\partial N} = -k_BT \ln\left[\prod_{i}^{modes} \frac{e^{-h\nu_i\beta/2}}{1-e^{-h\nu_i\beta}}\right] + \frac{\partial E_{DFT}}{\partial N} + k_BT \ln\left(\frac{\theta}{\theta_*}\right); \quad \theta_* = 1-\theta_A-\theta_B-\theta_C$$

μ° (includes ZPE); activity

$$\therefore \boxed{\mu = \mu^\circ + k_BT \ln\left(\frac{\theta}{\theta_*}\right)}$$; For TS, do not include reaction coordinate

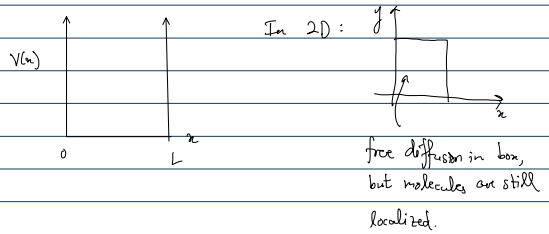
Caution: This approach fails for low frequency modes ($\nu_i \lesssim 150\,cm^{-1}$). } Low frequencies imply low diffusion barriers – anharmonic



Crude correction: take all modes w/ $\nu_i \lesssim 150\,cm^{-1}$, and assume entropies are all equal to a mode w/ $\nu = 150\,cm^{-1}$ (!?)

(b) Better treatments of low freq. modes

(i) Particle-in-a-box (for adsorbate)



In 2D, $E_{n,l} = \frac{\pi^2\hbar^2}{2m}\left[\frac{n^2}{L_x^2} + \frac{l^2}{n_y^2}\right]$ $n,l = 1,2,\ldots$

$$q_{trans}\ (\text{per molecule}) = \sum_{n=1}^{\infty}\sum_{l=1}^{\infty} \exp\left[\frac{-\pi^2\hbar^2}{2mk_BT}\left(\frac{n^2}{L_x^2}+\frac{l^2}{L_y^2}\right)\right]$$

$$\approx \int_0^{\infty}\int_0^{\infty} e^{\frac{-\pi^2\hbar^2 n^2}{2mk_BTL^2}} e^{\frac{-\pi^2\hbar^2 l^2}{2mk_BTL^2}} dn\,dl \quad ; \text{ if } L_x = L_y = L$$

$$= \frac{2\pi mk_BT}{h^2} A_{box} \qquad A_{box} = L^2 = \text{area of free diffusion for each adsorbate}$$

$$Q_{trans} = q_{trans,box}^{N} \quad ; \quad \mu_{trans} = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} A_{box}\right]$$

define this for the area that you care around the active site

$$\boxed{\mu^\circ = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} A_{box}\right]}$$

Replace low frequency trans modes w/ this
everything else is same

Useful for: zeolites, surfaces @ high coverages

(ii) Two-dimensional ideal gases (simple adsorption/desorption, weakly adsorbates)

2D ideal gases: $q_{trans} = A\frac{2\pi mk_BT}{h^2}$ (per molecule)

$$Q_{tot} = \frac{q_{trans}^{N}}{N!} q_{vib}^{N} q_{rot}^{N} e^{-E_{DFT}/k_BT}$$

↖ Identify for low freq. modes & high freq. modes

gives $\frac{A}{N} = \frac{a_s}{\theta}$; $\theta = \frac{N}{N_s}$

$$\mu = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} a_s\right] - k_BT \ln(q_{vib}\, q_{rot}) + \frac{\partial E_{DFT}}{\partial N}$$

↳ includes ZPE

$+k_BT \ln\theta$

$$\boxed{\mu = \mu^\circ + k_BT \ln\theta}$$

↖ No θ_a because i.g. have no volume (so no site blocking term)

→ Not correct for real molecules @ high coverages

→ still approx. true if $\boxed{\theta_a \ll 1}$; $\mu = \mu^\circ + k_BT \ln\left(\frac{\theta}{\theta_a}\right)$

Rough correction:

change $A \to A - N a_{ads}$

($N a_{ads}$: excluded from the area)

then $\frac{A}{N} \to \frac{A - N a_{ads}}{N} = \frac{N_s a_s - N a_{ads}}{N}$

$$= a_s \frac{\left(1 - \frac{a_{ads}}{a_s}\theta\right)}{\theta}$$

$\Rightarrow \mu^\circ$ is the same,

$$\mu = \mu^\circ + k_B T \ln \frac{\theta}{\theta_*} \quad ; \quad \theta_* = 1 - \frac{a_{ads}}{a_s}\theta$$

If the coverage is getting too high ($\theta_* < 1$), consider particle in a box (or diff. model))

Hws

Monday, 30 January 2023 8:13 PM

Question 1

(a) Particle in a box → ground state energy

From class, we know TISE solution for two particles in PIB is:

$$\phi_1 = \sqrt{\frac{2}{L}} \sin\left(\frac{n_1 \pi x_1}{L}\right)$$

$$\phi_2 = \sqrt{\frac{2}{L}} \sin\left(\frac{n_2 \pi x_2}{L}\right)$$

; Shown from separation of variables & applying BC's

where $\psi_{n_1,n_2} = \phi_1(x)\,\phi_2(x)$

Since we are dealing w/ fermions (w/ half-integer spins), we can use the slater determinant to obtain the new wave function:

$$\psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_1(x_1) & \phi_2(x_1) \\ \phi_1(x_2) & \phi_2(x_2) \end{vmatrix}$$

$$\psi(x_1,x_2) = \frac{1}{\sqrt{2}}\left[\phi_1(x_1)\,\phi_2(x_2) - \phi_1(x_2)\,\phi_2(x_1)\right]$$

$$\boxed{\psi(x_1,x_2) = \frac{1}{\sqrt{2}}\left[\sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_1}{L}\right)\cdot\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_2}{L}\right) - \sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_2}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_1}{L}\right)\right]}$$

(b) Expectation energy of the system: $E_{GS} = \dfrac{\iint \psi^* \hat{H} \psi \, dx_1\, dx_2}{\iint \psi^* \psi \, dx_1\, dx_2}$

Using the $\psi(x_1,x_2)$ expression from part (a), we can solve for E_{GS}:

$$\hat{H} = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x_1^2} \quad ; \text{ for 2 particles: } \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x_1^2} - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_2^2} = \hat{H}$$

First derivative: $\dfrac{\partial \psi}{\partial x_1} = \dfrac{1}{\sqrt{2}}\left[+\sqrt{\dfrac{2}{L}}\cos\left(\dfrac{n_1\pi x_1}{L}\right)\cdot\dfrac{n_1\pi}{L}\cdot\sqrt{\dfrac{2}{L}}\sin\left(\dfrac{n_2\pi x_2}{L}\right) - \dfrac{2}{L}\cos\left(\dfrac{n_2\pi x_1}{L}\right)\dfrac{n_2\pi}{L}\cdot\sin\left(\dfrac{n_1\pi x_2}{L}\right)\right]$

$$= \frac{\sqrt{2}}{L}\cdot\frac{n_1\pi}{L}\cos\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) - \frac{\sqrt{2}}{L}\frac{n_2\pi}{L}\cos\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

$$\frac{\partial^2\psi}{\partial x_1^2} = -\frac{\sqrt{2}}{L}\cdot\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) + \frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

Similarly

$$\frac{\partial^2\psi}{\partial x_2^2} = -\frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right) + \frac{\sqrt{2}}{L}\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)$$

Since ψ has no complex parts, $\psi^* = \psi$

$$\therefore \quad \psi^* = \frac{1}{\sqrt{2}}\left[\underbrace{\frac{2}{L}\cdot\sin\left(\frac{n_1\pi x_1}{L}\right)\cdot\sin\left(\frac{n_2\pi x_2}{L}\right)}_{A_1} - \underbrace{\frac{2}{L}\cdot\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)}_{A_2}\right]$$

$$\hat{H}\psi = \frac{-\hbar^2}{2m}\left[\frac{\partial^2\psi}{\partial x_1^2} + \frac{\partial^2\psi}{\partial x_2^2}\right] = \frac{\hbar^2}{2m}\left[\underbrace{-\frac{\sqrt{2}}{L}\cdot\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)}_{B_1} + \underbrace{\frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)}_{B_2} \underbrace{-\frac{\sqrt{2}}{L}\frac{n_2^2\pi^2}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)}_{B_3} + \underbrace{\frac{\sqrt{2}}{L}\frac{n_1^2\pi^2}{L^2}\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)}_{B_4}\right]$$

Normalization:

$$\int_0^L\int_0^L \psi^*\psi \, dx_1\, dx_2 = \int_0^L\int_0^L \frac{1}{2}\cdot\left[\frac{2}{L^2}\sin^2\left(\frac{n_1\pi x_1}{L}\right)\cdot\sin^2\left(\frac{n_2\pi x_1}{L}\right) + \frac{2}{L^2}\sin^2\left(\frac{n_1\pi x_2}{L}\right)\sin^2\left(\frac{n_2\pi x_2}{L}\right) - \frac{8}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)\right]\cdot dx_1\, dx_2$$

$$= \frac{1}{2}\left[\int_0^L \frac{4}{L^2}\sin^2\left(\frac{n_1\pi x_1}{L}\right)\cdot dx_1 \int_0^L \frac{1}{2}\cdot dx_2\right] + \frac{1}{2}\left[\int_0^L \frac{4}{L^2}\sin^2\frac{n_2\pi x_2}{L}\cdot dx_2 \int_0^L \sin^2\frac{n_1\pi x_1}{L}\cdot dx_1\right] - \frac{1}{2}\left[\int_0^L \frac{8}{L^2}\sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\cdot dx_1 \int_0^L \sin\left(\frac{n_2\pi x_2}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\cdot dx_2\right]$$

(a) (b)

$$= \frac{1}{2}\cdot\frac{\cancel{4}}{\cancel{L^2}}\cdot\frac{\cancel{L^2}}{\cancel{4}} + \frac{1}{2}\cdot\frac{\cancel{4}}{\cancel{L^2}}\cdot\frac{\cancel{L^2}}{\cancel{4}}$$

$$= 1 \ \checkmark$$

$$\int_0^L\int_0^L \psi^*\hat{H}\psi \cdot dx_1\, dx_2 = \int_0^L\int_0^L \frac{\sqrt{2}}{\sqrt{2}}\cdot\frac{\hbar^2}{2m}\left[-A_1\cdot B_1 + A_1\cdot B_2 - A_1\cdot B_3 + A_1\cdot B_4 + A_2\cdot B_1 - A_2\cdot B_2 + A_2\cdot B_3 - A_2\cdot B_4\right]\cdot dx_1\, dx_2$$

Solving for individual terms:

$$\underbrace{\left(\frac{n_1^2\pi^2}{L^2}\right)}_{C_1}\iint -A_1\cdot B_1 = C_1\int_0^L\int_0^L \sin^2\left(\frac{n_1\pi x_1}{L}\right)\sin^2\left(\frac{n_2\pi x_2}{L}\right)\cdot dx_1\, dx_2 \quad ;$$

$$= C_1\int_0^L \sin^2\left(\frac{n_1\pi x_1}{L}\right)\cdot dx_1 \int_0^L \sin^2\left(\frac{n_2\pi x_2}{L}\right)\cdot dx_2$$

$$= C_1\left(\frac{L}{2}\cdot\frac{L}{2}\right)$$

$$= \frac{n_1^2\pi^2}{L^2}\cdot\frac{L^2}{4} = -\frac{n_1^2\pi^2}{4}$$

$\because \sin(n\pi) = 0$ for $n = 0, \pm1, \pm2$

$$\left(\frac{n_1^2\pi^2}{L^2}\right)\iint A_1\cdot B_2 = \int_0^L \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_1\pi x_2}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)\cdot dx_1\,dx_2 \Rightarrow \int_0^L \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_1}{L}\right)\cdot dx_1$$

→ Since sin is an even function → This integral evaluated at 0 to L for integer n's will evaluate to 0

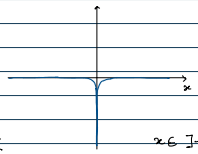
Similarly solving for individual terms we only get $-A_1B_1$, $-A_1B_2$, $-A_2B_2$ & $-A_2B_1$ to survive. → each corresponding to an n_1n_2 combination

That yields us w/ the desired answer:

$$E_{CIS} = \frac{-\hbar^2}{2mL^3}\left[\frac{-2(n_1\pi)^2}{4} - \frac{2(n_2\pi)^2}{4}\right] = \varepsilon_1 + \varepsilon_2 \quad ; \quad \varepsilon_i = \frac{n_i^2\pi^2\hbar^2}{2mL^2}$$

Question 2

δ function @ $V_{ext}(x) = -\alpha\delta(x)$



BC's: $\psi \to 0$ @ $x \to \pm\infty$

$$\psi(x) = \frac{\sqrt{m\alpha}}{\hbar}\exp\left(\frac{-m\alpha}{\hbar^2}|x|\right) \quad ; \quad E = \frac{-m\alpha^2}{2\hbar^2} \qquad x \in]-\infty,\infty$$

↳ These are given solutions to the full TISE

Using a DFT approach:

$$E[\rho] = \int_{-\infty}^{\infty}\frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx - \alpha\int_{-\infty}^{\infty}\delta(x)\rho\cdot dx \rightarrow \text{Our energy functional}$$

Given constraint: $\int_{-\infty}^{\infty}\rho(x)\cdot dx = 1$ (Assuming dealing w/ N particles)

↳ Constructing the lagrangian functional w/ the given constraint:

$$F[\rho] = E[\rho] - \mu\left(\int_{-\infty}^{\infty}\rho\cdot dx - N\right) = \int f(\rho(x))\cdot dx - \alpha\rho(0) + \mu N$$

We know for functionals:

$$\frac{\delta F[\rho(x)]}{\delta\rho(x)} = \frac{\partial f}{\partial\rho} - \nabla\frac{\partial f}{\partial\nabla_\rho} \quad ; \text{ where } f \text{ is the integrand as shown above}$$

Since f does not have any derivatives of ρ: $\frac{\delta F}{\delta\rho} = \frac{\partial f}{\partial\rho}$

$$\frac{\delta F}{\delta\rho} = 0 = \frac{\partial}{\partial\rho}\left(\frac{\pi^2\hbar^2}{6m}\rho^3 - \mu\rho\right) - \frac{\partial}{\partial\rho}(\alpha\rho(x)) + \frac{\partial}{\partial\rho}(\mu N)$$

$$0 = \frac{\pi^2\hbar^2}{2m}\rho^2 - \mu$$

$$\rho_{min} = \sqrt{\frac{2m\mu}{\pi^2\hbar^2}}$$

Plugging into constraint:

$$\int_{-\infty}^{\infty}\sqrt{\frac{2m\mu}{\pi^2\hbar^2}}\cdot dx = N$$

$$\frac{2m\mu}{\pi^2\hbar^2}\int_{-\infty}^{\infty}\cdot dx = N$$

This integral diverges and cannot be evaluated

This integral does not converge, therefore cannot find an expression for ρ_{min} via DFT approach.

This issue could be due to the unbounded box sides which means that the e^- can be spread across the infinite distance, which is an unphysical system.

Question 3

$V_{ext} = \frac{1}{2}kx^2 \quad ; \quad x \in]-\infty,\infty[$

(a) Energy functional:

$$E[\rho] = \int_{-\infty}^{\infty}\frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \int_{-\infty}^{\infty}\frac{1}{2}kx^2\rho(x)\cdot dx$$

(b) Constraint: $\int_{-\infty}^{\infty}\rho(x)\cdot dx = N$ (Normalization)

$$\therefore F[\rho] = E[\rho] - \mu\left(\int_{-\infty}^{\infty}\rho(x)\cdot dx - N\right) = \int\overbrace{\left(\frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx + \frac{1}{2}kx^2\rho - \mu\rho\right)}^{f}\cdot dx + \mu N$$

Minimizing functional $\frac{\delta F}{\delta\rho} = 0$;

$$\frac{\delta F}{\delta\rho} = \frac{\partial f}{\partial\rho} - \nabla\frac{\partial f}{\partial\nabla_\rho} = \frac{\partial}{\partial\rho}\left(\frac{\pi^2\hbar^2}{6m}\rho^3 + \frac{1}{2}kx^2\rho - \mu\rho\right) + \frac{\partial}{\partial\rho}(\mu N)$$

$$0 = \frac{\pi^2\hbar^2}{2m}\rho^2 + \frac{1}{2}kx^2 - \mu$$

$$\frac{\delta F}{\delta p} = \frac{\partial f}{\partial p} - \nabla \cdot \frac{\partial f}{\partial \nabla p} = \frac{\partial}{\partial p}\left(\frac{\pi^2\hbar^2}{6m}p^3 + \frac{1}{2}kx^2p - \mu p\right) + \frac{\partial}{\partial p}(\mu N)$$

$$0 = \frac{\pi^2\hbar^2}{2m}p^2 + \frac{1}{2}kx^2 - \mu$$

$$p_{min} = \left[\left(\mu - \frac{1}{2}kx^2\right)\frac{2m}{\pi^2\hbar^2}\right]^{1/2}$$

$$\boxed{p_{min} = \frac{\sqrt{2m}}{\pi\hbar}\left(\mu - \frac{1}{2}kx^2\right)^{1/2}} \rightarrow \text{For physical solutions } \mu - \frac{1}{2}kx^2 > 0 \Rightarrow \mu > \frac{1}{2}kx^2$$

Solving for μ:

$$\int_{-\infty}^{\infty} p_{min}(x)\cdot dx = N$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sqrt{2m}}{\pi\hbar}\left(\mu - \frac{1}{2}kx^2\right)^{1/2}\cdot dx = N$$

$$\Rightarrow \int_{-\infty}^{\infty}\left(\mu - \frac{1}{2}kx^2\right)^{1/2}dx = \frac{N\pi\hbar}{\sqrt{2m}} \quad ; \quad \text{Since } \mu > \frac{1}{2}kx^2 \Rightarrow x^2 > \frac{2\mu}{k} \Rightarrow -\sqrt{\frac{2\mu}{k}} < x < \sqrt{\frac{2\mu}{k}}$$

The integration limits change to:

$$\int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}}\left(\mu - \frac{1}{2}kx^2\right)^{1/2} = \frac{N\pi\hbar}{\sqrt{2m}}$$

Using Wolfram Alpha:

$$\left[\frac{\mu}{2\sqrt{\frac{1}{2}k}}\cdot\arcsin\left(\frac{\sqrt{\frac{1}{2}k}\,x}{\sqrt{\mu}}\right) + \frac{x}{2}\sqrt{\mu - \frac{1}{2}kx^2}\right]_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} = \frac{N\pi\hbar}{\sqrt{2m}}$$

$$\Rightarrow \frac{\pi\mu}{2\sqrt{\frac{1}{2}k}} = \frac{N\pi\hbar}{\sqrt{2m}}$$

$$\therefore \boxed{\mu = \frac{N\hbar\sqrt{k}}{\sqrt{m}}}$$

Plugging back into p_{min}: $p_{min} = \left[\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)\frac{2m}{\pi^2\hbar^2}\right]^{1/2}$

(c) $E[p_{min}] = \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \frac{\pi^2\hbar^2}{6m}p_{min}^3\,dx + \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}} \frac{1}{2}kx^2 p_{min}(x)\cdot dx$; The bounds are changed to exclude unphysical regions; $\mu = \frac{N\hbar\sqrt{k}}{\sqrt{m}}$

$$= \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}}\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}\cdot\frac{\pi^3\hbar^3}{6m}\cdot\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)^{3/2} + \int_{-\sqrt{\frac{2\mu}{k}}}^{\sqrt{\frac{2\mu}{k}}}\frac{1}{2}kx^2\cdot\left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\cdot\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}} - \frac{1}{2}kx^2\right)^{1/2}\cdot dx$$

Using Wolfram Alpha

$$= \frac{\sqrt{2}}{8}\frac{1}{\pi\hbar}\sqrt{m}\cdot\frac{8\pi}{8\sqrt{\frac{1}{2}k}}\cdot\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}}\right)^2 + \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\cdot\frac{\pi\left(\frac{N\hbar\sqrt{k}}{\sqrt{m}}\right)^2}{8\sqrt{\frac{1}{2}k}}$$

$$= \frac{1}{4}\sqrt{\frac{k}{m}}\cdot\frac{N^2\hbar k}{m} + \frac{\pi}{4\pi\sqrt{k}}\cdot\frac{N^2\hbar^2 k}{m}$$

$$\boxed{E[p_{min}] = \frac{N^2\hbar\sqrt{k}}{2\sqrt{m}}}$$

$$\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}\cdot\frac{\pi^3\hbar^3}{6m}\cdot\frac{8\pi a^2}{8\sqrt{b}} + \frac{\pi a^2}{8\sqrt{b}}\sqrt{\frac{2m}{\pi^2\hbar^2}}$$

$$\frac{\pi a^2}{8\sqrt{b}}\left(\frac{\sqrt{2m}}{\hbar\pi} + \frac{\sqrt{2m}}{\pi\hbar}\right)$$

$$= \frac{\pi a^2}{4\sqrt{8b}}\cdot\frac{\sqrt{2m}}{\pi\hbar}$$

$$= N^2\hbar^2\frac{k}{m}\cdot\frac{1}{4}\cdot\frac{\sqrt{2}}{\sqrt{k}}\cdot\frac{\sqrt{2m}}{\hbar}$$

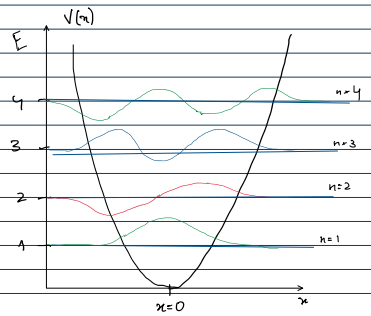
$$= \frac{1}{2}N^2\hbar\frac{\sqrt{k}}{\sqrt{m}}$$

(d) Harmonic Oscillator : $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$; $n = 0, 1, 2, 3\ldots$; $\omega = \sqrt{\frac{k}{m}}$

$$\psi_n(x) = N_n e^{-\beta^2x^2/2}H_n(\beta x) \quad ; \quad n = 0,1,2,3$$

where $\beta = \sqrt{\frac{m\omega}{\hbar}}$

$H_n(x) \rightarrow$ hermite polynomial



N	DFT	Harmonic
1	$\frac{1}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{3}{2}\hbar\sqrt{\frac{k}{m}}$
2	$2\hbar\sqrt{\frac{k}{m}}$	$\frac{5}{2}\hbar\sqrt{\frac{k}{m}}$
3	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{7}{2}\hbar\sqrt{\frac{k}{m}}$
4	$8\hbar\sqrt{\frac{k}{m}}$	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$

DFT overapproximates the energy

		Energy
3	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$	$\frac{7}{2}\hbar\sqrt{\frac{k}{m}}$
4	$8\hbar\sqrt{\frac{k}{m}}$	$\frac{9}{2}\hbar\sqrt{\frac{k}{m}}$
5	$\frac{25\hbar}{2}\sqrt{\frac{k}{m}}$	$\frac{11}{2}\hbar\sqrt{\frac{k}{m}}$

Question 4

Non-interacting & spherical symmetry

(a) Doing similar analysis:

$$F[\rho] = E[\rho] - \mu\left(\int \rho\cdot d^3r - N\right) = \int\left(\rho^{5/3}\cdot k - \gamma\cdot\frac{\rho}{r} - \mu\rho\right)4\pi r^2\cdot dr + \mu N$$

$$\frac{\delta F}{\delta\rho} = 0 = \frac{\partial}{\partial\rho}\left(\left(\rho^{5/3}\cdot k - \frac{\gamma\rho}{r} - \mu\rho\right)4\pi r^2\right) + \cancel{\frac{\partial}{\partial\rho}(\mu N)}^{\,0}$$

$$0 = 4\pi\left(k\frac{5}{3}\rho^{2/3}r^2 - \gamma r - \mu r^2\right)$$

$$\frac{5}{3}kr^2\cdot\rho^{2/3} = \gamma r + \mu r^2$$

$$\rho^{2/3} = \frac{3}{5}\left(\frac{\gamma}{r}+\mu\right)\frac{1}{k}$$

$$\rho_{min} = \left(\frac{3}{5k}\left(\frac{\gamma}{r}+\mu\right)\right)^{3/2}$$

Finding μ: $\int_0^{\gamma/|\mu|}\rho_{min}\cdot d^3r = N$; For physical solutions, $\left(\mu+\frac{\gamma}{r}\right)>0$

$$\frac{\gamma}{r} > -\mu \quad\rightarrow\quad r > -\frac{\gamma}{\mu}$$

Since r cannot be negative and γ has to also be positive, $\mu<0$; $\mu=-|\mu|$ s.t. $r<\frac{\gamma}{|\mu|}$

$$\int_0^{\gamma/|\mu|}\left(\frac{3}{5k}\right)^{3/2}\cdot\left(\frac{\gamma}{r}-|\mu|\right)^{3/2}\cdot 4\pi r^2\cdot dr = N$$

$$\int_0^{\gamma/|\mu|}\left(\frac{\gamma}{r}-|\mu|\right)^{3/2}r^2\cdot dr = \frac{N\cdot\left(\frac{5k}{3}\right)^{3/2}}{4\pi}$$

Using wolfram Alpha:

$$\frac{\pi\gamma^3}{4\gamma^6(|\mu|)^{3/2}} = \frac{N}{4\pi}\left(\frac{5k}{3}\right)^{3/2}$$

$$(|\mu|)^{2/3} = \frac{1}{4}\frac{\pi^2\gamma^3}{N}\left(\frac{3}{5k}\right)^{3/2}$$

$$|\mu| = \left(\frac{1}{4}\frac{\pi^2}{N}\right)^{3/3}\cdot\gamma^3\cdot\frac{3}{5k}$$

$$|\mu| = \frac{\gamma^3}{5k}\left(\frac{\pi^2}{4N}\right)^{2/3}$$

Since $\mu=-|\mu|$;

$$\boxed{\mu = -\frac{\gamma^2 3}{5k}\left(\frac{\pi^2}{4N}\right)^{2/3}}$$

(b) $E[\rho_{min}] = ?$

Plugging in μ in ρ_{min} expression:

$$\rho_{min} = \left[\frac{3}{5k}\frac{\gamma}{r} - \frac{3}{5k}|\mu|\right]^{3/2}$$

$$\rho_{min} = \left(\frac{3}{5k}\left(\frac{\gamma}{r} - \frac{\gamma^2 3}{5k}\left(\frac{\pi^2}{4N}\right)^{2/3}\right)\right)^{3/2}$$

$$\rho_{min} = \left[\frac{3\gamma}{5k\cdot r} - \frac{9\gamma^2}{25k^2}\left(\frac{\pi^2}{4N}\right)^{2/3}\right]^{3/2}$$

$$E[\rho_{min}] = k\int_0^{\gamma/|\mu|}\left(\frac{3}{5k}\frac{\gamma}{r} - \frac{3}{5k}|\mu|\right)^{5/2}4\pi r^2\cdot dr - \gamma\int_0^{\gamma/|\mu|}\left(\frac{3}{5k}\frac{\gamma}{r} - \frac{3}{5k}|\mu|\right)^{3/2}\frac{1}{r}\cdot 4\pi r^2\cdot dr$$

$$= \left(\frac{3}{5k}\right)^{5/2}k\cdot 4\pi\int_0^{\gamma/|\mu|}\left(\frac{\gamma}{r}-|\mu|\right)^{5/2}r^2\cdot dr - \gamma\left(\frac{3}{5k}\right)^{3/2}4\pi\int_0^{\gamma/|\mu|}\left(\frac{\gamma}{r}-|\mu|\right)^{3/2}\cdot r\cdot dr$$

$$= \left(\frac{3}{5k}\right)^{5/2}\cdot k\cdot 4\pi\cdot\frac{5\pi\gamma^3}{16\sqrt{|\mu|}} - 4\pi\gamma\left(\frac{3}{5k}\right)^{3/2}\cdot\frac{3\pi\gamma^2}{8\sqrt{|\mu|}}$$

$$= \frac{4\pi\gamma^3}{\sqrt{|\mu|}}\left(\frac{3}{5k}\right)^{3/2}\left(\frac{3}{5k}\cdot k\cdot\frac{5\pi}{16} - \frac{3\pi}{8}\right)$$

$$\left(\frac{5\kappa}{16\sqrt{|\mu|}}\right) \cdots \left(\frac{5\kappa}{8\sqrt{|\mu|}}\right)$$

$$= \frac{4\pi\delta^3}{\sqrt{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\left(\frac{3}{8\kappa}\cdot\frac{k\cdot 8\pi}{16} - \frac{3\pi}{8}\right)$$

$$= \frac{4\pi\delta^3}{\sqrt{|\mu|}}\left(\frac{3}{5\kappa}\right)^{3/2}\left(-\frac{2\pi}{k_2}\right) \qquad \frac{r^2}{3s} = \frac{2^{-3}}{2\times 2} = -\frac{7}{6}$$

$$E[\rho_{min}] = -\frac{\pi^2\delta^3}{2}\left(\frac{3}{5\kappa}\right)^{3/2}\cdot\left(\frac{5\kappa}{3\delta^2}\right)^{1/2}\left[\frac{\pi^2}{4N}\right]^{1/3} \qquad \frac{(5\kappa)^{1/2}}{(5\kappa)^{3/2}} = 5\kappa^{-2/2} \qquad \frac{3}{1}-\frac{2}{3} = \frac{7}{3}$$

$$2+\frac{2}{3}$$

$$\boxed{E[\rho_{min}] = -\frac{3}{10\kappa}\delta^2\pi^{2/3}\cdot\left(\frac{\pi^2}{4N}\right)^{1/3}}$$

$$E[\rho_{min}] \propto \left(\frac{1}{N}\right)^{1/3}$$

(c) 2e⁻ per orbit

Solution to TISE for H_2 atom: ; $n \geq l \geq |m|$

$$\psi_{nlm} = R_{nl}(r)\,Y_l^m(\theta,\phi)$$

$$E_n = -\frac{\hbar^2}{2m_e a_0^2}\cdot\frac{1}{n^2} \; ; \; a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} \; ; \; E_1 = -13.61\ eV$$

ψ_{100} → corresponds to ground state.

$E_n \propto \frac{1}{n^2}$ while our prediction has $E_n \propto \frac{1}{N^3}$

ψ_{210} → n=2 w/ azimuthal no. or p-state

e⁻ will fill up n, then l & the m

HW 2

Wednesday, 15 February 2023 3:16 PM

Question 1

As shown in class, for a periodic dirac-comb w/ periodicity of $n = 0, \pm 1, \pm 2 \ldots$

$0 < x < a$, $V(x) = 0$: TISE : $\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$

$$\psi(x) = A\sin(kx) + B\cos(kx) \quad ; \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Applying Bloch's Theorem : $\psi(x+a) = e^{ika}\left[A\sin(k(x+a)) + B\cos(k(x+a))\right]$; $K = \frac{2\pi n}{P \cdot a}$

for $-a < x < 0$

Plugging in initial conditions of : @ $x = 0$, $\psi_{0^-} = \psi_{0^+}$ (continuity)

& $\left.\frac{d\psi}{dx}\right|_{0^+} - \left.\frac{d\psi}{dx}\right|_{0^-} = \frac{2m\alpha}{\hbar^2}\psi(0)$ ["cusp"]

and eliminating for A and B:

$$\cos(Ka) = \cos(ka) + \frac{m\alpha a}{\hbar^2}\frac{\sin(ka)}{ka} \quad \ldots \text{ Eq 1}$$

Equation 1 can be solved numerically : $m = \alpha = a = \hbar = 1$

$$\therefore \cos(K) = \cos(k) + \frac{\sin(k)}{k} \quad ; \quad K = \frac{2\pi n}{P} \quad ; \quad k = \sqrt{2E}$$

(a) → Solving for the specific k values given K values for the first band ($n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$) P+1 non-interacting fermions we obtain the following table:

For $P = 10$; $n = P + 1 = 11$ particles

$P = 20$; $n = P + 1 = 21$ particles

; A band is defined when we fill all discrete energy levels between $1 < \cos(K) < -1$

	P =	10			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	**33.4**
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	

	P =	20			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	**0.854**	
1	0.314	0.951	1.343	**0.902**	
-1	-0.314	0.951	1.343	**0.902**	
2	0.628	0.809	1.447	**1.047**	
-2	-0.628	0.809	1.447	**1.047**	
3	0.942	0.588	1.606	**1.290**	
-3	-0.942	0.588	1.606	**1.290**	
4	1.257	0.309	1.803	**1.625**	
-4	-1.257	0.309	1.803	**1.625**	
5	1.571	0.000	2.092	**2.188**	
-5	-1.571	0.000	2.092	**2.188**	**53.2**
6	1.885	-0.309	2.272	**2.581**	
-6	-1.885	-0.309	2.272	**2.581**	
7	2.199	-0.588	2.526	**3.190**	
-7	-2.199	-0.588	2.526	**3.190**	
8	2.513	-0.809	2.781	**3.867**	
-8	-2.513	-0.809	2.781	**3.867**	
9	2.827	-0.951	3.017	**4.551**	
-9	-2.827	-0.951	3.017	**4.551**	
10	3.142	-1.000	3.142	**4.935**	
-10	-3.142	-1.000	3.142	**4.935**	

(b) P=10, but first two bands are occupied

n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	33.4
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	
6	3.770	-0.809	4.051	8.205	
-6	-3.770	-0.809	4.051	8.205	
7	4.398	-0.309	4.619	10.668	
-7	-4.398	-0.309	4.619	10.668	
8	5.027	0.309	5.211	13.577	171.5
-8	-5.027	0.309	5.211	13.577	
9	5.655	0.809	5.806	16.855	
-9	-5.655	0.809	5.806	16.855	
10	6.2832	1.000	6.283	19.739	
-10	-6.283	1.000	6.283	19.739	

Total N = 21

(c) P=10, but first three bands are occupied.

	P =	10			
n	K	cos(K)	k	Energy	Discrete Band Energy
0	0.000	1.000	1.307	0.854	
1	0.628	0.809	1.447	1.047	
-1	-0.628	0.809	1.447	1.047	
2	1.257	0.309	1.803	2.581	
-2	-1.257	0.309	1.803	2.581	
3	1.885	-0.309	2.272	3.867	33.4
-3	-1.885	-0.309	2.272	3.867	
4	2.513	-0.809	2.781	3.867	
-4	-2.513	-0.809	2.781	3.867	
5	3.142	-1.000	3.142	4.935	
-5	-3.142	-1.000	3.142	4.935	
6	3.770	-0.809	4.051	8.205	
-6	-3.770	-0.809	4.051	8.205	
7	4.398	-0.309	4.619	10.668	
-7	-4.398	-0.309	4.619	10.668	
8	5.027	0.309	5.211	13.577	171.5
-8	-5.027	0.309	5.211	13.577	
9	5.655	0.809	5.806	16.855	
-9	-5.655	0.809	5.806	16.855	
10	6.2832	1.000	6.283	19.739	
-10	-6.283	1.000	6.283	19.739	
11	6.912	0.809	7.066	24.964	
-11	-6.912	0.809	7.066	24.964	
12	7.540	0.309	7.672	29.430	
-12	-7.540	0.309	7.672	29.430	
13	8.168	-0.309	8.286	34.329	
-13	-8.168	-0.309	8.286	34.329	517.0
14	8.796	-0.809	8.900	39.605	
-14	-8.796	-0.809	8.900	39.605	
15	9.425	-1.000	9.425	44.413	
-15	-9.425	-1.000	9.425	44.413	

Total N = 31

(d) As seen from the table below, the integral method (which was a trapezoid numerical integration)

(d) As seen from the table below, the integral method (which was a trapezoid numerical integration) are larger than energies calculated w/ discrete energy summation

P = 10

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy
0	0.000	1.000	1.307	0.854		
1	0.628	0.809	1.447	1.047		
-1	-0.628	0.809	1.447	1.047		
2	1.257	0.309	1.803	2.581		
-2	-1.257	0.309	1.803	2.581		
3	1.885	-0.309	2.272	3.867	33.4	36.8
-3	-1.885	-0.309	2.272	3.867		
4	2.513	-0.809	2.781	3.867		
-4	-2.513	-0.809	2.781	3.867		
5	3.142	-1.000	3.142	4.935		
-5	-3.142	-1.000	3.142	4.935		
6	3.770	-0.809	4.051	8.205		
-6	-3.770	-0.809	4.051	8.205		
7	4.398	-0.309	4.619	10.668		
-7	-4.398	-0.309	4.619	10.668		
8	5.027	0.309	5.211	13.577	171.5	174.9
-8	-5.027	0.309	5.211	13.577		
9	5.655	0.809	5.806	16.855		
-9	-5.655	0.809	5.806	16.855		
10	6.2832	1.000	6.283	19.739		
-10	-6.283	1.000	6.283	19.739		
11	6.912	0.809	7.066	24.964		
-11	-6.912	0.809	7.066	24.964		
12	7.540	0.309	7.672	29.430		
-12	-7.540	0.309	7.672	29.430		
13	8.168	-0.309	8.286	34.329	517.0	520.3
-13	-8.168	-0.309	8.286	34.329		
14	8.796	-0.809	8.900	39.605		
-14	-8.796	-0.809	8.900	39.605		
15	9.425	-1.000	9.425	44.413		
-15	-9.425	-1.000	9.425	44.413		

P = 20

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy	DFT Ground State Energy
0	0.000	1.000	1.307	**0.854**			
1	0.314	0.951	1.343	**0.902**			
-1	-0.314	0.951	1.343	**0.902**			
2	0.628	0.809	1.447	**1.047**			
-2	-0.628	0.809	1.447	**1.047**			
3	0.942	0.588	1.606	**1.290**			
-3	-0.942	0.588	1.606	**1.290**			
4	1.257	0.309	1.803	**1.625**			
-4	-1.257	0.309	1.803	**1.625**			
5	1.571	0.000	2.092	**2.188**			
-5	-1.571	0.000	2.092	**2.188**	53.2	55.7	59.1
6	1.885	-0.309	2.272	**2.581**			
-6	-1.885	-0.309	2.272	**2.581**			
7	2.199	-0.588	2.526	**3.190**			
-7	-2.199	-0.588	2.526	**3.190**			
8	2.513	-0.809	2.781	**3.867**			
-8	-2.513	-0.809	2.781	**3.867**			
9	2.827	-0.951	3.017	**4.551**			
-9	-2.827	-0.951	3.017	**4.551**			
10	3.142	-1.000	3.142	**4.935**			
-10	-3.142	-1.000	3.142	**4.935**			
11	3.456	-0.951	3.802	**7.228**			
-11	-3.456	-0.951	3.802	**7.228**			
12	3.770	-0.809	4.051	**8.205**			
-12	-3.770	-0.809	4.051	**8.205**			
13	4.084	-0.588	4.330	**9.374**			
-13	-4.084	-0.588	4.330	**9.374**			
14	4.398	-0.309	4.619	**10.668**			
-14	-4.398	-0.309	4.619	**10.668**			
15	4.712	0.000	4.913	**12.069**			
-15	-4.712	0.000	4.913	**12.069**	316.1	318.5	153.5
16	5.027	0.309	5.210	**13.572**			
-16	-5.027	0.309	5.210	**13.572**			
17	5.341	0.588	5.509	**15.175**			
-17	-5.341	0.588	5.509	**15.175**			
18	5.655	0.809	5.806	**16.855**			
-18	-5.655	0.809	5.806	**16.855**			
19	5.969	0.951	6.093	**18.562**			
-19	-5.969	0.951	6.093	**18.562**			
20	6.283	1.000	6.283185	**19.739**			
-20	-6.283	1.000	6.283185	**19.739**			

(e) DFT energy per unit cell for all cases:

For dirac comb, the DFT energy functional is given by: $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{na}^{na} \rho^3\,dx + \int_{-na}^{na} V_{ext}(x)\rho\,dx$; n = some integer

$$\frac{\delta F}{\delta \rho} = 0 \Rightarrow \rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}(\mu - V_{ext}(x))\right)^{1/2}$$

∴ Solving for ground state energy: $\frac{E_{GS}}{P} = \frac{\pi^2\hbar^2}{6ma^2} N_{cell}^3 + \frac{\alpha}{a} N_{cell}$; $m = a = \hbar = \alpha = 1$; $N_{cell} = \frac{N}{P}$

$$\frac{E_{GS}}{P} = \frac{\pi^2}{6} N_{cell}^3 + N_{cell}$$

The table below shows the calculated values for DFT compared to those calculated via the integral method

P = 10

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy	DFT Ground State Energy	% Difference
0	0.000	1.000	1.307	0.854				
1	0.628	0.809	1.447	1.047				
-1	-0.628	0.809	1.447	1.047				
2	1.257	0.309	1.803	2.581				
-2	-1.257	0.309	1.803	2.581				
3	1.885	-0.309	2.272	3.867	33.4	36.8	32.9	9.0%
-3	-1.885	-0.309	2.272	3.867				
4	2.513	-0.809	2.781	3.867				
-4	-2.513	-0.809	2.781	3.867				
5	3.142	-1.000	3.142	4.935				
-5	-3.142	-1.000	3.142	4.935				
6	3.770	-0.809	4.051	8.205				
-6	-3.770	-0.809	4.051	8.205				
7	4.398	-0.309	4.619	10.668				
-7	-4.398	-0.309	4.619	10.668				
8	5.027	0.309	5.211	13.577	171.5	174.9	173.3	1.0%
-8	-5.027	0.309	5.211	13.577				
9	5.655	0.809	5.806	16.855				
-9	-5.655	0.809	5.806	16.855				
10	6.2832	1.000	6.283	19.739				
-10	-6.283	1.000	6.283	19.739				
11	6.912	0.809	7.066	24.964				
-11	-6.912	0.809	7.066	24.964				
12	7.540	0.309	7.672	29.430				
-12	-7.540	0.309	7.672	29.430				
13	8.168	-0.309	8.286	34.329	517.0	520.3	521.0	0.1%
-13	-8.168	-0.309	8.286	34.329				
14	8.796	-0.809	8.900	39.605				
-14	-8.796	-0.809	8.900	39.605				
15	9.425	-1.000	9.425	44.413				
-15	-9.425	-1.000	9.425	44.413				

P = 20

n	K	cos(K)	k	Energy	Discrete Band Energy	Integral Band Energy	DFT Ground State Energy	% Difference
0	0.000	1.000	1.307	**0.854**				
1	0.314	0.951	1.343	**0.902**				
-1	-0.314	0.951	1.343	**0.902**				
2	0.628	0.809	1.447	**1.047**				
-2	-0.628	0.809	1.447	**1.047**				
3	0.942	0.588	1.606	**1.290**				
-3	-0.942	0.588	1.606	**1.290**				
4	1.257	0.309	1.803	**1.625**				
-4	-1.257	0.309	1.803	**1.625**				
5	1.571	0.000	2.092	**2.188**				
-5	-1.571	0.000	2.092	**2.188**	53.2	55.7	59.1	4.5%
6	1.885	-0.309	2.272	**2.581**				
-6	-1.885	-0.309	2.272	**2.581**				
7	2.199	-0.588	2.526	**3.190**				
-7	-2.199	-0.588	2.526	**3.190**				
8	2.513	-0.809	2.781	**3.867**				
-8	-2.513	-0.809	2.781	**3.867**				
9	2.827	-0.951	3.017	**4.551**				
-9	-2.827	-0.951	3.017	**4.551**				
10	3.142	-1.000	3.142	**4.935**				
-10	-3.142	-1.000	3.142	**4.935**				
11	3.456	-0.951	3.802	**7.228**				
-11	-3.456	-0.951	3.802	**7.228**				
12	3.770	-0.809	4.051	**8.205**				
-12	-3.770	-0.809	4.051	**8.205**				
13	4.084	-0.588	4.330	**9.374**				
-13	-4.084	-0.588	4.330	**9.374**				
14	4.398	-0.309	4.619	**10.668**				
-14	-4.398	-0.309	4.619	**10.668**				
15	4.712	0.000	4.913	**12.069**				
-15	-4.712	0.000	4.913	**12.069**	316.1	318.5	153.5	0.7%
16	5.027	0.309	5.210	**13.572**				
-16	-5.027	0.309	5.210	**13.572**				
17	5.341	0.588	5.509	**15.175**				
-17	-5.341	0.588	5.509	**15.175**				
18	5.655	0.809	5.806	**16.855**				
-18	-5.655	0.809	5.806	**16.855**				
19	5.969	0.951	6.093	**18.562**				
-19	-5.969	0.951	6.093	**18.562**				
20	6.283	1.000	6.283185	**19.739**				
-20	-6.283	1.000	6.283185	**19.739**				

The DFT values fall in between the discrete and and integral calculations and the difference b/w the integral band values decrease from P=10 to P=20. (i.e. DFT becomes more accurate)

Question 2

$V(x) = \alpha \sin\left(\frac{2\pi x}{a}\right)$ ← sin-wave potential

We know, $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{-na}^{na} \rho^3\,dx + \int_{-na}^{na} V(x)\,\rho\,dx$;

Lagrangian, $F[\rho] = E[\rho] - \mu\left(\int_{-na}^{na} \rho\,dx - N\right)$

For $\alpha = 5$,

$V(x) = 5\sin\left(\frac{2\pi x}{L}\right)$ → period of 1

$$F[\rho] = \frac{\pi^2}{6}\int_0^1 \rho^3\,dx + \int_0^1 5\sin(2\pi x)\,\rho\,dx - \mu\left(\int_0^1 \rho\,dx - N\right)$$

Solving this,

$$\rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\left(\mu - 5\sin\frac{2\pi x}{1}\right)\right)^{1/2}$$

$$\int_0^1 \rho_{min}(x)\,dx = N \quad \rightarrow \quad \text{Constraint}$$

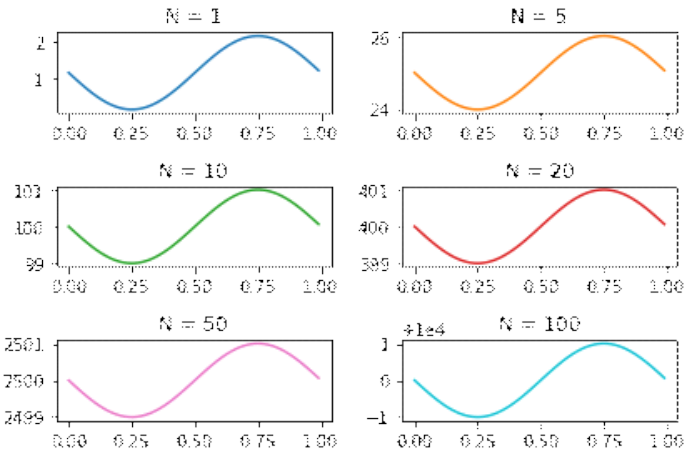
$$\therefore \int_0^1 \frac{2}{\pi^2}\left(\mu - 5\sin(2\pi x)\right) = N \quad \leftarrow \text{Solving this numerically for } \mu \text{ for each } N \text{ value} \quad \rightarrow$$

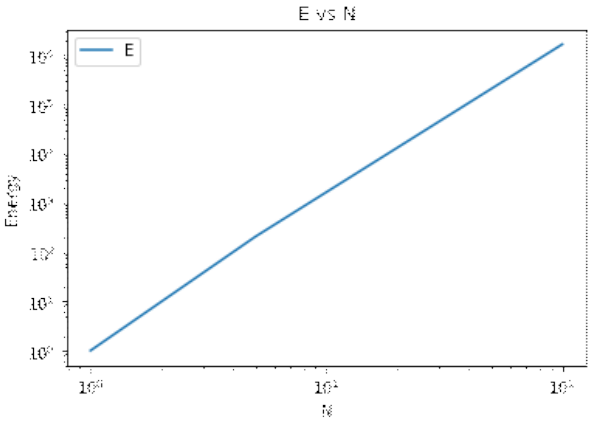
These μ values are then used to find ρ_{min}

↓

That is used to find $E[\rho_{min}]$ by evaluating the integral numerically

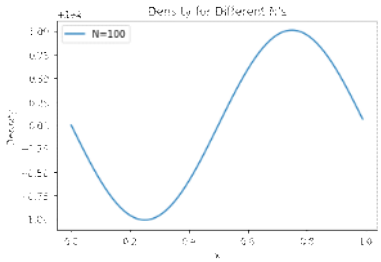
Thus, we can calculate ρ_{min} & then $E[\rho_{min}]$ ← These values are shown below.

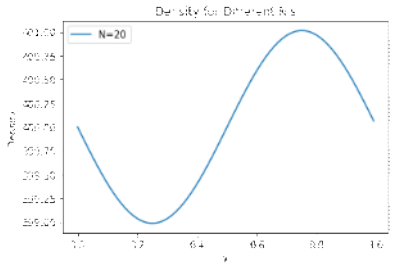




Graph above shows the periodic nature of the densities (plotted for $0 < x < 1$ to showcase the sin curve behaviour)

Log-Log plot of E vs. N. This shows a linear relationship → ie. power law behaviour b/w E and N





HW3

Wednesday, 1 March 2023 1:38 PM

Question 1

Ab-initio phase diagram for H-adsorption on Pt(111) (surface & sub-surface layers)

$$E_{ads} = \left(\underbrace{E^{nH}_{DFT} - E^{Pt(111)}_{DFT}}_{\text{XS surface energy}} - \frac{n}{2} E^{H_2}_{DFT} \right) \frac{1}{n}$$

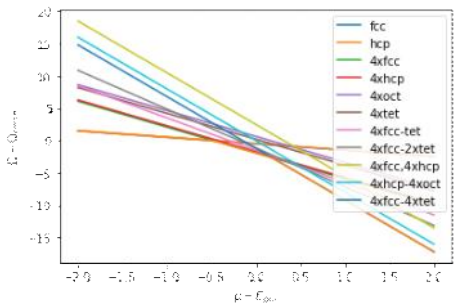
+ XS surface energy : $(E_{DFT} - E^{clean}_{DFT})$ → instead of $(\mu_{Pt} \cdot N_{Pt})$
+ XS surface entropy = 0
  - ↳ One way to better approximate this would be to calculate configurational entropy, i.e. counting all possible arrangements of H adsorbate.
+ Plot : Ω vs. μ (x-axis : $(\mu - E_{gas})$)
+ Plot : T, P scale

+ ZPE's on surface → assume each H-atom has harmonic vibrational freq. of 450, 450 and 900 cm^{-1}
+ Assuming surface is in equilibrium w/ bulk phases $[\mu_i = \mu_i^{gas}$, $\mu_i = \mu_i^{bulk\,metal}]$

$$\gamma = (U - \cancel{TS} - \Sigma \mu_i N_i) \frac{1}{A} = (U - \mu_H n_H - \mu_{Pt} N_{Pt}) \frac{1}{A}$$

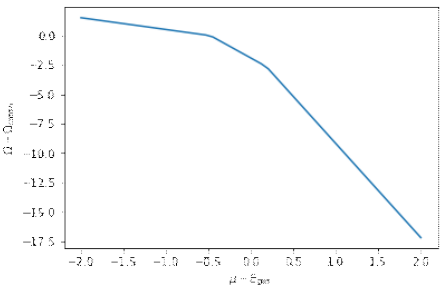
$U \approx (n_H \cdot E_{ads}) + ZPE$; $ZPE = \Sigma \frac{1}{2} \hbar \nu_i$

$\gamma A = (E_{ads} + ZPE - \mu_H n_H) = \Omega$ ← varying μ_H will give values of Ω



This phase diagram shows that certain phase plane and subsurface structures are thermodynamically stable and adsorption on these surfaces will be prefered.

↳ The lowest $(\Omega - \Omega_{clean})$ values can be extrapolated and plotted as shown on the figure on the right



T, P relations : $\mu_H = U^H_{tot} + k_B T - T S^H_{tot}$ (i.g.)

$$U^H_{tot} = U_{vib} + U_{rot} + U_{trans}$$

$$= \sum \left(\frac{1}{2} \hbar \nu_i + \frac{h\nu_i}{e^{\beta h \nu_i} - 1} \right) + \underbrace{k_B T}_{\text{Linear}} + \frac{3}{2} k_B T \quad ; \beta = \frac{1}{k_B T}$$

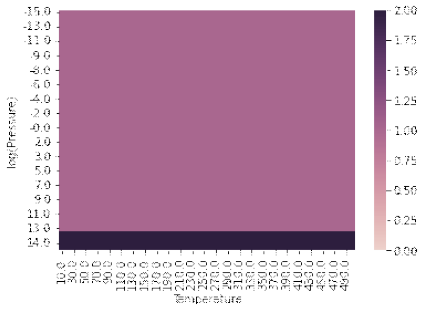
ZPE — Thermal excitation

$$S^H_{tot} = S_{vib} + S_{rot} + S_{trans}$$

$$= \sum_i -\frac{h\nu_i}{2T} - k_B \ln(1 - e^{-h\nu_i \beta}) + \frac{1}{T} \sum_i \left(\frac{1}{2} h\nu_i + \frac{h\nu_i}{e^{\beta h \nu_i} - 1} \right) + \left[k_B \ln \left[\left(\frac{m}{2\pi\beta\hbar^2} \right)^{3/2} \frac{k_B T}{P} \right] + \frac{5}{2} k_B T \right] n_H + k_B \ln \left(\frac{I}{\sigma \beta \hbar^2} \right) + k_B$$

Given a T & P array, μ_H can be calculated and this value can be associated w/ the lowest $(\Omega - \Omega_{clean})$ value and then corresponding H coverage (ML).

↳ This can be used to create a heat map.



There seems to be a bug w/ my code which I could not figure out.

↓

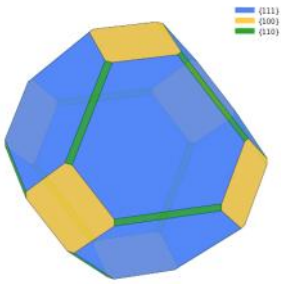
The H-ML should increase w/ pressure and this trend is shown in the graph above, however temp. variation is not explored properly

↓

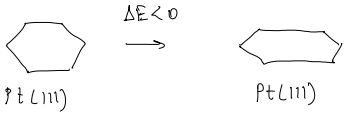
From intuition, the ML should decrease to 0 at higher temp since the system has a much higher energy

Question 2

(a) Wulff diagram is shown



(b) For a favorable interaction, the nanoparticle shape will be "wetted" i.e. become squished



(c) Shape of the nanoparticle change w/ T and P

↳ Using given T & P, a similar approach can be followed where $\Omega - \Omega_{clean}$ can be calculated to given an associated γ value which can be added to the $E_{surf.}$ to obtain new relationship.

↳ Sample calculation is shown in the attached jupyter notebook → could not figure out bug in Q1 which inhibits me from answering this question

HW4

Tuesday, 4 April 2023 3:35 PM

Question 1

Approach 1

Fully quasi-steady state: (Case IV derived in class)

$r_1 = r_2 \Rightarrow k_1 P_A' \theta_* - k_{-1}\theta_A = k_2\theta_A - k_{-2}\theta_B$

$r_2 = -r_3 \Rightarrow k_2\theta_A - k_{-2}\theta_B = k_{-3}\theta_B - k_3 P_B'\theta_*$

$\theta_A + \theta_B + \theta_C = 1$

QE steps 1 & 3:

$(k_2 \sim k_{-2}) << (k_{-1} \sim k_1) \sim (k_3 \sim k_{-3})$

Dividing the equations by k_1:

$$\frac{k_1 P_A' \theta_*}{k_1} - \frac{k_{-1}\theta_A}{k_1} = \frac{k_2}{k_1}\theta_A - \frac{k_{-2}}{k_1}\theta_B \quad (k_2, k_{-2} \to 0)$$

$$P_A'\theta_* - \frac{k_{-1}}{k_1}\theta_A = 0$$

$$\boxed{\theta_A = \frac{k_1}{k_{-1}} P_A' \theta_*}$$

Similarly,

$$\frac{k_2}{k_1}\theta_A - \frac{k_{-2}}{k_1}\theta_B = \frac{k_{-3}}{k_1}\theta_B - \frac{k_3}{k_1} P_B'\theta_* \quad (k_2, k_{-2} \to 0)$$

$$0 = k_{-3}\theta_B - k_3 P_B'\theta_*$$

$$0 = \frac{k_{-3}}{k_3}\theta_B - P_B'\theta_*$$

$$\therefore \boxed{\theta_B = \frac{k_3}{k_{-3}} P_B'\theta_*}$$

The obtained θ_A and θ_B expressions are equal to those obtain when step 1 and step 3 are at QE

↓

This much mean that r_1, r_3 are QE

as shown in case 1 in class

Approach 2

(a) dimensional rate expressions:

$r_1 = r_2 \Rightarrow k_1 P_A' \theta_* - k_{-1}\theta_A = k_2\theta_A - k_{-2}\theta_B$

$r_2 = -r_3 \Rightarrow k_2\theta_A - k_{-2}\theta_B = k_{-3}\theta_B - k_3 P_B'\theta_*$

$\theta_A + \theta_B + \theta_C = 1$

Divide by k_{-1}

$r_1 = r_2 \Rightarrow \frac{k_1}{k_{-1}} P_A'\theta_* - \theta_A = \frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B \quad \cdots (1)$

$r_2 = -r_3 \Rightarrow \frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B = \frac{k_{-3}}{k_{-1}}\theta_B - \frac{k_3}{k_{-1}} P_B'\theta_* \quad \cdots (2)$

↑ Non-dimensional equations

(b) $\theta_A = \theta_{A_1} + \theta_{A_2}$

$\theta_B = \theta_{B_1} + \theta_{B_2}$

$\theta_* = \theta_{*_1} + \theta_{*_2}$

Plugging in to the non-dimensional equations:

(1) $\frac{k_1}{k_{-1}} P_A'(\theta_{*_1} + \theta_{*_2}) - (\theta_{A_1} + \theta_{A_2}) = \frac{k_2}{k_{-1}}(\theta_{A_1} + \theta_{A_2}) - \frac{k_{-2}}{k_{-1}}(\theta_{B_1} + \theta_{B_2})$

$\Rightarrow \frac{k_1}{k_{-1}} P_A'\theta_{*_1} - \theta_{A_1} + \frac{k_1}{k_{-1}} P_A'\theta_{*_2} - \theta_{A_2} = \frac{k_2}{k_{-1}}\theta_{A_1} - \frac{k_{-2}}{k_{-1}}\theta_{B_1} + \frac{k_2}{k_{-1}}\theta_{A_2} - \frac{k_{-2}}{k_{-1}}\theta_{B_2}$

$\Rightarrow \boxed{\frac{k_1}{k_{-1}} P_A'\theta_{*_1} + \theta_{A_1}\left(-1 - \frac{k_2}{k_{-1}}\right) + \frac{k_{-2}}{k_{-1}}\theta_{B_1} = \theta_{A_2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}} P_A'\theta_{*_2} - \frac{k_{-2}}{k_{-1}}\theta_{B_2}}$

$\theta_{A_2} << \theta_{A_1}$

$\theta_{B_2} << \theta_{B_1}$

$\theta_{*_2} << \theta_{*_1}$

(2) $\frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B = \frac{k_{-3}}{k_{-1}}\theta_B - \frac{k_3}{k_{-1}} P_B'\theta_*$

$\frac{k_2}{k_{-1}}(\theta_{A_1} + \theta_{A_2}) - \frac{k_{-2}}{k_{-1}}(\theta_{B_1} + \theta_{B_2}) = -\frac{k_{-3}}{k_{-1}}(\theta_{B_1} + \theta_{B_2}) - \frac{k_3}{k_{-1}} P_B'(\theta_{*_1} + \theta_{*_2})$

$$\frac{k_1}{k_{-1}}\theta'_{A1} - \frac{k_{-1}}{k_1}\theta'_{B1} + \frac{k_{-2}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_2}{k_{-1}}\theta'_{A2} + \frac{k_2}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}P_B'$$

③ $\theta'_{A1} + \theta'_{A2} + \theta'_{B1} + \theta'_{B2} + \theta'_{*1} + \theta'_{*2} = 1$

$$(\theta'_{A1} + \theta'_{B1} + \theta'_{*1}) + (\theta'_{A2} + \theta'_{B2} + \theta'_{*2}) = 1$$

(c) From approach #1: $(k_2 \sim k_{-2}) << (k_1 \sim k_{-1}) \sim (k_3 \sim k_{-3})$

① $\frac{k_1}{k_{-1}}P_A'\theta'_{*1} + \theta'_{A1}\left(-1 - \frac{k_2}{k_{-1}}\right) + \frac{k_{-2}}{k_{-1}}\theta'_{B1} = \theta'_{A2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}}P_A'\theta'_{*2} - \frac{k_{-2}}{k_{-1}}\theta'_{B2}$

$$\frac{k_1}{k_{-1}}P_A'\theta'_{*1} - \theta'_{A1} = \theta'_{A2} - \frac{k_1}{k_{-1}}P_A'\theta'_{*2}$$

For order 1: $\frac{k_1}{k_{-1}}P_A'\theta'_{*1} - \theta'_{A1} = 0 \Rightarrow \theta'_{A1} = \frac{k_1}{k_{-1}}P_A'\theta'_{*1}$

② $\frac{k_2}{k_{-1}}\theta'_{A1} - \frac{k_{-1}}{k_1}\theta'_{B1} + \frac{k_{-2}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_2}{k_{-1}}\theta'_{A2} + \frac{k_2}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}$

$$\frac{k_{-3}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}$$

For order 1: $\frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B1}$

③ For site balance: $\underbrace{(\theta'_{A1} + \theta'_{B1} + \theta'_{*1})}_{=1} + \underbrace{(\theta'_{A2} + \theta'_{B2} + \theta'_{*2})}_{=0} = 1$

$\theta'_{A1} + \theta'_{B1} + \theta'_{*1} = 1$

∴ For first order:

$\theta'_{A1} = \frac{k_1}{k_{-1}}P_A'\theta'_{*1}$; $\frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B1}$

$$\frac{k_1}{k_{-1}}P_A'\theta'_{*1} + \frac{k_3}{k_{-3}}P_B'\theta'_{*1} + \theta'_{*1} = 1$$

$$\theta'_{*1} = \frac{1}{\left(\frac{k_1}{k_{-1}}P_A' + \frac{k_3}{k_{-3}}P_B' + 1\right)}$$

This is the first order approximation

For second-order expression, we cannot make the simplification assumptions → Therefore have to solve for the earlier solution.

For second order: ① $0 = \theta'_{A2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}}P_A'\theta'_{*2} - \frac{k_{-2}}{k_{-1}}\theta'_{B2}P_B' \Rightarrow \theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2} - k_{-2}\theta'_{B2}P_B' = 0 \Rightarrow \theta'_{B2} = \frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{P_B'k_{-2}}$

② $0 = -\frac{k_2}{k_1}\theta'_{A2} + \frac{k_2}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2} \Rightarrow -k_2\theta'_{A2} + k_{-2}\theta'_{B2} - k_{-3}\theta'_{B2} - k_3\theta'_{*2} = 0 \Rightarrow -k_2\theta'_{A2} + (k_{-2}-k_{-3})\left(\frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{P_B'k_{-2}}\right) - k_3\theta'_{*2} = 0$

$$\Rightarrow -k_2\theta'_{A2} + \frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B'k_{-2}}\theta'_{A2} + \frac{(k_{-2}-k_{-3})}{k_{-2}}k_1P_A'\theta'_{*2} - k_3\theta'_{*2} = 0$$

$$\theta'_{A2}\underbrace{\left(-k_2 + \frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B'k_{-2}}\right)}_{A_1} + \theta'_{*2}\underbrace{\left[\frac{(k_{-2}-k_{-3})}{k_{-2}}k_1P_A' - k_3\right]}_{A_2} = 0$$

$$\theta'_{A2} = \frac{-\theta'_{*2}[A_2]}{(A_1)}$$

③ $\theta'_{A2} + \theta'_{B2} + \theta'_{*2} = 0$

Plugging everything back in expression ②:

$$-\frac{\theta'_{*2}[A_2]}{[A_1]} + \frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{k_{-2}} + \theta'_{*2} = 0$$

$$-\frac{\theta_{*2}[A_2]}{[A_1]} - \frac{(k_{-1}+k_2)}{k_{-2}}\frac{[A_2]}{[A_1]}\theta_{*2} + \frac{k_1 P_A'\theta_{*2}}{k_{-2}} + \theta_{*2} = 0 \quad \Rightarrow \boxed{\theta_{*2} = 0}$$

Plugging in θ_{*2} will give $\theta_{B2} = f(\theta_{A2})$ only!

$$\therefore \quad \theta_{A_2} = -\theta_{*2}\frac{\left[\frac{(k_{-2}-k_{-3})}{k_{-2}}\frac{k_1 P_A'}{} - k_3\right]}{\left[-k_2 + \frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B' k_{-2}}\right]} \quad ; \quad \theta_{B2} = \frac{\theta_{A_2}(k_{-1}+k_2) + k_1 P_A'\theta_{*2}}{P_B' k_{-2}}$$

(d) $r_1 = r_2 = r_3$

$r_2 = k_2\theta_A - k_{-2}\theta_B$

For first order:

$$r_2^{(1)} = k_2\theta_{A1} - k_{-2}\theta_{B1} \quad ; \quad \theta_{A1} = \frac{k_1}{k_{-1}}P_A'\theta_{*1} \quad ; \quad \frac{k_3}{k_{-1}}P_B'\theta_{*1} = -\frac{k_{-3}}{k_{-1}}\theta_{B1}$$

$$= \frac{k_2 k_1}{k_{-1}}P_A'\theta_{*1} - k_{-2}\frac{k_3}{k_{-3}}P_B'\theta_{*1}$$

$$r_2^{(1)} = \left[k_1 K_1 P_A' - k_{-2}K_3 P_B'\right]\theta_{*1} \quad ; \quad \theta_{*1} = \frac{1}{\left(\frac{k_1}{k_{-1}}P_A' + \frac{k_3}{k_{-3}}P_B' + 1\right)}$$

This first order approximation matches the QE expressions obtained in approach 1 → Exact solution

(e) First order solution represents step 1 and step 3 as QE

↳ This means that the forward & reverse rates are much faster than that of step 2 so it therefore "appears" that step 1 & 2 are at equilibrium

↳ The first order perturbation represents that exact qE solution

↳ Given this analysis, the net rate should be first order (since second order solution gives the trivial case)